Thinking Mathematics Grade 7

Theory – Worked examples – 250 practice and challenge problems

Student book

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ThinkAlgo Tip.

- Each chapter contains: **theory** (with explanations and proofs), **worked examples** with guided solutions, and **50 problems**: *Core practice* followed by *Extension and challenge*.
- Problems marked ⋆ are hard, ⋆⋆ very hard.
- Try each problem on your own first, then compare with the solution.

1 Algebra and Equations

The big idea. Algebra is *the language of unknown numbers*. Instead of saying at length "some number, add 5, then double it", we simply write $2(x+5)$. The letter x (called a **variable** or an **unknown**) holds the place of a number that we do not know yet or that may change.
In this topic you will practise four core skills:

1. **Translate** words into expressions and formulas ("more than", "times", "sum", "difference"...).
2. **Substitute** numbers to evaluate an expression, or to check whether an equation is true or false.
3. **Solve equations** with the *balance principle*: whatever you do to one side, do exactly the same to the other side.
4. **Spot patterns**: sequences, tables of values, ratios, and the order of numbers.

Every word problem follows the same route: *read carefully* → *choose an unknown* → *write an expression or an equation* → *solve* → *check* → *answer*.

① Variables, algebraic expressions and formulas

1. Variables and expressions. An **algebraic expression** is built from numbers, letters (variables) and the operations $+$, $-$, $\times$, $\div$, powers and brackets. Examples: $3x-1$; $2(n-1)$; $120+30m$; xy.
Short-hand. Between a number and a letter, or between two letters, we leave out the multiplication sign: $3x$ means $3\cdot x$; xy means $x\cdot y$; $2(n-1)$ means $2\cdot(n-1)$. The number is written before the letter.
2. Value of an expression. To evaluate an expression for given values of the variables, we *replace* each letter by its number (best put in brackets), then calculate in the correct order of operations.
Example: for $x=3$, $3x-1=3\cdot 3-1=9-1=8$; $2x^2=2\cdot 3^2=2\cdot 9=18$.
3. Translating words into expressions.

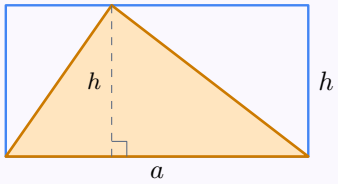
Words	Expression
5 more than x; x increased by 5	$x+5$
5 less than x; x decreased by 5	$x-5$
3 times x; triple x	$3x$
half of x; x divided by 2	$\frac{x}{2}$ or $x\div 2$
start with 120, add 30 each time, m times	$120+30m$
three consecutive whole numbers, the smallest is n	$n,\ n+1,\ n+2$

4. A formula is an equation that tells us how to calculate one quantity from other quantities. Useful geometry formulas (side lengths a, b, c):

- Perimeter of a triangle: $P=a+b+c$ (the sum of the three sides).

- Rectangle with length a and width b: perimeter $P = 2(a + b)$, area $A = a \cdot b$.
- Triangle with base a and height h: area $A = \dfrac{a \cdot h}{2}$.
- Rectangular prism $a \times b \times c$: total surface area $S = 2(ab + bc + ca)$, volume $V = abc$.

Why is it true? Why is the area of a triangle half of (base $\times$ height)? Draw a rectangle with one side equal to the base a of the triangle and the other side equal to the height h. The height splits the triangle into two right triangles; each right triangle is exactly half of a smaller rectangle (in the picture, the shaded and white parts match in pairs). So the triangle covers exactly half of the big rectangle: $A = \dfrac{a \cdot h}{2}$.



Similarly, the perimeter of a rectangle is $a + b + a + b = 2a + 2b = 2(a + b)$ because a rectangle has two equal long sides and two equal short sides. □

ThinkAlgo Tip. When substituting, *always put the number in brackets*, especially a negative number: for $x = -2$, $x^2 = (-2)^2 = 4$, while $3x = 3 \cdot (-2) = -6$. Brackets protect you from sign mistakes.

Common mistake. $3x$ with $x = 4$ is $3 \cdot 4 = 12$, *not* "34". And x^2 (which is $x \cdot x$) is different from $2x$ (which is $x + x$): for $x = 5$, $x^2 = 25$ but $2x = 10$.

ThinkAlgo Tip. Write everything in terms of a few letters. Consider a list in which every number (from the third on) is the sum of the two before it. If the first two numbers are a and b, the next ones are

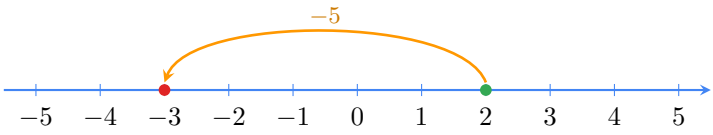
$$a, \quad b, \quad a + b, \quad b + (a + b) = a + 2b, \quad (a + b) + (a + 2b) = 2a + 3b, \ \ldots$$

For instance, with $a = 11$, $b = 20$ the fifth number is $2 \cdot 11 + 3 \cdot 20 = 22 + 60 = 82$, just as in the list $11, 20, 31, 51, 82$. Writing the numbers in terms of a and b turns a question about the list into an equation.

Note for parents. Play the "number machine" game with your child: you say a rule ("multiply by 3, then subtract 1"), your child writes it as the expression $3x - 1$, and then you take turns feeding numbers into the machine. This game makes the letter x feel friendly instead of scary.

② Integers, order of operations and the distributive law

1. Integers. The numbers $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$ are called integers. On the number line, a number further to the left is smaller. The opposite of a is $-a$; we have $a + (-a) = 0$.



Example: $2 + (-5) = -3$: starting at 2, move 5 steps to the left and arrive at -3.

2. Adding and subtracting integers.

- Two numbers with the same sign: add the sizes and keep the common sign: $(-11) + (-26) = -37$.
- Two numbers with different signs: subtract the smaller size from the larger one and take the sign of the number with the larger size: $11 + (-37) = -26$.
- Subtracting means adding the opposite: $a - b = a + (-b)$. Example: $-26 - 11 = -26 + (-11) = -37$.

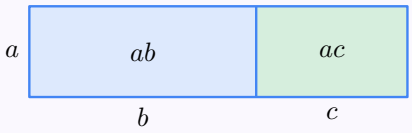
3. Order of operations. Brackets first → powers → multiplication and division (left to right) → addition and subtraction (left to right). Example: $6 + 2 \cdot 3 = 6 + 6 = 12$ (not $8 \cdot 3$).

4. The distributive law. $a(b+c) = ab+ac$ and $a(b-c) = ab-ac$. *Examples:* $3(T+4) = 3T+12$; $20 \cdot 3 - 20 \cdot 2.25 = 20 \cdot (3 - 2.25) = 20 \cdot 0.75 = 15$.

5. Simplifying expressions. Add "like terms": $3x+5x = 8x$; $(a+1)+(b+1)+(c+1)+(d+1) = (a+b+c+d)+4$.

6. Points in the coordinate plane. The point (x, y) has x-coordinate x and y-coordinate y. Moving k units to the right increases x by k; moving k units up increases y by k (moving left or down decreases them).

Why is it true? Why is $a(b+c) = ab+ac$? Take a rectangle with width a and length $b+c$. Its area is $a(b+c)$. But we can also cut it into two smaller rectangles with areas ab and ac. It is the same area, so $a(b+c) = ab+ac$.



□

Common mistake. $3(T+4)$ is *not* equal to $3T+4$: you must multiply 3 by *each* term inside the brackets, $3(T+4) = 3T+3 \cdot 4 = 3T+12$. Likewise $-(a-b) = -a+b$ (change the sign of *every* term in the brackets).

③ Equations and the balance principle

1. Equations. A statement of equality containing an unknown, such as $2(x+8)=38$, is called an **equation**. A number that makes both sides equal when it replaces the unknown is called a **solution**. To *solve an equation* means to find its solutions.
2. The balance principle. Imagine the equation as a pair of scales in balance. The scales stay balanced if we:

- add (or subtract) *the same number* on both sides;
- multiply (or divide) both sides by *the same non-zero number*.

A familiar consequence is the **rule for moving terms**: when a term moves from one side to the other, its *sign changes*.



Example: two identical boxes plus a 3 kg weight balance 11 kg: $2x+3=11$. Remove the 3 kg weight from both pans: $2x=11-3=8$. Halve both pans: $x=8\div 2=4$. Each box weighs 4 kg.
3. Substitution on scales. If A on one pan balances B, then on any other pair of scales we may replace A by B and the scales stay balanced (just like replacing an expression by an equal expression).
4. Always check. After solving, substitute the solution into the original equation to make sure both sides are equal.

Why is it true? Why does a term change sign when it moves across? From $2x+3=11$, subtract 3 from both sides (the scales stay balanced): $2x+3-3=11-3$, that is, $2x=11-3$. Looking at the result, the term $+3$ on the left seems to have "jumped" to the right and become -3. In fact nothing "jumps": we only subtracted the same number from both sides. □

Common mistake. Forgetting to change the sign: from $x+11=-26$, writing $x=-26+11$ is **wrong**. The correct step is $x=-26-11=-37$.

ThinkAlgo Tip. In a multiple-choice question with an equation, you can *substitute each option in turn* into the equation. It is slower but very safe, and it is also the best way to check.

④ Solving word problems with equations

Four steps for a word problem.

1. **Choose an unknown**: call the quantity you want (or a related one) x; write its unit and any condition (for example $x>0$, x is a whole number).
2. **Express** the other quantities in terms of x.
3. **Write an equation** from the conditions of the problem and **solve** it.
4. **Compare** with the conditions, **check**, and **answer** the actual question.

Common types.

- **Consecutive integers:** $n,\ n+1,\ n+2$ (or, more neatly, $n-1,\ n,\ n+1$). Consecutive even

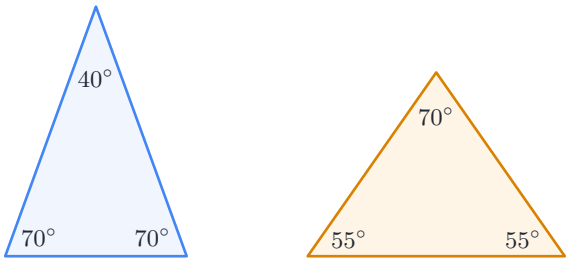
(odd) numbers differ by 2: $n,\ n+2,\ n+4$.

- **Sums in pairs:** three objects a, b, c weighed two at a time give $a+b$, $b+c$, $c+a$. Adding the three results: $(a+b)+(b+c)+(c+a)=2(a+b+c)$, so *the total of the three objects is half of the sum of the results.* To find one object, subtract the pair formed by the other two from the total.
- **Angles in a triangle:** the three angles of a triangle add up to 180°; an isosceles triangle has two equal base angles.
- **Average:** average = sum of the values ÷ number of values; average price = total cost ÷ total mass.

Two small examples.

a) An isosceles triangle with a 70° angle. The 70° angle may be a *base angle* or the *apex angle*, so we must look at two cases:

- If 70° is a base angle: the base angles are 70° and 70°, and the apex angle is $180°-70°-70°=40°$.
- If 70° is the apex angle: the two base angles are equal and add up to $180°-70°=110°$, so each is 55°.



b) Average price. You buy 300 g of type I candy for \$3 and 200 g of type II candy for \$4. The average price of 100 g of candy is $(3+4)\div(300+200)\cdot 100=7\div 500\cdot 100=1.4$, that is, \$1.40. You must *not* average the two prices \$1 and \$2 per 100 g (that gives \$1.50, which is wrong), because different amounts of the two candies were bought.

Why is it true? Three consecutive integers: the middle one equals the sum divided by 3. Call the three numbers $n-1$, n, $n+1$. Their sum is $(n-1)+n+(n+1)=3n$ (the -1 and $+1$ cancel). So the middle number is $n=\text{sum}\div 3$. For example, if the sum is 153, the middle number is 51 and the three numbers are $50, 51, 52$. □

ThinkAlgo Tip. Choosing the *middle number* as the unknown usually gives the neatest equation. For "sums in pairs" problems, do not rush to use three unknowns: *add everything up* first.

Common mistake. Solving the equation is not the end of the problem! Re-read the question: does it ask for the *largest* number, the *lightest* object, or the *total*...? Many students find the correct x but choose the wrong option because they answer a different question.

Note for parents. When your child is stuck, ask: "What don't we know yet? Let's call it x. Now, what does the problem tell us about x?" Let your child say it in words first, and only then write it as an equation.

5 Patterns, tables of values and arithmetic sequences

1. Arithmetic sequences. A list of numbers in which neighbouring numbers always differ by the same number d (called the **common difference**) is called an arithmetic sequence. If the first term is a, then

$$\text{the } n\text{-th term } = a + (n-1) \cdot d.$$

Example: the sequence $7, 10, 13, 16, \ldots$ has $a = 7$, $d = 3$; its n-th term is $7 + 3(n-1) = 3n + 4$.

2. From a table of values to an expression. If each time n increases by 1 the value increases by d, the expression has the form $d \cdot n +$ (some number). Find "some number" by substituting $n = 1$.

n	1	2	3	4	5
value	4	7	10	13	16

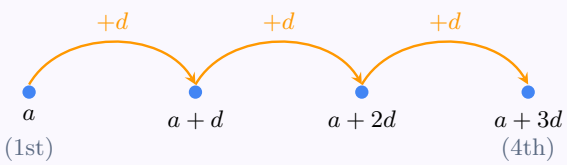
Each time n increases by 1, the value increases by 3, so the expression has the form $3n + b$. Substituting $n = 1$: $3 + b = 4$, so $b = 1$. Hence value $= 3n + 1$.

3. Is a number in the sequence? The number N belongs to the sequence $a,\ a+d,\ a+2d, \ldots$ exactly when $N-a$ is divisible by d (and $N \geq a$). For a sequence counting down, $a,\ a-d,\ a-2d, \ldots$, look at $a - N$ instead.

4. Common terms of two sequences; repeating events. If two arithmetic sequences with differences d_1 and d_2 have a common term, then their common terms are exactly $\text{lcm}(d_1, d_2)$ apart. Similarly, events that repeat every p minutes and every q minutes happen together again every $\text{lcm}(p, q)$ minutes.

ThinkAlgo Tip. *An example of repetition.* The first bus returns to the station every 12 minutes, the second bus every 20 minutes. At 7:00 a.m. both are at the station. The first bus returns after $12, 24, 36, 48, \mathbf{60}, \ldots$ minutes; the second after $20, 40, \mathbf{60}, \ldots$ minutes. The two buses meet at the station again every $\text{lcm}(12, 20) = 60$ minutes, that is, at 8:00 a.m., 9:00 a.m., ... Similarly, if the two sequences $7, 10, 13, \ldots$ (difference 3) and $2026, 2021, 2016, \ldots$ (difference 5) have a common term, then their common terms are $\text{lcm}(3, 5) = 15$ apart.

Why is it true? Why is the n-th term $a + (n-1)d$? Going from the 1st term to the n-th term, we cross $n - 1$ "gaps", and each gap adds d. It is like walking from the 1st lamp post to the 5th: you only pass 4 gaps.



From the 1st term to the 4th term there are 3 steps of $+d$, so the 4th term is $a + 3d$. □

Common mistake. Checking a formula against only *one* column of the table is not enough. For example, for the table $1, 3, 5, 7, 9$, both $3n - 2$ and $2n - 1$ give the value 1 when $n = 1$, but for $n = 2$ we get $3n - 2 = 4 \neq 3$. Test *at least two or three columns.*

⑥ Ratios, units of measurement and uniform motion

1. Directly proportional quantities. If one scoop of food feeds 8 fish, then 2 scoops feed 16 fish and 4 scoops feed 32 fish: when the number of scoops is multiplied by some number, the number of fish is multiplied by the same number. We say y is **directly proportional** to x with constant k if $y = k \cdot x$. The usual method is the **unit method** (work out the value for 1 part first, then multiply).

2. Uniform motion. With s the distance, v the speed and t the time:

$$s = v \cdot t, \qquad t = \frac{s}{v}, \qquad v = \frac{s}{t}.$$

Use matching units: if v is in km/h, then s is in km and t is in hours.

Two people moving *for the same length of time*: whoever is k times as fast covers k times the distance.

Running in opposite directions around a circle: from one meeting to the next, the *total* distance run by the two people is exactly one lap. If the first person is k times as fast as the second, then during each such stretch the first runs $\frac{k}{k+1}$ of a lap and the second runs $\frac{1}{k+1}$ of a lap.

3. Converting units of time. 1 minute = 60 seconds; 1 hour = 60 minutes = $60 \times 60 = 3\,600$ seconds; 1 day = 24 hours; 1 week = 7 days. Therefore 1 week = 7×24 hours = $7 \times 24 \times 60$ minutes = $7 \times 24 \times 60 \times 60$ seconds.

4. Money. Number of bills = total amount ÷ value of one bill. Difference after n times = $n \times$ (higher price − lower price).

ThinkAlgo Tip. Read a product such as $60 \times 60 \times 24 \times 7$ as a *chain of unit conversions*, from right to left: 7 (days in a week) ×24 (hours in a day) ×60 (minutes in an hour) ×60 (seconds in a minute). The result is the number of seconds in one week.

Why is it true? Why can we work out the difference "per time" and then multiply? By the distributive law: $20 \times 3 - 20 \times 2.25 = 20 \times (3 - 2.25) = 20 \times 0.75 = 15$. Both ways always give the same result, but the second is much quicker to do in your head. □

Common mistake. In motion problems, read carefully "how much has been travelled" and "how much is left". After half of a 510 km trip, the distance left is $510 \div 2 = 255$ km, not 510 km.

⑦ Equivalent fractions and a fraction of a quantity

1. The basic property. Multiplying (or dividing) the numerator and the denominator of a fraction by the same non-zero number gives an equal fraction: $\frac{3}{7} = \frac{3 \cdot 9}{7 \cdot 9} = \frac{27}{63}$.

2. Recognising equal fractions. For $b \neq 0$, $d \neq 0$: $\frac{a}{b} = \frac{c}{d}$ exactly when $a \cdot d = b \cdot c$ ("cross-multiplying").

3. Fractions with the same numerator. If $\frac{1}{p} = \frac{1}{q}$ (or $\frac{m}{p} = \frac{m}{q}$ with $m \neq 0$), then $p = q$. Two equal fractions with equal numerators also have equal denominators.

4. A fraction of a quantity. To find $\frac{m}{n}$ of a number A, calculate $A \cdot \frac{m}{n} = A \div n \cdot m$. *Example:* $\frac{3}{7}$ of 168 km is $168 \div 7 \cdot 3 = 24 \cdot 3 = 72$ km. The rest corresponds to $1 - \frac{3}{7} = \frac{4}{7}$.

Why is it true? Why does $\frac{1}{p} = \frac{1}{q}$ imply $p = q$? Cross-multiply: $1 \cdot q = p \cdot 1$, that is, $q = p$. In other words: if a cake is cut into p pieces and another identical cake into q pieces, and the pieces are the same size, then the numbers of pieces must be the same. □

⑧ Comparing and reasoning about order

1. The transitive property. If $a < b$ and $b < c$, then $a < c$. This lets us line up several numbers: $a < b < c < d$.

2. The "set it equal to k" technique. When several expressions are equal, for example $a - 2 = b + 2 = c + 1 = d - 3$, call the common value k:

$$a = k + 2, \qquad b = k - 2, \qquad c = k - 1, \qquad d = k + 3.$$

Now every number is compared with the same reference point k; ordering the "extras" $-2 < -1 < +2 < +3$ we get $b < c < a < d$.



3. Logical reasoning about order. For sentences like "Anna owns more cats than Ben", translate them into symbols ($A > B$), write down all the conditions and link them with the transitive property. A statement that "*must be true*" has to be true in *every* situation that fits the problem; a single counter-example is enough to reject it.

ThinkAlgo Tip. *A reasoning example.* Anna is taller than Ben; Chloe is shorter than Ben; David is taller than Chloe. Who is the shortest? In symbols (each letter is one child's height): $A > B$, $B > C$, $D > C$. From $A > B > C$ and $D > C$ we see that A, B and D are all greater than C: **Chloe is the shortest**. But we do *not* know whether David is taller or shorter than Anna (for instance, if Anna is 1.6 m, Ben 1.5 m and Chloe 1.2 m, then David being 1.7 m or 1.3 m both fit all the conditions), so the statement "David is the tallest" is *not necessarily* true.

Common mistake. Seeing that $a - 2$ and $b + 2$ are equal, many students conclude too quickly that "a is smaller because something is subtracted". The opposite is true: a must *lose* 2 to equal b *plus* 2, so a is exactly 4 more than b.

Note for parents. Use real objects: "If your bag loses 2 marbles, it has the same number as my bag plus 2 marbles. Which bag has more?" Let your child check with marbles. A concrete picture keeps the child from being fooled by plus and minus signs.

⑨ Worked examples

Example 1 (Translating words and evaluating an expression)

a) Evaluate the expression $A = 2x^2 - 3x + 1$ for $x = -2$.
b) Emma has \$50 in her money box. Every week she puts in another \$20. Write an expression for the amount (in dollars) in the money box after t weeks, then work out that amount after 6 weeks.

Guided solution.

Step 1. Part a: substitute $x = -2$, remembering the brackets

$$A = 2 \cdot (-2)^2 - 3 \cdot (-2) + 1.$$

Step 2. Calculate in order: powers → multiplication → addition and subtraction
$(-2)^2 = (-2) \cdot (-2) = 4$, so $2 \cdot 4 = 8$. $3 \cdot (-2) = -6$, so $-3 \cdot (-2) = +6$. Hence $A = 8 + 6 + 1 = 15$.

Step 3. Part b: find the pattern with a small table

Weeks	0	1	2	t
Amount (dollars)	50	$50 + 20$	$50 + 2 \cdot 20$	$50 + 20t$

The starting amount stays the same; the added part is 20 times the number of weeks. Expression: $50 + 20t$ (dollars).

Step 4. Substitute $t = 6$
$50 + 20 \cdot 6 = 50 + 120 = 170$ (dollars).

Common mistake. Do not write $(50 + 20) \cdot t$: the starting amount is there only *once*; it must not be multiplied by the number of weeks.

Answer: a) $A = 15$; b) $50 + 20t$ (dollars); after 6 weeks there is \$170.

Example 2 (Solving equations with the balance principle)

Solve the equations: **a)** $5x - 7 = 2x + 11$; **b)** $\dfrac{x}{12} = \dfrac{5}{4}$.

Guided solution.

Step 1. Part a: collect the terms with x on the left
Subtract $2x$ from both sides: $5x - 2x - 7 = 11$, that is, $3x - 7 = 11$.

Step 2. Collect the numbers on the right
Add 7 to both sides: $3x = 11 + 7 = 18$.

Step 3. Divide both sides by 3
$x = 18 \div 3 = 6$.

Check: Left side: $5 \cdot 6 - 7 = 30 - 7 = 23$. Right side: $2 \cdot 6 + 11 = 12 + 11 = 23$. Both sides are equal.

Step 4. Part b: use a common denominator (or cross-multiply)

We have $\frac{5}{4} = \frac{5 \cdot 3}{4 \cdot 3} = \frac{15}{12}$. The fractions $\frac{x}{12}$ and $\frac{15}{12}$ have the same denominator and are equal, so $x = 15$. (Cross-multiplying also works: $4x = 12 \cdot 5 = 60$, $x = 60 \div 4 = 15$.)

Recall. $\frac{a}{b} = \frac{c}{d}$ exactly when $ad = bc$ $(b, d \neq 0)$.

Answer: a) $x = 6$; **b)** $x = 15$.

Example 3 (Consecutive even numbers)

The sum of four consecutive even numbers is 116. Find the largest of the four numbers.

Guided solution.

Step 1. Choose an unknown

Let the smallest even number be n (n is even). Consecutive even numbers differ by 2, so the four numbers are $n,\ n+2,\ n+4,\ n+6$.

Step 2. Write an equation

$n + (n+2) + (n+4) + (n+6) = 116$, that is, $4n + 12 = 116$.

Step 3. Solve

$4n = 116 - 12 = 104$, $n = 104 \div 4 = 26$. The four numbers are $26,\ 28,\ 30,\ 32$.

Check: $26 + 28 + 30 + 32 = 116$. Correct.

Step 4. Answer the actual question

The question asks for the *largest* number: it is 32.

ThinkAlgo Tip. Mental shortcut: the four numbers are equally spaced, so their average $116 \div 4 = 29$ lies exactly halfway between the two middle numbers 28 and 30. The four numbers follow at once.

Answer: 32

Example 4 (Weighing in pairs)

Three friends, Anna, Ben and Chloe, step on a scale two at a time. Anna and Ben weigh 72 kg; Anna and Chloe weigh 78 kg; Ben and Chloe weigh 80 kg. How many kilograms does each child weigh?

Guided solution.

Step 1. Choose an unknown and express the other quantities with it

Let Anna's mass be x (kg), $x > 0$. Since Anna and Ben weigh 72 kg, Ben weighs $72 - x$ (kg). Since Anna and Chloe weigh 78 kg, Chloe weighs $78 - x$ (kg).

Step 2. Form an equation from the third weighing

Ben and Chloe weigh 80 kg, so

$$(72 - x) + (78 - x) = 80.$$

Step 3. Solve the equation

Simplify: $72 + 78 - x - x = 150 - 2x$, so $150 - 2x = 80$. Move terms across: $2x = 150 - 80 = 70$,

hence $x = 70 \div 2 = 35$.

Step 4. Find the masses of the other two children

Anna weighs 35 kg; Ben weighs $72 - 35 = 37$ (kg); Chloe weighs $78 - 35 = 43$ (kg).

> **ThinkAlgo Tip. Method 2.** Let the three masses be a, b, c (kg). Add $a + b = 72$, $a + c = 78$, $b + c = 80$ side by side (each letter appears twice): $2(a + b + c) = 230$, so $a + b + c = 115$. Hence $c = 115 - 72 = 43$, $b = 115 - 78 = 37$, $a = 115 - 80 = 35$.

Check: $35 + 37 = 72$, $35 + 43 = 78$, $37 + 43 = 80$. All three weighings match.

Answer: Anna 35 kg, Ben 37 kg, Chloe 43 kg.

Example 5 (Arithmetic sequences)

Consider the sequence 4, 7, 10, 13, ... (each number is 3 more than the one before). **a)** Write an expression for the n-th term. **b)** Find the 50th term. **c)** Is 2026 in the sequence? If so, which term is it?

Guided solution.

Step 1. The n-th term

First term $a = 4$, common difference $d = 3$: the n-th term is $4 + 3(n - 1) = 4 + 3n - 3 = 3n + 1$.

> **Recall.** The n-th term of an arithmetic sequence $= a + (n - 1)d$.

Step 2. The 50th term

$3 \cdot 50 + 1 = 151$.

Step 3. Test the number 2026

We need a whole number n with $3n + 1 = 2026$, that is, $3n = 2025$. Since $2025 \div 3 = 675$ exactly, $n = 675$.

Check: $3 \cdot 675 + 1 = 2025 + 1 = 2026$. Correct.

> **Common mistake.** Do not forget to check that the division is exact. For instance, 2027 is *not* in the sequence because 2026 is not divisible by 3.

Answer: a) $3n + 1$; b) 151; c) yes, it is the 675th term.

Example 6 (Motion, fractions and unit conversion)

A train travels at a constant speed of 72 km/h on a 360 km route. It has already covered $\frac{2}{3}$ of the route. How much longer (in hours and minutes) does the train need to reach its destination?

Guided solution.

Step 1. Distance already covered

$\frac{2}{3}$ of 360 km is $360 \div 3 \cdot 2 = 120 \cdot 2 = 240$ km.

Step 2. Distance left

$360 - 240 = 120$ km. (Or: the remaining part is $1 - \frac{2}{3} = \frac{1}{3}$ of the route, $360 \div 3 = 120$ km.)

Step 3. Time left

Recall. $s = v \cdot t$, so $t = \dfrac{s}{v}$ (time = distance ÷ speed).

Let t (hours) be the time the train still needs. Then $72 \cdot t = 120$, so $t = 120 \div 72 = \dfrac{120}{72} = \dfrac{5}{3}$ hours $= 1\dfrac{2}{3}$ hours.

Step 4. Convert to hours and minutes

$\dfrac{2}{3}$ hour $= \dfrac{2}{3} \cdot 60 = 40$ minutes. So $t = 1$ hour 40 minutes.

Check: In 1 hour 40 minutes $= \dfrac{5}{3}$ hours, the train covers $72 \cdot \dfrac{5}{3} = 24 \cdot 5 = 120$ km, exactly the distance left.

Answer: 1 hour 40 minutes.

Example 7 (A balance-scale problem)

Both scales below are balanced. Identical shapes have equal weights.



How many circles ○ balance two squares □□?

Guided solution.

Step 1. Write each scale as an equation

Let S, T, C be the weights of a square, a triangle and a circle. Scale 1: $S = T + C$. Scale 2: $S + C = 3T$.

Step 2. Substitute scale 1 into scale 2

On scale 2, replace the square by "one triangle and one circle" (allowed by scale 1): $T + C + C = 3T$.

Step 3. Remove identical things from both pans

Remove one triangle from each pan: $C + C = T + T$, that is, $2C = 2T$, so $C = T$: a circle weighs the same as a triangle.

Step 4. Answer

$S = T + C = C + C = 2C$, so $2S = 4C$: two squares balance 4 circles.

Check: Try $C = T = 1$, $S = 2$: scale 1: $2 = 1 + 1$; scale 2: $2 + 1 = 3$. Both are correct.

Answer: 4 circles.

Example 8 (Ordering with the "set it equal to k" technique)

The positive integers a, b, c (all greater than 5) satisfy $\frac{2}{a+3} = \frac{2}{b-1} = \frac{2}{c+5}$. Arrange a, b, c in increasing order.

Guided solution.

Step 1. Equal fractions with the same numerator have equal denominators

The three fractions all have numerator 2 and are equal, so their denominators are equal: $a+3 = b-1 = c+5$.

> **Recall.** If $\frac{m}{p} = \frac{m}{q}$ with $m \neq 0$, then $p = q$.

Step 2. Call the common value k

$a+3 = k \Rightarrow a = k-3$; $\quad b-1 = k \Rightarrow b = k+1$; $\quad c+5 = k \Rightarrow c = k-5$.

Step 3. Compare with the reference point k

$k-5 < k-3 < k+1$, so $c < a < b$.

Check: Take $k = 12$: $a = 9$, $b = 13$, $c = 7$. Then $\frac{2}{12} = \frac{2}{12} = \frac{2}{12}$ and indeed $7 < 9 < 13$.

Answer: $c < a < b$.

Core practice

Problem 1

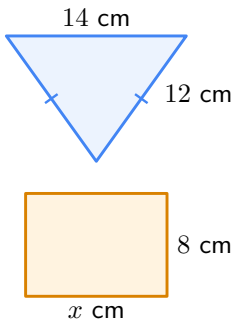
Carrie drives a car at a constant speed of 85 km/h on a 510 km trip. She has now covered exactly half of the distance. How many more hours does she need to finish the trip?

(A) 5 hours (B) 425 hours (C) 12 hours (D) 1.5 hours (E) 3 hours

Your work:

Problem 2

In the diagram, the triangle has one side of length 14 cm and two equal sides of length 12 cm each; the rectangle has width 8 cm and length x cm. The perimeter of the triangle equals the perimeter of the rectangle. Find x.



(A) 8 (B) 10 (C) 11 (D) 14 (E) 15

Your work:

Problem 3

Let $x = 3$. Which of the following equations is true?

(A) $2x = 5$ (B) $3x - 1 = 8$ (C) $x + 5 = 3$ (D) $7 - x = 2$ (E) $6 + 2x = 14$

Your work:

Problem 4

The sum of three consecutive integers is 153. The largest of the three integers is

(A) 52 (B) 50 (C) 53 (D) 54 (E) 51

Your work:

Problem 5

One scoop of fish food can feed 8 goldfish. How many goldfish can 4 such scoops of fish food feed?

(A) 12 (B) 16 (C) 8 (D) 64 (E) 32

Your work:

Problem 6

If $x = 4$ and $y = 3x$, the value of y is

(A) 12 (B) 24 (C) 7 (D) 81 (E) 4

Your work:

Problem 7

Each bus ride costs Alex \$2.25, while Sam pays \$3.00 per ride (they use different routes). If each of them rides the bus 20 times, how much less does Alex pay than Sam in total?

(A) \$25 (B) \$10 (C) \$15 (D) \$45 (E) \$60

Your work:

Problem 8

Which expression produces the values in the second row of the table, given the values of n in the first row?

n	1	2	3	4	5
Value	1	3	5	7	9

(A) $3n - 2$ (B) $2(n - 1)$ (C) $n + 4$ (D) $2n$ (E) $2n - 1$

Your work:

Problem 9

When two numbers are added, the result is -26. One of the numbers is 11. The other number is

(A) -37 (B) 37 (C) -15 (D) 15 (E) -48

Your work:

Problem 10

Mikhail has \$10 000, all in \$50 bills. How many \$50 bills does he have?

(A) 1 000 (B) 200 (C) 1 250 (D) 500 (E) 2 000

Your work:

Problem 11

The product $60 \times 60 \times 24 \times 7$ equals

(A) the number of minutes in seven weeks

(B) the number of hours in sixty days

(C) the number of seconds in seven hours

(D) the number of seconds in one week

(E) the number of minutes in twenty-four weeks

Your work:

Problem 12

Each of a, b, c, d is a positive integer greater than 3, and

$$\frac{1}{a-2} = \frac{1}{b+2} = \frac{1}{c+1} = \frac{1}{d-3}.$$

Which ordering of these four numbers is correct?

(A) $a < b < c < d$ (B) $c < b < a < d$ (C) $b < a < c < d$

(D) $d < a < c < b$ (E) $b < c < a < d$

Your work:

Problem 13

A rectangle has length x and width y. A triangle has base 16 and corresponding height x. If the area of the rectangle equals the area of the triangle, then the value of y is

(A) 16 (B) 4 (C) 8 (D) 12 (E) 32

Your work:

Problem 14

The symbol $\begin{array}{|c|c|}\hline 3 & 4 \\ \hline 5 & 6 \\ \hline \end{array}$ is evaluated as $3 \times 6 + 4 \times 5 = 38$. If $\begin{array}{|c|c|}\hline 2 & 6 \\ \hline 1 & \\ \hline \end{array}$ is evaluated as 16, then the number that should be placed in the empty space is

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Your work:

Problem 15

Three pumpkins are weighed two at a time in all possible ways. The weights of the pairs are 12 kg, 13 kg and 15 kg. How much does the lightest pumpkin weigh?

(A) 4 kg (B) 5 kg (C) 6 kg (D) 7 kg (E) 8 kg

Your work:

Problem 16

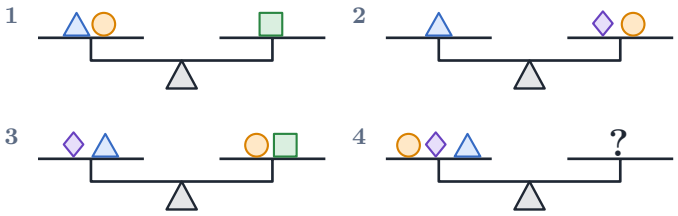
Rosie is learning to save money. Today she has \$120 in her savings account. From now on, every time she makes a deposit she will add exactly \$30. If Rosie makes m such deposits, which expression correctly gives the amount of money (in dollars) in her account?

(A) $120 + m$ (B) $30m$ (C) $30 + 120m$ (D) $150m$ (E) $120 + 30m$

Your work:

Problem 17

All four scales below are balanced. Identical shapes have equal masses. One possible replacement for the "?" that keeps the fourth scale balanced is



(A) △□ (B) ◇△ (C) ○□ (D) □◇ (E) △○

Your work:

Problem 18

The number that goes in the box to make $\frac{3}{7} = \frac{\square}{63}$ true is

(A) 27 (B) 9 (C) 59 (D) 63 (E) 3

Your work:

Problem 19

The sum of four numbers is T. Each of the four numbers is increased by 1. The four new numbers are added together, and the sum is then multiplied by 3. The final result is

(A) $3T + 3$ (B) $3T + 4$ (C) $3T + 12$ (D) $T + 12$ (E) $12T$

Your work:

Problem 20

Two isosceles triangles each have at least one angle measuring $70°$. In the first triangle, the measure (in degrees) of each of the other two angles is even. In the second triangle, the measure of each of the other two angles is odd. Let S be the sum of the two equal angles of the first triangle and T the sum of the two equal angles of the second triangle. The value of $S + T$ is

(A) $280°$ (B) $250°$ (C) $220°$ (D) $200°$ (E) $300°$

Your work:

Problem 21

Andrea is on a 6-day kayak trip with a total distance of 168 km. At the end of the third day she has covered $\frac{3}{7}$ of the total distance. How many kilometres per day must she average for the rest of the trip?

(A) 29 (B) 24 (C) 27 (D) 32 (E) 26

Your work:

Problem 22

A dot starts at the point $(20, 19)$ of the coordinate plane. On the first move it travels 1 unit horizontally or vertically, that is, to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$ or $(20, 18)$. On the second move it travels 2 units (in either direction) *perpendicular* to the first move. From then on, each move is 1 unit longer than the previous one (3 units, 4 units, 5 units, ...) and is always perpendicular to the move just before it. The dot stops after 10 moves. Which of the following final positions is **not** possible?

(A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Your work:

Problem 23

Gail works as a server in a restaurant. On Saturday she gets up at 6:30 a.m., starts work at x a.m. and finishes at x p.m. (the same day). How long does Gail work on Saturday?

(A) $24 - 2x$ hours (B) $12 - x$ hours (C) $2x$ hours

(D) 0 hours (E) 12 hours

Your work:

Problem 24

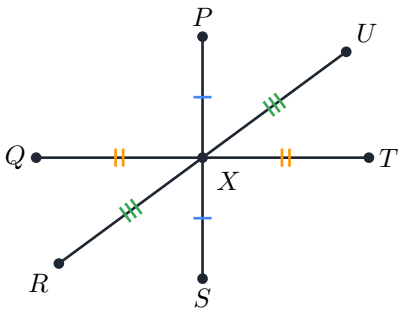
An $8 \times 8 \times n$ rectangular box (n a positive integer) is built from $1 \times 1 \times 1$ cubes. Let A be the surface area of the box and B the combined surface area of all the small cubes that make it up. The sum of all values of n for which $\frac{B}{A}$ is an integer is

(A) 86 (B) 90 (C) 70 (D) 78 (E) 96

Your work:

Problem 25

Three bus routes cross at the stop X as shown (segments with the same number of tick marks have equal length). Every 12 minutes, bus A completes a trip $P \to X \to S \to X \to P$. Every 20 minutes, bus B completes a trip $Q \to X \to T \to X \to Q$. Every 28 minutes, bus C completes a trip $R \to X \to U \to X \to R$. At 1:00 p.m., buses A, B and C leave P, Q and R respectively; each bus drives at a constant speed and turns around instantly at the end of its route. The buses run continuously until 11:00 p.m. Between 5:00 p.m. and 10:00 p.m., how many times are two or more buses at X at the same moment?



(A) 18 (B) 19 (C) 20

(D) 21 (E) 22

Your work:

Problem 26

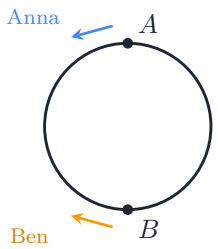
Steve starts at 7 and counts forward by 3, getting the list $7, 10, 13, \ldots$ Dave starts at 2026 and counts backward by 5, getting the list $2026, 2021, 2016, \ldots$ Which of the following numbers appears in both lists?

(A) 1009 (B) 1006 (C) 1003 (D) 1001 (E) 1011

Your work:

Problem 27*

On a circular track, Anna stands at point A and Ben stands at point B directly opposite A (segment AB is a diameter). Anna runs counterclockwise and Ben runs clockwise. They run at constant but different speeds. After a while they notice that every time they pass each other, it is always at the same three places on the track. The ratio of their speeds is



(A) 3 : 2 (B) 3 : 1 (C) 4 : 1
(D) 2 : 1 (E) 5 : 2

Your work:

Problem 28

The list $11, 20, 31, 51, 82$ is an example of an increasing list of five positive integers in which the first and second numbers add to the third, the second and third add to the fourth, and the third and fourth add to the fifth. How many such lists have 124 as the fifth number?

(A) 10 (B) 7 (C) 9 (D) 6 (E) 8

Your work:

Problem 29

Kathy owns more cats than Alice and more dogs than Bruce. Alice owns more dogs than Kathy but fewer cats than Bruce. Which of the following statements **must** be true?

(A) Bruce owns the fewest cats.

(B) Bruce owns the most cats.

(C) Kathy owns the most cats.

(D) Alice owns the most dogs.

(E) Kathy owns the fewest dogs.

Your work:

Problem 30

Chloe bought 300 g of candy A for \$5.00 and x g of candy B for \$7.00. She worked out that the average price of all the candy she bought was \$1.50 per 100 g. What is the value of x?

(A) 525 (B) 600 (C) 500 (D) 450 (E) 900

Your work:

Extension and challenge

Problem 31

The road from Mr Henry's home to the city is 180 km long. He rides a motorbike at a constant speed, and after 3 hours he has covered $\frac{2}{3}$ of the distance.

a) Find the speed of the motorbike.

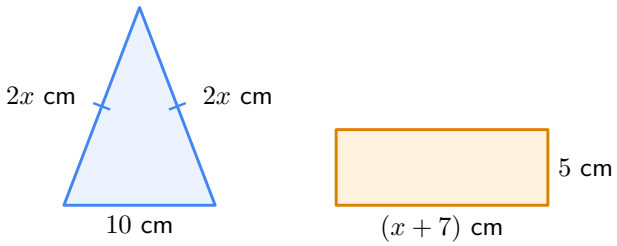
b) If he keeps the same speed, how much longer will it take him to reach the city?

c) If he wants to arrive 30 minutes earlier than planned in part b), at what speed must he ride for the rest of the way?

Your work:

Problem 32

An isosceles triangle has a base of 10 cm and two equal sides of $2x$ cm each. A rectangle has length $(x + 7)$ cm and width 5 cm. The perimeter of the triangle equals the perimeter of the rectangle.



a) Write an expression in x for the perimeter of each shape.

b) Find x, then find the side lengths of the triangle and the area of the rectangle.

Your work:

Problem 33 *

We use the rule of Problem 14 again: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a \cdot d + b \cdot c$.

a) Evaluate $\begin{vmatrix} -3 & 4 \\ 5 & 2 \end{vmatrix}$.

b) Find x if $\begin{vmatrix} x & 2 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & x \\ 5 & 2 \end{vmatrix}$.

c) Find all integers x such that $\begin{vmatrix} x & x \\ x & x \end{vmatrix} = 50$.

Your work:

Problem 34

a) The sum of five consecutive odd numbers is 155. Find the five numbers.

b) Leo says: "I have found four consecutive odd numbers whose sum is 150." Is Leo right or wrong? Why?

Your work:

Problem 35

Every 2 scoops of fish food are enough to feed 12 fish for one day.

a) How many fish can 5 scoops of food feed for one day?

b) Mia's fish tank has 18 fish. A box of food contains 60 scoops. For how many days will the box feed the whole tank?

Your work:

Problem 36

Each bus ride costs Lucy $2.50. The bus company also sells a monthly pass for $72 with unlimited rides.

a) Write an expression for the amount (in dollars) Lucy pays if she makes n rides in a month without a monthly pass.

b) What is the smallest number of rides in a month for which the monthly pass is cheaper than paying per ride?

Your work:

Problem 37

Look at the table below (the values increase steadily as n increases):

n	1	2	3	4	5
Value	2	7	12	17	22

a) Write an expression in n for the row of values.

b) Find the value for $n = 40$.

c) For which n is the value 302?

d) Is any value in the table (continued forever) equal to 2026?

Your work:

Problem 38

a) The product $60 \times 24 \times 365$ is the number of which time unit in a non-leap year (with 365 days)? Work out the product.

b) How many days, hours, minutes and seconds are there in one million seconds?

Your work:

Problem 39

Four positive integers a, b, c, d satisfy

$$\frac{3}{a-5} = \frac{3}{b+1} = \frac{3}{c-2} = \frac{3}{d+4} \quad \text{and} \quad a+b+c+d = 50.$$

a) Arrange a, b, c, d in increasing order.

b) Find the value of each number.

Your work:

Problem 40

There are four bags of rice with different weights. They are weighed two at a time in all possible ways, giving six results:

7 kg, 8 kg, 9 kg, 10 kg, 11 kg, 12 kg.

a) Find the total weight of the four bags.

b) Each bag weighs a whole number of kilograms. Find the weight of each bag.

Your work:

Problem 41

The three scales below are balanced; identical shapes have equal masses.



Scale 1 Scale 2 Scale 3

a) How many □ shapes balance one ○?

b) How many □ shapes balance one ◇?

c) One ◇ and one ○ are placed on the left pan. If only △ shapes may be used on the right pan, how many △ are needed?

d) The four objects △, ○, □, ◇ (one of each kind) have a total mass of 50 g. Find the mass of each object.

Your work:

Problem 42 *

In this problem we only consider triangles in which every angle measures (in degrees) a *whole number*. Two isosceles triangles are said to be of the *same type* if they have the same three angle measures.

a) Prove that there is no triangle whose three angles all have odd measures.

b) How many types of isosceles triangle have all three angles with even measures?

c) How many types of isosceles triangle have exactly two angles with odd measures? Is there an isosceles triangle with exactly one angle of odd measure?

d) How many types of isosceles triangle are there altogether?

Your work:

Problem 43 **

A dot starts at the origin $O(0, 0)$ and moves by the rule of Problem 22: the first move is 1 unit horizontally or vertically; each later move is 1 unit longer than the previous one and perpendicular to the move just before it (either direction allowed).

a) Explain why after exactly 10 moves the dot cannot be back at O.

b) Find the smallest n ($n \geq 1$) such that after exactly n moves the dot can be back at O. Show one way to do it.

c) Can the dot be back at O after exactly 8 moves?

Your work:

Problem 44 ★★

A $4 \times 6 \times n$ rectangular box (n a positive integer) is built from $1 \times 1 \times 1$ unit cubes. Let A be the surface area of the box and B the combined surface area of all the unit cubes.

a) Write A, B and the ratio $\dfrac{B}{A}$ as expressions in n.

b) Prove that $\dfrac{B}{A} < 7.2$ for every n.

c) Find all values of n for which $\dfrac{B}{A}$ is an integer.

Your work:

Problem 45 ★★

Two bus routes cross at the stop X. Bus A runs back and forth along the route P–X–S; each arm (PX or XS) takes exactly 4 minutes. Bus B runs along the route Q–X–T; each arm takes exactly 6 minutes. At 1:00 p.m., bus A leaves P and bus B leaves Q; the buses turn around instantly at the ends of their routes and run continuously.

a) Prove that buses A and B are never at X at the same time.

b) A bus C is added on the route R–X–U; each arm takes 10 minutes, and it leaves R at 1:00 p.m. Between 1:00 p.m. and 7:00 p.m., at which times are two buses at X together?

c) Suppose each arm of bus C's route takes m minutes (m any positive integer). Prove that bus C cannot both meet bus A at X (at some moment) and meet bus B at X (at some moment).

Your work:

Problem 46 ⋆

Sam counts forward $7, 10, 13, \ldots$ (adding 3 each time) and stops before going past 2026. Tom counts backward $2026, 2021, 2016, \ldots$ (subtracting 5 each time) and stops before going below 7.

a) How many numbers appear in both lists?

b) Find the sum of all the numbers in part a.

c) Zoe writes the list of positive multiples of 4: $4, 8, 12, \ldots$ How many numbers appear in all three lists (Sam's, Tom's and Zoe's)? What is the largest of them?

Your work:

Problem 47 ⋆⋆

Emma and Noah both start at point A on a circular track; each runs at a constant speed.

a) They run in opposite directions, with the ratio of speeds Emma : Noah equal to $5 : 3$. At how many different points of the track do they meet?

b) If they run in the *same direction* (still with ratio $5 : 3$), at how many different points does Emma overtake Noah?

c) Find all speed ratios $p : q$ (with $p > q$, where p and q are coprime positive integers) such that, running in opposite directions, they meet at exactly 10 different points.

Your work:

Problem 48 ⋆⋆

We call an *increasing* list of positive integers a *Fibo list* if, from the third number on, each number is the sum of the two numbers just before it (for example $11, 20, 31, 51, 82$ as in Problem 28).

a) Find all Fibo lists of 6 numbers whose last number is 100.

b) How many Fibo lists of 5 numbers have last number 200?

c) Prove that there is no Fibo list of 6 numbers whose last number is 30.

Your work:

Problem 49 ⋆

Four friends Anna, Ben, Chloe and David each have at least one marble. We know that:

(1) Anna and Ben together have as many marbles as Chloe and David together.

(2) Anna and David together have more marbles than Ben and Chloe together.

(3) Ben has more marbles than Anna and Chloe together.

a) Put the four friends in order from fewest to most marbles.

b) Altogether the four friends have 30 marbles. What is the greatest number of marbles Chloe can have? In that case, how many marbles does each friend have?

Your work:

Problem 50

A shop sells candy A at \$12 per kilogram and candy B at \$20 per kilogram.

a) How many grams of each kind must be mixed to get 1 kg of mixture costing \$15 per kilogram (the mixture is worth exactly the total value of the candy put in)?

b) How many grams of candy C, costing \$30 per kilogram, must be added to the 1 kg mixture of part a to get a new mixture costing \$18 per kilogram?

c) Prove that if candy A and candy B (both present) are mixed in any proportion, the price of the mixture is always more than \$12 and less than \$20 per kilogram.

Your work:

2 Geometry and Measurement

The big idea. Geometry is the study of *shapes*; measurement attaches *numbers* to those shapes: lengths, angle measures, perimeters, areas and volumes. Almost every problem in this topic is solved with a handful of very simple ideas that are used over and over again:

- **Angles have a fixed "budget":** two supplementary angles always add up to 180°, the three angles of a triangle always add up to 180°, and a full turn is 360°. Knowing some of the angles, we can work out the rest.
- **Areas can be cut, rearranged and subtracted:** a complicated shape is split into familiar pieces (rectangles, triangles, trapezoids), or computed as "big shape minus the extra bits".
- **Ratios instead of numbers:** "what percentage is shaded" only requires comparing the shaded part with the whole shape – often without knowing any real sizes.
- **Shape does not change when turned or flipped:** rotating, reflecting or rolling a cube only changes *position*, never *size* or *shape*.
- **A box is built in layers:** volume = base area × height.

Whenever you meet a geometry problem, always **draw the figure and write the given numbers on it**, then ask yourself: "What do I know? What is missing? Which formula connects the two?".

① The coordinate plane

The coordinate system Oxy consists of two number lines that are perpendicular at the origin O:

- the horizontal axis Ox is called the **x-axis**; its positive direction points to the *right*;
- the vertical axis Oy is called the **y-axis**; its positive direction points *up*.

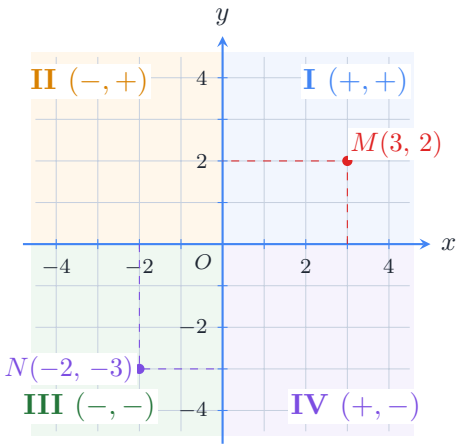
Each point M of the plane corresponds to exactly one pair of numbers (x_0, y_0), written $M(x_0, y_0)$:

- x_0 is the **x-coordinate**: drop a perpendicular from M to the x-axis; its foot is at the number x_0;
- y_0 is the **y-coordinate**: draw a perpendicular from M to the y-axis; its foot is at the number y_0.

The two axes divide the plane into four **quadrants**, numbered counterclockwise:

	Quadrant I	Quadrant II	Quadrant III	Quadrant IV
Signs of (x, y)	$(+, +)$	$(-, +)$	$(-, -)$	$(+, -)$

Distances along horizontal and vertical lines. Two points with the same y-coordinate (on the same horizontal line) are a distance apart equal to the difference of their x-coordinates (larger minus smaller). Similarly, two points with the same x-coordinate are a distance apart equal to the difference of their y-coordinates.



ThinkAlgo Tip. Read coordinates in the order "**across first, then up or down**": start at O, move right (positive) or left (negative) to get the x-coordinate; then move up (positive) or down (negative) to get the y-coordinate. For example, to reach $N(-2, -3)$: go 2 squares left, then 3 squares down.

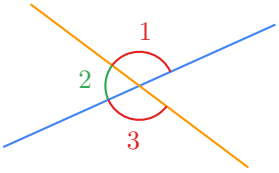
Common mistake. Swapping the two numbers is a very common mistake: $(3, -1)$ and $(-1, 3)$ are two *different* points (one in quadrant IV, the other in quadrant II). The x-coordinate is *always* written first.

Mini example. Let $A(1, 1)$, $B(5, 1)$, $C(5, 4)$. Since A and B have the same y-coordinate 1, $AB = 5 - 1 = 4$. Since B and C have the same x-coordinate 5, $BC = 4 - 1 = 3$. Angle B is a right angle (one side horizontal, one side vertical), so the area of triangle ABC is $\dfrac{4 \cdot 3}{2} = 6$ (square units).

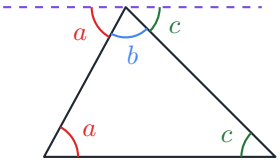
Note for parents. You can turn the coordinate plane into a "treasure hunt" on squared paper: you say "left 2, up 3", your child marks the point and writes its coordinates $(-2, 3)$. Then swap roles: your child gives directions and you draw. After a few rounds, the order "x first, y second" becomes automatic.

② Angles: supplementary, vertically opposite, angle sum of a triangle

- A **right angle** is 90°, a **straight angle** (half a turn) is 180°, a **full turn** is 360°.
- **Two adjacent supplementary angles** (sharing one arm, with the other two arms forming a straight line) add up to 180°.
- **Two vertically opposite angles** (each arm of one is the extension of an arm of the other) are **equal**.
- **The three angles of a triangle add up to** 180°**.** Hence: third angle = 180° − (sum of the two known angles).
- An **exterior angle of a triangle** (supplementary to an interior angle) equals **the sum of the two interior angles not adjacent to it**.
- In a right triangle, the two acute angles add up to 90°.



$\widehat{1}+\widehat{2}=180°$ (supplementary)
$\widehat{1}=\widehat{3}$ (vertically opposite)



$a+b+c=180°$

Why is it true? **Vertically opposite angles are equal.** In the left figure, $\widehat{1}+\widehat{2}=180°$ (supplementary) and $\widehat{3}+\widehat{2}=180°$ (also supplementary). Angles $\widehat{1}$ and $\widehat{3}$ are both supplementary to the same angle $\widehat{2}$, so $\widehat{1}=180°-\widehat{2}=\widehat{3}$.

The angles of a triangle add up to $180°$**.** Through the top vertex of the triangle, draw the line parallel to the base (dashed). For parallel lines, *alternate angles* are equal: the angle on the left at the top vertex equals a, and the angle on the right equals c. The three angles a, b, c at the top vertex together form a straight angle, so $a+b+c=180°$.

Exterior angle. The exterior angle at a vertex is supplementary to the interior angle there: exterior angle $=180°-$ interior angle. But $180°-$ interior angle is exactly the sum of the other two interior angles. So the exterior angle equals the sum of the two interior angles not adjacent to it. □

ThinkAlgo Tip. In "find the angle" problems, write the measure of every angle you find **on the figure** immediately. An angle you have just found is often the key to the next one (supplementary, vertically opposite, angle sum, ...).

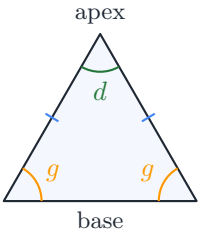
Common mistake. Do not confuse *vertically opposite* angles (equal) with *supplementary* angles (sum $180°$). Vertically opposite angles sit "across from each other" at the intersection point and share no arm; adjacent supplementary angles sit "next to each other" and share an arm.

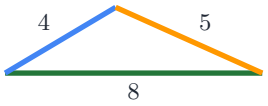
③ Isosceles triangles, equilateral triangles and the triangle inequality

- An **isosceles triangle** has two equal sides (the *legs*); the third side is the *base*, and the angle between the two legs is the *apex angle*.
- In an isosceles triangle, **the two base angles are equal**. So if the apex angle is d and each base angle is g, then

$$d+2g=180°, \qquad d=180°-2g, \qquad g=\frac{180°-d}{2}.$$

- A base angle is always **acute** (less than $90°$), because the two base angles are equal and their sum must be less than $180°$.
- An **equilateral triangle** has three equal sides and three equal angles, each $180°\div 3=60°$.
- **Triangle inequality:** in a triangle, each side is always *less* than the sum of the other two. To check whether three lengths form a triangle, it is enough to check: *longest side* $<$ *sum of the other two.*





$8 < 4 + 5$: a triangle exists

Why is it true? **Why are the base angles equal?** Let M be the midpoint of the base BC of the isosceles triangle ABC ($AB = AC$). Triangles ABM and ACM have $AB = AC$, $BM = CM$ and the common side AM, so they are congruent (side – side – side). Hence the corresponding angles $\widehat{B}$ and $\widehat{C}$ are equal.

Why does the triangle inequality hold? The shortest path between two points is the straight segment. Going from B to C "via" A (that is, travelling $BA + AC$) is always longer than going straight along BC. If the longest side were equal to or greater than the sum of the other two, the two short sides could not "reach each other": they would lie flat along the long side or fail to meet, and no triangle would be formed. □

Common mistake. When a problem only says "an isosceles triangle has *an angle* of 50°", we do **not yet know** whether it is the apex angle or a base angle. We must consider **both cases**:

- 50° is the apex angle: the base angles are $(180° - 50°) \div 2 = 65°$ and 65°;
- 50° is a base angle: the other base angle is also 50°, and the apex angle is $180° - 2 \cdot 50° = 80°$.

Only when the given angle is *greater than or equal to* 90° must it be the apex angle.

④ Rotations, reflections and rolling cubes

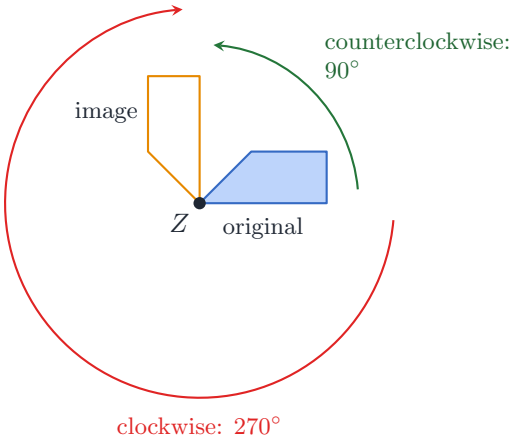
A **rotation** about a point Z (the *centre of rotation*) through an angle α in a given direction (clockwise or counterclockwise) sends each point M to a point M' such that $ZM' = ZM$ and $\widehat{MZM'} = \alpha$.

- Every point of the figure turns through **the same angle**, in the same direction; distances to the centre do not change.
- A rotation keeps the shape and size of the figure; the figure is only "turned", never "flipped over".
- Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise. A 360° rotation brings the figure back to where it started.
- **How to measure the angle of rotation:** choose a segment starting at the centre Z on the original figure, find the matching segment on the new figure, and measure the angle that segment has swept through *in the given direction*.

A **reflection in a line** (flipping over the line, like a mirror) also keeps sizes, but **reverses the orientation** of the figure: the letter P seen in a mirror becomes a backwards P.

In the coordinate plane (centre of rotation at the origin O):

Rotation 90° counterclockwise	$(x, y) \to (-y, x)$
Rotation 180°	$(x, y) \to (-x, -y)$
Rotation 90° clockwise	$(x, y) \to (y, -x)$
Reflection in the x-axis / in the y-axis	$(x, y) \to (x, -y)$ / $(x, y) \to (-x, y)$



Why is it true? Why is a 270° clockwise rotation the same as a 90° counterclockwise one? A full turn is 360°. After going 270° one way, only $360° - 270° = 90°$ more would bring us back to the start; so the end position is exactly where we would arrive by going 90° the other way. In the figure: to turn the horizontal side (pointing right) into the vertical side (pointing up), we can rotate 90° counterclockwise, or rotate 270° clockwise (through pointing down, pointing left, and only then up). □

Rolling a cube. When a cube is rolled, each face is carried along unchanged. So a face with a painted pattern:

- may appear in another position (front, top, side, ...);
- may be *turned* by 90°, 180° or 270° within its own plane;
- **but never** becomes a mirror image, changes size, or "gains" extra paint.

ThinkAlgo Tip. For cube problems, "mark" one special feature of the pattern (for example: the right angle of the black triangle sits at *a corner* of the face) and check that feature in every option. Cutting out a paper square, drawing the pattern on it and turning it is also a very effective check.

5 Units of measurement and the number line

Units of length. Each unit is 10 times the next one:

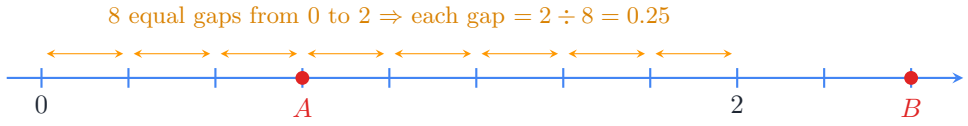
$$1 \text{ km} = 10 \text{ hm}, \quad 1 \text{ hm} = 10 \text{ dam}, \quad 1 \text{ dam} = 10 \text{ m},$$

$$1\text{ m} = 10\text{ dm}, \quad 1\text{ dm} = 10\text{ cm}, \quad 1\text{ cm} = 10\text{ mm}.$$

The most used relations: $1\text{ m} = 100\text{ cm} = 1000\text{ mm}$; $1\text{ km} = 1000\text{ m}$.
Units of area go up by a factor of 100: $1\text{ m}^2 = 100\text{ dm}^2 = 10\,000\text{ cm}^2$.
Units of volume go up by a factor of 1000: $1\text{ m}^3 = 1000\text{ dm}^3$, $1\text{ dm}^3 = 1000\text{ cm}^3 = 1$ litre.
Conversion rule: from a *larger* unit to a *smaller* unit, **multiply**; from a *smaller* unit to a *larger* unit, **divide**.
Number line with equally spaced ticks. If we know the numbers at two ticks and count k *gaps* (not k ticks) between them, then

$$\text{value of one gap} = \frac{\text{difference of the two numbers}}{k}.$$

Why is it true? Why is $1\text{ m}^2 = 10\,000\text{ cm}^2$ and not 100 cm^2? One square metre is the area of a square with side 1 m, i.e. side 100 cm. That square can be divided into 100 rows of 100 squares of side 1 cm, so it contains $100 \times 100 = 10\,000$ squares of 1 cm^2. Likewise, a cube of side 1 dm = 10 cm consists of $10 \times 10 \times 10 = 1000$ small cubes of 1 cm^3. □



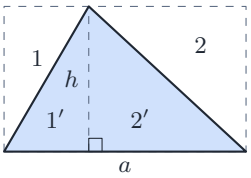
Mini example. On the number line above, A is three gaps from 0, so $A = 3 \cdot 0.25 = 0.75$; B is ten gaps from 0, so $B = 10 \cdot 0.25 = 2.5$.

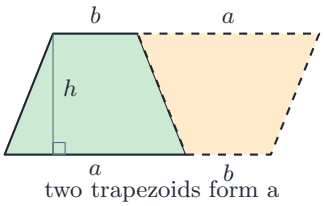
Common mistake. Counting *ticks* instead of *gaps*: from tick 0 to tick 2 in the figure above there are 9 ticks but only 8 gaps. Between two numbers there is always one gap fewer than ticks (like trees in a row: 9 trees have 8 spaces between them).

⑥ Perimeter and area of plane figures

Figure	Perimeter	Area
Rectangle (length a, width b)	$P = 2 \cdot (a + b)$	$S = a \cdot b$
Square (side a)	$P = 4 \cdot a$	$S = a \cdot a = a^2$
Triangle (base a, height h to that base)	sum of the three sides	$S = \frac{a \cdot h}{2}$
Parallelogram (base a, height h)	2·(sum of two adjacent sides)	$S = a \cdot h$
Trapezoid (parallel sides a, b; height h)	sum of the four sides	$S = \frac{(a+b) \cdot h}{2}$

The **height** of a triangle (parallelogram, trapezoid) is the segment drawn *perpendicular* from the vertex (or from the opposite side) to the line containing the base.
Working backwards: from $S = \frac{a \cdot h}{2}$ we get $h = \frac{2 \cdot S}{a}$ and $a = \frac{2 \cdot S}{h}$; from the perimeter of a square, side $= P \div 4$.





two trapezoids form a parallelogram with base $a + b$

Why is it true? Area of a triangle. Place the triangle inside the rectangle with the same base a and the same height h (left figure). The height splits the rectangle into two smaller rectangles; each smaller rectangle is cut by a diagonal into two equal triangles: the part inside the triangle ($1'$, $2'$) equals the part outside it (1, 2). So the triangle is exactly half of the rectangle: $S = \dfrac{a \cdot h}{2}$.
Area of a parallelogram. Cut off a right triangle at one end and slide it to the other end: we get a rectangle with the same base a and the same height h: $S = a \cdot h$.
Area of a trapezoid. Two identical trapezoids, one turned upside down, fit together into a parallelogram with base $a+b$ and height h (right figure). One trapezoid is half of it: $S = \dfrac{(a+b) \cdot h}{2}$. □

ThinkAlgo Tip. Two triangles with **the same base** (or equal bases) and **the same height** have equal areas, even if they look very different. Consequence: the segment joining a vertex to the midpoint of the opposite side divides a triangle into two parts of equal area.

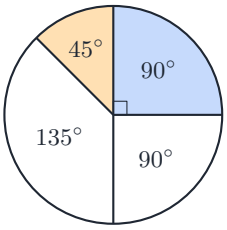
Common mistake.

- The height must be **perpendicular** to the base; a slanted side is *not* the height.
- Forgetting to divide by 2 in the area formulas of the triangle and the trapezoid.
- Mixing up units: perimeter is measured in cm, area in cm^2. A square with perimeter 36 cm has side 9 cm and area 81 cm^2 – not 36 cm^2.

⑦ Parts of an area: fractions, percentages, circles and rearranging

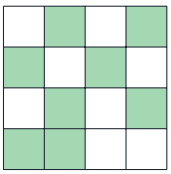
- If a figure is divided into n **equal** parts and m parts are shaded, the shaded part is $\dfrac{m}{n}$ of the figure, i.e. $\dfrac{m}{n} \cdot 100\%$.
- **Circles and sectors.** One turn around the centre is 360°. A sector with central angle $n°$ is $\dfrac{n}{360}$ of the area of the circle. For example: a semicircle corresponds to 180°, a quarter circle to 90° (a right angle at the centre).
- **Spinners:** the pointer stops at random, all directions being equally likely, so the chance of stopping in a region equals the fraction of the area (i.e. of the angle) of that region.
- **Cut and rearrange:** split the figure into familiar pieces and add their areas. **Complement:** required area = area of the big figure − area of the extra pieces.
- **Systematic counting of figures:** when asked how many figures (triangles, parallelograms, ...) appear in a diagram, sort them by *size* (made of 1 piece, 2 pieces, 3 pieces, ...) and count

each type so that nothing is missed or counted twice.



blue: $\frac{90}{360} = \frac{1}{4} = 25\%$

orange: $\frac{45}{360} = \frac{1}{8} = 12.5\%$



8 of 16 squares: $\frac{8}{16} = 50\%$

Why is it true? Why is a sector with angle n° equal to $\frac{n}{360}$ of the circle? Imagine cutting the circle into 360 tiny sectors, each with a central angle of 1°. These tiny sectors are identical (turning the circle by 1° about its centre makes one fit exactly onto the next), so they have equal areas, each $\frac{1}{360}$ of the circle. A sector with angle n° is made of n such tiny sectors. □

Common mistake. You may only count "m parts out of n" when the parts are **equal**. If the pieces have different sizes, reduce them to one kind of piece (for example by cutting them further into equal pieces) or compute the area of each piece.

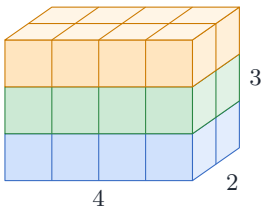
Note for parents. Cut a cake (or a paper circle) into 4, then into 8 equal parts so your child can see: 1 quarter = 90°, 1 eighth = 45°, and half of a quarter is an eighth. The idea "halving the angle halves the area" becomes very concrete.

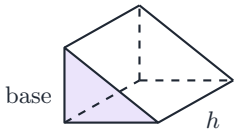
⑧ Solids: volume and surface area

- A **rectangular box** (rectangular prism) with dimensions a, b, c:

$$V = a \cdot b \cdot c, \qquad S_{\text{lateral}} = 2 \cdot (a + b) \cdot c, \qquad S_{\text{total}} = 2 \cdot (ab + bc + ca).$$

- A **cube** with side a: $V = a^3$, $S_{\text{total}} = 6 \cdot a^2$.
- A **right prism** (with a triangular, quadrilateral, ... base): $V = S_{\text{base}} \cdot h$, where h is the height of the prism (the length of a lateral edge).
- The number of 1 cm × 1 cm × 1 cm cubes needed to fill a box exactly equals the volume of the box in cm^3.





each layer $4 \cdot 2 = 8$ cubes, 3 layers: $8 \cdot 3 = 24$ cubes $V = S_{\text{base}} \cdot h$

Why is it true? **Why is $V = a \cdot b \cdot c$?** Stack 1 cm^3 cubes in layers. The bottom layer is a rectangle of a rows $\times$ b columns, containing $a \cdot b$ cubes (exactly the base area). Stack c such layers and we have $a \cdot b \cdot c$ cubes. The argument "each layer = base area, number of layers = height" works for every right prism, so $V = S_{\text{base}} \cdot h$. In particular, for a prism whose base is a right triangle with legs a, b: two such prisms fit together into an $a \times b \times h$ box, so each prism has volume $\dfrac{a \cdot b}{2} \cdot h$. $\square$

ThinkAlgo Tip. The surface area of a box is the total area of its 6 faces, which come in 3 pairs of equal opposite faces. When small blocks are joined into a big block, the faces that are "pressed together" are no longer on the surface, so the surface area of the big block is *less* than the total surface area of the small blocks.

Example 1

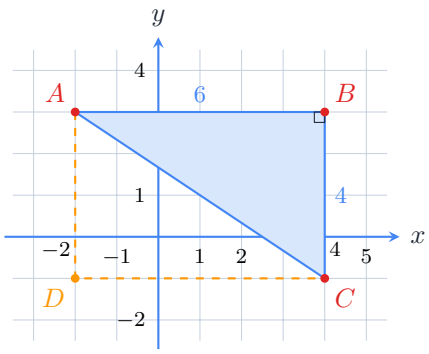
In the coordinate plane Oxy, consider the points $A(-2, 3)$, $B(4, 3)$ and $C(4, -1)$.

a) In which quadrant does each of the points A, B, C lie?

b) Find the lengths AB, BC and the area of triangle ABC.

c) Find the coordinates of the point D such that $ABCD$ is a rectangle.

Guided solution.



Step 1. Use the signs of the coordinates to find the quadrant

$A(-2, 3)$: negative x, positive y $\Rightarrow$ quadrant II. $B(4, 3)$: both positive $\Rightarrow$ quadrant I. $C(4, -1)$: positive x, negative y $\Rightarrow$ quadrant IV.

Step 2. Find the lengths of the horizontal and vertical sides

A and B have the same y-coordinate 3, so AB is horizontal and $AB = 4 - (-2) = 4 + 2 = 6$ (units). B and C have the same x-coordinate 4, so BC is vertical and $BC = 3 - (-1) = 3 + 1 = 4$.

Step 3. Find the area of the triangle

Side AB is horizontal and side BC is vertical, so angle B is a right angle. The area of a right triangle is half the product of the two legs:

$$S_{ABC} = \frac{AB \cdot BC}{2} = \frac{6 \cdot 4}{2} = 12 \text{ (square units)}.$$

Step 4. Find the point D

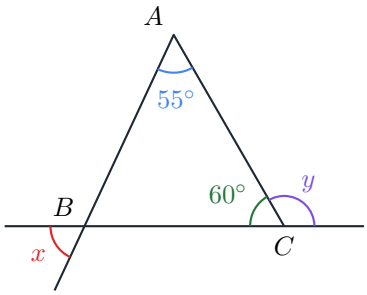
In rectangle $ABCD$, D is "opposite" B: D must be vertically in line with A (same x-coordinate -2) and horizontally in line with C (same y-coordinate -1). So $D(-2, -1)$, in quadrant III.

Check: Rectangle $ABCD$ has area $6 \cdot 4 = 24$, exactly twice the area of triangle ABC (the diagonal AC cuts the rectangle into two equal halves).

Answer: **a)** A: **II**, B: **I**, C: **IV**. **b)** $AB = 6$, $BC = 4$, $S = 12$. **c)** $D(-2, -1)$.

Example 2

In the figure below, triangle ABC has $\widehat{A} = 55°$, $\widehat{C} = 60°$. Sides AB and CB are extended beyond B; side BC is extended beyond C. Find the measures x and y.



Guided solution.

Analysis. Angle x lies *outside* the triangle, but it is *vertically opposite* to the angle $\widehat{ABC}$ of the triangle (the arms of x are the extensions of BC and of BA). So we find $\widehat{ABC}$ first.

Step 1. **Find angle B of the triangle**

Recall. The three angles of a triangle add up to 180°.

$\widehat{ABC} = 180° - (55° + 60°) = 180° - 115° = 65°$.

Step 2. **Find x**

Recall. Vertically opposite angles are equal.

Angle x and angle $\widehat{ABC}$ are vertically opposite, so $x = 65°$.

Step 3. **Find y**

Angle y is supplementary to angle $\widehat{ACB}$, so $y = 180° - 60° = 120°$.

Check: By the exterior angle property, y must equal the sum of the two non-adjacent interior angles: $55° + 65° = 120°$. It matches!

Answer: $x = 65°$, $y = 120°$.

Example 3

a) An isosceles triangle has an angle of 40°. Find the other two angles (consider all possibilities).

b) An isosceles triangle has an angle of 100°. Find the other two angles.

c) Which of the triples of lengths (3, 4, 8) and (3, 5, 7) forms a triangle? An isosceles triangle has two sides of lengths 4 cm and 9 cm. Find its perimeter.

Guided solution.

Step 1. Part a – two cases

Recall. An isosceles triangle has two equal base angles; the three angles add up to 180°.

Case 1: 40° *is the apex angle.* The base angles are equal and add up to $180° - 40° = 140°$, so each is $140° \div 2 = 70°$. The triangle has angles 40°, 70°, 70°.
Case 2: 40° *is a base angle.* The other base angle is also 40°; the apex angle is $180° - 40° - 40° = 100°$. The triangle has angles 40°, 40°, 100°.

Step 2. Part b – only one case

100° cannot be a base angle: otherwise the two base angles alone would add up to $200° > 180°$. So 100° is the apex angle, and each base angle is $(180° - 100°) \div 2 = 40°$.

Step 3. Part c – use the triangle inequality

Recall. Three lengths form a triangle exactly when the longest is less than the sum of the other two.

(3, 4, 8): $8 > 3 + 4 = 7$, *no* triangle. (3, 5, 7): $7 < 3 + 5 = 8$, a triangle exists.
An isosceles triangle with two sides 4 and 9 has its third side equal to 4 or to 9:

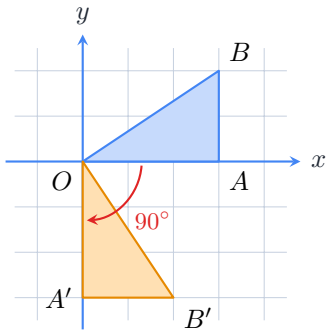
- (4, 4, 9): $9 > 4 + 4 = 8$, rejected;
- (4, 9, 9): $9 < 4 + 9 = 13$, fine. Perimeter $= 4 + 9 + 9 = 22$ cm.

Answer: a) 70° and 70°, or 40° and 100°. b) 40° and 40°. c) only (3, 5, 7); perimeter 22 cm.

Example 4

Triangle OAB with $O(0, 0)$, $A(3, 0)$, $B(3, 2)$ is rotated about the origin O *clockwise* and becomes triangle $OA'B'$ with $A'(0, -3)$, $B'(2, -3)$. What is the angle of rotation? If we rotate counterclockwise instead, through what angle must we turn to get the same result?

Guided solution.



Step 1. Choose a segment starting at the centre of rotation

Segment OA lies along the ray Ox, pointing *right*. Its image is OA' with $A'(0, -3)$, lying on the y-axis and pointing *down*.

Step 2. Measure the angle swept clockwise

A clock hand going from the "3 o'clock" direction (right) to the "6 o'clock" direction (down) sweeps

a right angle. So the clockwise angle of rotation is $90°$.

Step 3. Convert to the other direction

Recall. Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise.

Counterclockwise: $360° - 90° = 270°$.

Check: By the table in section 4, a $90°$ clockwise rotation sends (x, y) to $(y, -x)$: $A(3, 0) \to (0, -3) = A'$ and $B(3, 2) \to (2, -3) = B'$. Correct.

Answer: $90°$ clockwise, equivalent to $270°$ counterclockwise.

Example 5

On a number line, the ticks are equally spaced. The first tick is at 0, the seventh tick is at 3 (that is, six gaps away from 0). Point M is at the tick three gaps from 0, point N at the tick nine gaps from 0.

a) Find the numbers at M and N, and compute $M \div N$.

b) If the unit on the line is the metre, convert M and N to centimetres.

Guided solution.



Step 1. Find the value of one gap

From 0 to 3 there are 6 equal gaps, so each gap is worth $3 \div 6 = 0.5$.

Step 2. Find M and N

M is three gaps from 0: $M = 3 \cdot 0.5 = 1.5$. N is nine gaps from 0: $N = 9 \cdot 0.5 = 4.5$.

Step 3. Compute the quotient

$M \div N = 1.5 \div 4.5 = \dfrac{15}{45} = \dfrac{1}{3} \approx 0.33$. (Quick view: N is 3 times as far from 0 as M, so $M \div N = \frac{1}{3}$.)

Step 4. Convert units

Recall. 1 m = 100 cm; from a larger unit to a smaller unit, multiply.

$1.5 \text{ m} = 1.5 \cdot 100 \text{ cm} = 150 \text{ cm}$; $4.5 \text{ m} = 450 \text{ cm}$.

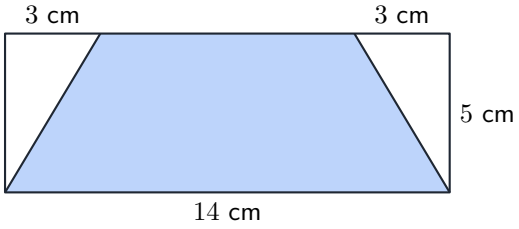
Answer: a) $M = 1.5$; $N = 4.5$; $M \div N = \frac{1}{3}$. b) 150 cm and 450 cm.

Example 6

A rectangle is 14 cm long and 5 cm wide. On its top side, two points are marked, each 3 cm from a top corner; they are joined to the two bottom corners to form the shaded trapezoid (see figure).

a) Find the area of the trapezoid in two ways.

b) A triangle has area 24 cm^2 and a base of length 8 cm. Find the height to that base.



Guided solution.

Step 1. **Method 1 – the trapezoid formula**

Recall. $S_{\text{trapezoid}} = \dfrac{(a+b)\cdot h}{2}$, where a, b are the parallel sides and h is the height.

Longer parallel side $a = 14$ cm. Shorter parallel side $b = 14 - 3 - 3 = 8$ cm. The height equals the width of the rectangle: $h = 5$ cm.

$$S = \frac{(14+8)\cdot 5}{2} = \frac{22\cdot 5}{2} = \frac{110}{2} = 55 \text{ (cm}^2\text{)}.$$

Step 2. **Method 2 – big shape minus the extra pieces**

Area of the rectangle: $14\cdot 5 = 70$ (cm^2). The two unshaded parts are right triangles, each with legs 3 cm and 5 cm and area $\dfrac{3\cdot 5}{2} = 7.5$ (cm^2). Area of the trapezoid: $70 - 2\cdot 7.5 = 70 - 15 = 55$ (cm^2).

Step 3. **Part b – find the height by working backwards**

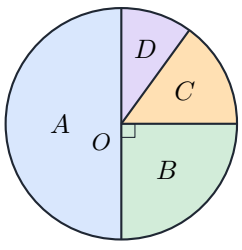
From $S = \dfrac{a\cdot h}{2}$ we get $h = \dfrac{2\cdot S}{a} = \dfrac{2\cdot 24}{8} = \dfrac{48}{8} = 6$ (cm).

Check: $\dfrac{8\cdot 6}{2} = 24$. Correct.

Answer: a) 55 cm^2. b) 6 cm.

Example 7

A circular spinner with centre O is divided into four regions A, B, C, D: region A is a semicircle, region B is a quarter circle, and the rest is split into regions C and D so that the central angle of C is one and a half times the central angle of D. What percentage of the spinner's area does each region take up? About what percentage of spins will stop in region C?



Guided solution.

Step 1. **Central angles of regions A and B**

The whole circle corresponds to 360°. Region A (a semicircle) corresponds to 180°, region B (a

quarter) to 90°.

Step 2. Central angles of regions C and D

Regions C and D share what is left: $360° - 180° - 90° = 90°$.

Let the central angle of region D be x (degrees). "The angle of C is one and a half times the angle of D" means the angle of C is $1.5x$. The two angles add up to 90°, so we get the equation

$$x + 1.5x = 90.$$

Collecting like terms: $2.5x = 90$, so $x = 90 \div 2.5 = 36$. Hence the angle of D is 36° and the angle of C is $1.5 \cdot 36° = 54°$.

Step 3. Convert to percentages

Recall. A sector with central angle $n°$ is $\dfrac{n}{360}$ of the circle.

A: $\dfrac{180}{360} = 50\%$; B: $\dfrac{90}{360} = 25\%$; C: $\dfrac{54}{360} = 0.15 = 15\%$; D: $\dfrac{36}{360} = 0.1 = 10\%$.

Check: $50\% + 25\% + 15\% + 10\% = 100\%$, and $54° = 1.5 \cdot 36°$. Correct.

Answer: A: 50%, B: 25%, C: 15%, D: 10%; about 15% of spins stop in region C.

Example 8

a) How many cubes with edge 1 cm are needed to fill a 5 cm × 3 cm × 2 cm box? Find the total surface area of the box.

b) A right prism has a base that is a right triangle with legs 3 cm and 4 cm, and the prism is 10 cm high. Find its volume. If a prism with the same base has volume 90 cm^3, what is its height?

Guided solution.

Step 1. Part a – count layer by layer

Recall. $V_{\text{box}} = a \cdot b \cdot c$; $S_{\text{total}} = 2 \cdot (ab + bc + ca)$.

The bottom layer has $5 \cdot 3 = 15$ cubes; there are 2 layers, so we need $15 \cdot 2 = 30$ cubes (volume 30 cm^3).

The three pairs of opposite faces have areas $5 \cdot 3 = 15$, $3 \cdot 2 = 6$, $5 \cdot 2 = 10$ (cm^2), so $S_{\text{total}} = 2 \cdot (15 + 6 + 10) = 2 \cdot 31 = 62$ (cm^2).

Step 2. Part b – base area times height

Recall. $V_{\text{prism}} = S_{\text{base}} \cdot h$.

The base is a right triangle: $S_{\text{base}} = \dfrac{3 \cdot 4}{2} = 6$ (cm^2). Volume: $V = 6 \cdot 10 = 60$ (cm^3).

With volume 90 cm^3: $h = V \div S_{\text{base}} = 90 \div 6 = 15$ (cm).

Check: $6 \cdot 15 = 90$. Correct.

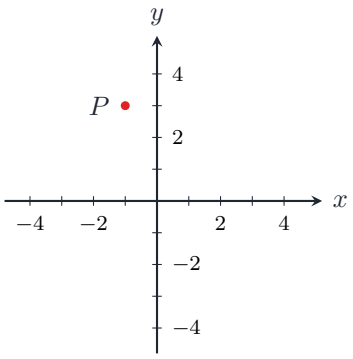
Answer: a) 30 cubes; 62 cm^2. b) 60 cm^3; 15 cm.

Note for parents. After each example, ask your child to *say in words* the formula just used and *why* it is true (for example: "a triangle is half of the rectangle with the same base and height"). Understanding the "why" means less rote memorisation and fewer mistakes on unfamiliar figures.

Core practice

Problem 1

In the coordinate plane below, each tick on the two axes represents 1 unit. Which pair of numbers could be the coordinates of point P?



(A) (1, 3) (B) (1, −3) (C) (−3, 1) (D) (3, −1) (E) (−1, 3)

Your work:

Problem 2

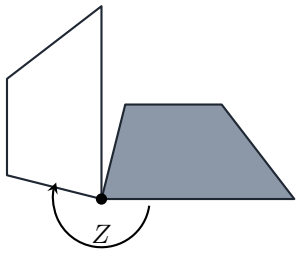
Two angles of a triangle measure 25° and 70°. The third angle of the triangle measures

(A) 85° (B) 105° (C) 65° (D) 95° (E) 75°

Your work:

Problem 3

Rotating the shaded quadrilateral about point Z *clockwise* (in the direction of the arrow in the figure) produces the unshaded quadrilateral. The angle of rotation is approximately

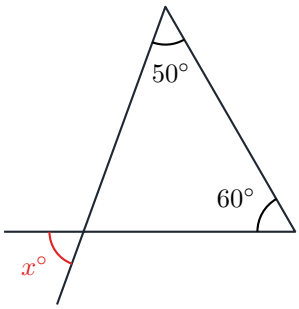


(A) 180° (B) 270° (C) 360° (D) 45° (E) 135°

Your work:

Problem 4

In the figure, two sides of the triangle are extended as shown. The value of x is

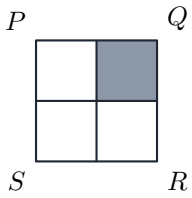


(A) 110 (B) 50 (C) 10 (D) 60 (E) 70

Your work:

Problem 5

Square $PQRS$ is divided into four equal smaller squares as shown. What percentage of the area of square $PQRS$ is shaded?

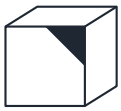


(A) 33% (B) 40% (C) 25% (D) 50% (E) 15%

Your work:

Problem 6

A cube has part of *exactly one face* painted black as shown (the paint is a right triangle at a corner of the face, whose two legs are half the edge of the cube). The other five faces are not painted. After the cube is rolled, which of the following could be the same cube?

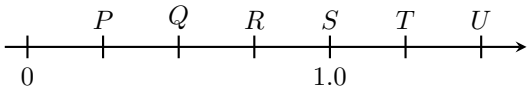


(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Your work:

Problem 7

On a number line the ticks are equally spaced. The numbers P, Q, R, S, T, U are marked as shown. Which of the following is closest to the value of $R \div U$?



(A) 0.25 (B) 0.50 (C) 0.75 (D) 1.25 (E) 1.50

Your work:

Problem 8

One angle of an isosceles triangle measures 50°. The other two angles of this triangle could be

(A) 50° and 90° (B) 40° and 50° (C) 50° and 80°

(D) 30° and 100° (E) 60° and 70°

Your work:

Problem 9

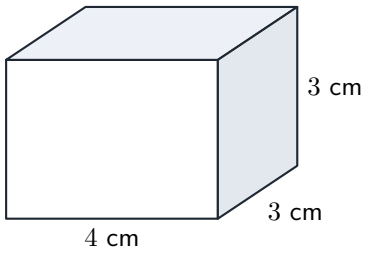
Ben measures a rope and finds that it is 3.5 m long. How many centimetres is that?

(A) 350 (B) 30.5 (C) 3.05 (D) 3.5 (E) 305

Your work:

Problem 10

Anna wants to build the solid rectangular box shown from small cubes measuring 1 cm×1 cm×1 cm. How many small cubes does Anna need?



(A) 10 (B) 12 (C) 33 (D) 66 (E) 36

Your work:

Problem 11

A square garden has a perimeter of 36 m. The area of the garden, in m^2, is

(A) 24 (B) 81 (C) 36 (D) 1296 (E) 324

Your work:

Problem 12

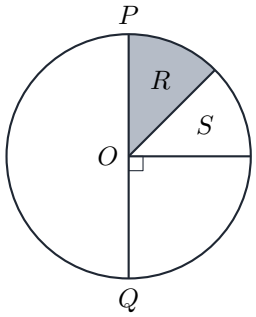
Two isosceles triangles each have at least one angle of 70°. In the first triangle, each of the two remaining angles measures an even number of degrees. In the second triangle, each of the two remaining angles measures an odd number of degrees. Let S be the sum of the two equal angles of the first triangle and T the sum of the two equal angles of the second triangle. The value of $S + T$ is

(A) 280° (B) 250° (C) 220° (D) 200° (E) 300°

Your work:

Problem 13

On the circular spinner with centre O shown, PQ is a diameter, the horizontal radius is perpendicular to PQ at O, and the sectors R and S have equal areas. About what percentage of spins will stop in the shaded region?

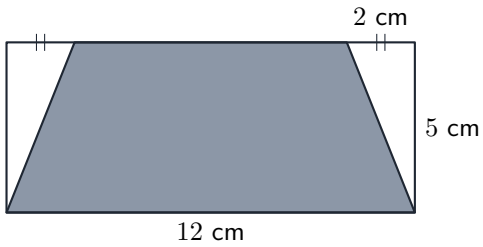


(A) 50% (B) 22.5% (C) 25% (D) 45% (E) 12.5%

Your work:

Problem 14

In the rectangle below (12 cm long, 5 cm wide), the two segments marked alike on the top side are each 2 cm long. The area of the shaded region is

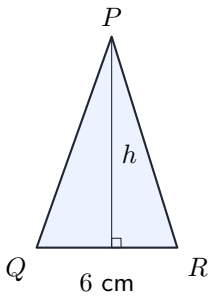


(A) 60 cm^2 (B) 20 cm^2 (C) 30 cm^2 (D) 40 cm^2 (E) 50 cm^2

Your work:

Problem 15

Triangle PQR has area 27 cm^2 and base $QR = 6$ cm. The height h of triangle PQR (to the base QR) is



(A) 9 cm (B) 18 cm (C) 4.5 cm (D) 2.25 cm (E) 7 cm

Your work:

Problem 16

A rectangular garden has a length that is 6 m more than *twice* its width. The perimeter of the garden is 120 m. How many metres wide is the garden?

(A) 8 m (B) 18 m (C) 27 m (D) 38 m (E) 22 m

Your work:

Problem 17

The letter P is written in a 2×2 grid of squares as shown: 

. A sequence of rotations about the centre of the grid and reflections in the two lines through the centre of the grid (the horizontal line and the vertical line) is carried out, and the result is: 

.

If *exactly the same sequence of transformations* is applied to the grid 

, which picture do we get?

(A) 

(B) 

(C) 

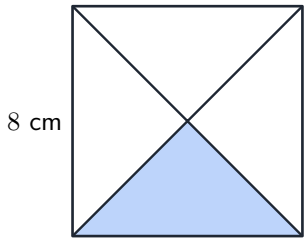
(D) 

(E) 

Your work:

Problem 18

In a square with side 8 cm, the two diagonals are drawn. What is the area of the shaded region?

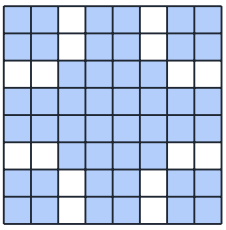


(A) 4 cm^2 (B) 8 cm^2 (C) 16 cm^2 (D) 56 cm^2 (E) 64 cm^2

Your work:

Problem 19

The grid below is made of identical small squares. What percentage of the area of the whole grid is shaded?

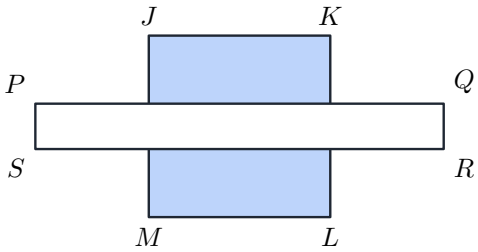


(A) 84% (B) 80% (C) 90% (D) 75% (E) 66%

Your work:

Problem 20

$JKLM$ is a square and $PQRS$ is a rectangle; they overlap as shown in the diagram. Given that JK is parallel to PQ, $JK = 8$ and $PS = 2$, find the total area of the shaded regions.



(A) 32 (B) 16 (C) 56 (D) 48 (E) 62

Your work:

Problem 21

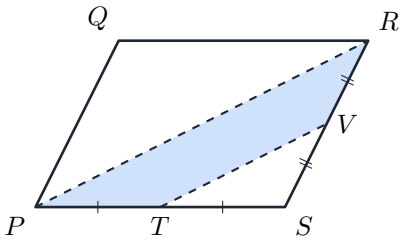
A triangle can be built with side lengths 4, 5, 8, but *no* triangle can be built with side lengths 4, 5, 10. Using only the lengths 2, 3, 5, 7, 11 (each length may be used more than once), how many different triangles with *exactly two equal sides* can be built?

(A) 8 (B) 5 (C) 20 (D) 10 (E) 14

Your work:

Problem 22

Parallelogram $PQRS$ has area 40. Points T and V are the midpoints of sides PS and RS respectively. Find the area of quadrilateral $PRVT$.

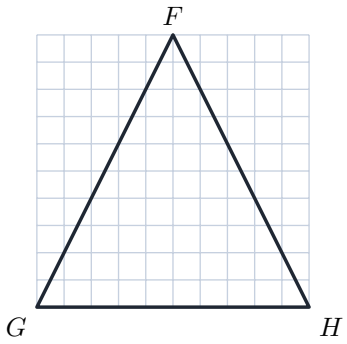


(A) 10 (B) 12 (C) 15 (D) 16 (E) 18

Your work:

Problem 23

On a 10×10 grid of squares (each square has side 1), triangle FGH is drawn as shown. Point P can be any of the 41 points where gridlines cross *inside* (not on the sides of) triangle FGH. For each position of P we get exactly three triangles: FPG, GPH, HPF. Of all these 123 triangles, how many have an area *exactly half* the area of triangle FGH?



(A) 5 (B) 7 (C) 3 (D) 11 (E) 9

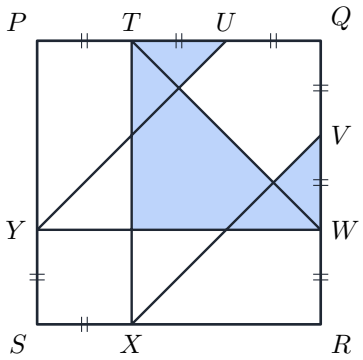
Your work:

Problem 24

Points T, U, V, W, X, Y lie on the sides of square $PQRS$ as shown, with

$$PT = TU = UQ = QV = VW = WR = XS = SY.$$

What fraction of the area of square $PQRS$ is shaded?

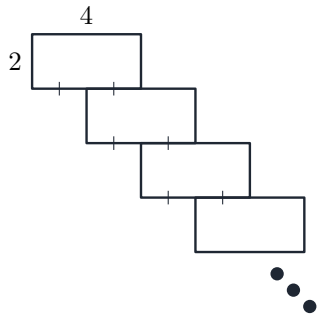


(A) $\frac{5}{18}$ (B) $\frac{1}{3}$ (C) $\frac{2}{9}$ (D) $\frac{1}{4}$ (E) $\frac{1}{6}$

Your work:

Problem 25

Rectangles measuring 4×2 are arranged by the following rule: from the second rectangle on, the top-left vertex of each rectangle is placed exactly at the midpoint of the bottom side of the rectangle just above it (as shown). When 10 rectangles in total are arranged in this way, what is the perimeter of the whole figure?



(A) 48 (B) 64 (C) 90 (D) 84 (E) 100

Your work:

Problem 26

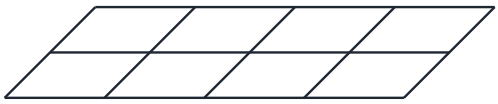
A quadrilateral has vertices $A(7, 6)$, $B(-5, 1)$, $C(-2, -3)$ and $D(10, 2)$ in the coordinate plane. The area of the quadrilateral (in square units) is

(A) 60 (B) 63 (C) 67 (D) 70 (E) 72

Your work:

Problem 27

The figure below is made of 8 identical small parallelograms joined together. Including these 8 small parallelograms, how many parallelograms are there in the figure altogether?

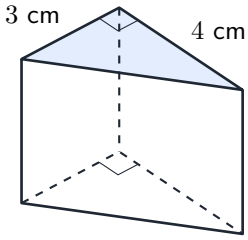


(A) 29 (B) 30 (C) 26 (D) 27 (E) 28

Your work:

Problem 28

A right triangular prism has volume 120 cm^3. Its base is a right triangle whose two legs are 3 cm and 4 cm (as shown). What is the height of the prism, in centimetres?



(A) 12 (B) 20 (C) 10 (D) 16 (E) 8

Your work:

Problem 29

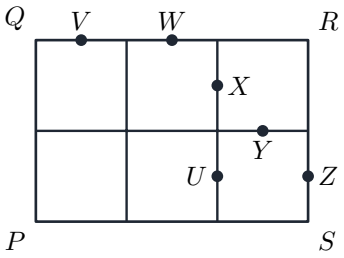
Small cubes with edge 1 are used to build a rectangular box of size $8 \times 8 \times n$ (n a positive integer). Let A be the surface area of the box and let B be the total surface area of all the small cubes (computed for each cube separately and then added up). Find the sum of all values of n for which $\frac{B}{A}$ is an integer.

(A) 86 (B) 90 (C) 70 (D) 78 (E) 96

Your work:

Problem 30

Rectangle $PQRS$ is made up of six identical squares. Points U, V, W, X, Y, Z are midpoints of sides of the squares, as shown. Which of the following triangles has the greatest area?



(A) PVU (B) PXZ (C) PVX (D) PYS (E) PQW

Your work:

Extension and challenge

Problem 31

In the coordinate plane Oxy, consider the point $A(-3, 2)$.

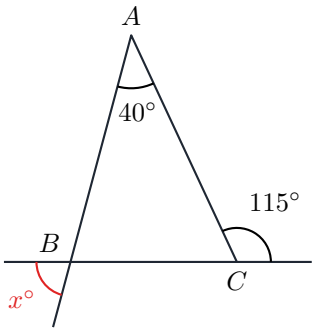
a) Let B be the reflection of A in the x-axis, C the reflection of A in the y-axis, and D the reflection of A in the origin O. Find the coordinates of B, C, D and state the quadrant of each point.

b) The four points A, C, D, B are the vertices of a rectangle. Find the perimeter and the area of that rectangle.

Your work:

Problem 32

In the figure, triangle ABC has $\widehat{BAC} = 40°$; side BC is extended in both directions and side AB is extended beyond B. The exterior angle at vertex C is 115°. The value of x is



(A) 65 (B) 75 (C) 105 (D) 115 (E) 140

Your work:

Problem 33

a) In an isosceles triangle, the apex angle is 4 times each base angle. Find the angles of the triangle.

b) An isosceles triangle has an angle of 36°. Find all possibilities for the other two angles. For each possibility, find the difference between the largest and the smallest angle of the triangle.

c) Is there an isosceles triangle with a base angle of 92°? Why?

Your work:

Problem 34 ⋆

In the coordinate plane, consider the point $A(4, 1)$.

a) Rotating A about the origin O through 90° counterclockwise gives the point B; rotating B a further 90° counterclockwise gives C; rotating C a further 90° counterclockwise gives E. Find the coordinates of B, C, E.

b) Rotating A about O through 270° *clockwise* gives which of these three points?

c) The quadrilateral $ABCE$ is a square. Find its area.

Your work:

Problem 35

Fill in the blanks:

a) 2.4 km = ... m; b) 75 mm = ... cm; c) 3 m^2 = ... cm^2;

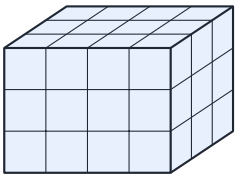
d) 2500 cm^3 = ... litres.

e) A rectangular board is 1.2 m long and 50 cm wide. Find the area of the board in cm^2 and in m^2.

Your work:

Problem 36*

Anna built a 4 cm × 3 cm × 3 cm box from 36 small cubes of edge 1 cm (as in Problem 10). Anna paints all the outside faces of the box and then takes the small cubes apart. How many small cubes have exactly 3 painted faces? Exactly 2? Exactly 1? No painted face?



Your work:

Problem 37

A rectangle has the same perimeter as a square of area 81 cm^2. The length of the rectangle is twice its width. The area of the rectangle is

(A) 81 cm^2 (B) 72 cm^2 (C) 54 cm^2 (D) 36 cm^2 (E) 64 cm^2

Your work:

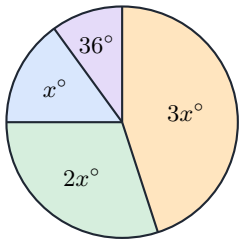
Problem 38 *

A circular spinner is divided into four regions with central angles 36°, x°, $2x$° and $3x$° (see figure).

a) Find x.

b) What percentage of the spinner's area does each region take up?

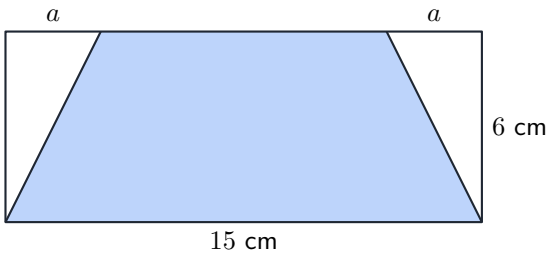
c) About what percentage of spins will *not* stop in the largest region?



Your work:

Problem 39

A rectangle is 15 cm long and 6 cm wide. On its top side, two segments of the same length a cm are taken at the two ends, and the two right triangles at the top corners are cut off, leaving the shaded trapezoid shown. The area of the trapezoid is 72 cm^2. Find a and the length of the shorter parallel side of the trapezoid.



Your work:

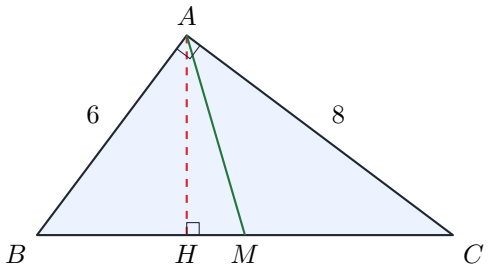
Problem 40

Triangle ABC has a right angle at A, with $AB = 6$ cm, $AC = 8$ cm and $BC = 10$ cm.

a) Find the area of triangle ABC.

b) Find the length of the height AH drawn from A to side BC.

c) Let M be the midpoint of BC. Find the area of triangle ABM.



Your work:

Problem 41

(Extension of Problem 16.) We only consider rectangles whose two side lengths are *positive integers* (in cm); a square also counts as a rectangle.

a) Find all rectangles whose area (in cm^2) is numerically *equal* to their perimeter (in cm).

b) Find all rectangles whose area is numerically *twice* their perimeter.

Your work:

Problem 42 *

(Extension of Problem 17.) The letter P is written in the top-right cell of a 2×2 grid: 

. We may use three transformations, as many times as we like and in any order:

- Q: rotate the grid 90° clockwise about its centre;
- D: reflect the grid in the *vertical* line through the centre;
- N: reflect the grid in the *horizontal* line through the centre.

a) Draw the result of doing D and then N. Can that result be obtained using only Q?

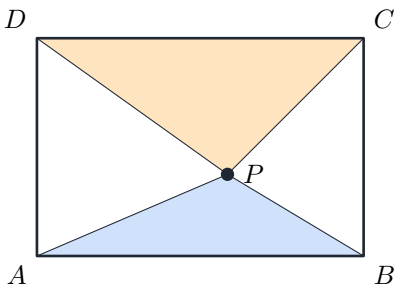
b) Show that doing D and then Q has the same effect as a reflection in one of the *diagonals* of the grid. Which diagonal?

c) How many different pictures can be obtained altogether (including the starting picture)? Explain why there cannot be more.

Your work:

Problem 43 *

(Extension of Problem 18.) Rectangle $ABCD$ has $AB = 12$ cm and $BC = 8$ cm. Point P lies inside the rectangle. (We write S_{XYZ} for the area of triangle XYZ.)



a) Prove that wherever P is, we always have $S_{PAB} + S_{PCD} = 48$ cm^2.

b) Given $S_{PAB} = 18$ cm^2 and $S_{PBC} = 20$ cm^2, find S_{PCD}, S_{PDA} and the distances from P to the four sides of the rectangle.

c) Is there a point P inside the rectangle with $S_{PAB} = 20$, $S_{PBC} = 30$ and $S_{PCD} = 30$ (cm^2)?

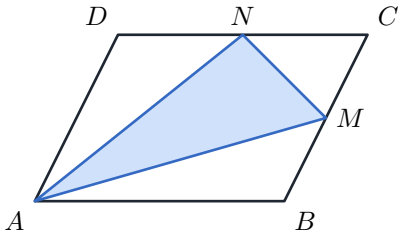
Your work:

Problem 44 *

(Extension of Problem 22.) Parallelogram $ABCD$ has area 48 cm^2.

a) Let M and N be the midpoints of BC and CD respectively. Find the area of triangle AMN.

b) Keep N as the midpoint of CD, but take M on side BC with $BM = 3\,MC$. Find the area of triangle AMN.



Your work:

Problem 45 *

(Extension of Problem 21.) Consider triangles whose three side lengths are positive integers and whose perimeter is 15 (two triangles with the same three side lengths count as one).

a) Let c be the longest side. Prove that $5 \le c \le 7$.

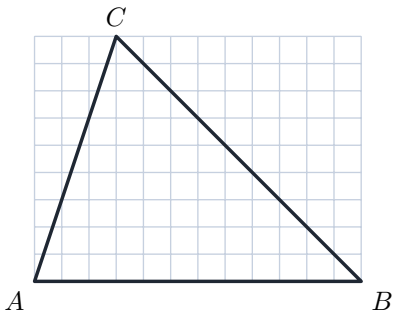
b) How many such triangles are there? List them.

c) How many of them are isosceles? How many have *exactly* two equal sides?

Your work:

Problem 46 ★★

(Extension of Problem 23.) On a square grid (each square has side 1), triangle ABC has vertices $A(0,0)$, $B(12,0)$, $C(3,9)$. Exactly 43 points where gridlines cross lie *inside* the triangle (not on its sides). For each such point P we get three triangles PAB, PBC, PCA.



a) Find the area of triangle ABC.

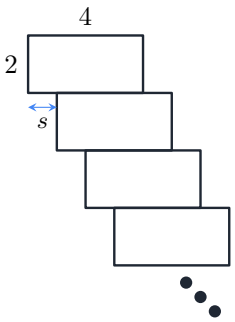
b) Find the point P for which the three triangles PAB, PBC, PCA have equal areas.

c) Of all the $3 \cdot 43 = 129$ triangles, how many have an area of exactly $\frac{1}{3}$ of the area of triangle ABC? For how many positions of P is at least one of the three triangles like that?

Your work:

Problem 47

(Extension of Problem 25.) Again we arrange 4×2 rectangles in a "staircase" as in Problem 25, but this time the top-left vertex of each rectangle (from the second one on) is placed on the bottom side of the rectangle just above it, at distance s from that rectangle's bottom-left vertex, where s is 1, 2 or 3. (Problem 25 is the case $s = 2$.)



a) For $s = 1$ and 10 rectangles, find the perimeter of the whole figure.

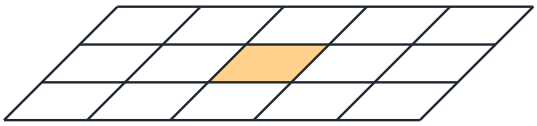
b) Find a formula for the perimeter in terms of the number of rectangles n and the distance s.

c) Can we choose $s \in \{1, 2, 3\}$ and a number of rectangles n so that the perimeter is exactly 100? Find all the ways.

Your work:

Problem 48 *

(Extension of Problem 27.) The figure below consists of 15 identical small parallelograms arranged in 3 rows of 5. The small parallelogram in the very middle is shaded.



a) How many parallelograms are there in the figure altogether?

b) How many parallelograms contain the shaded small parallelogram?

c) How many parallelograms are made of *exactly* 4 small parallelograms?

Your work:

Problem 49 * *

(Extension of Problems 28 and 29.) A $4 \times 5 \times 6$ rectangular box is built from 120 small cubes with edge 1. All the outside faces of the box are painted, and then the small cubes are separated.

a) How many small cubes have exactly 3 painted faces? exactly 2? exactly 1? no painted face at all?

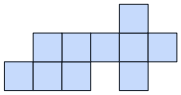
b) What is the total number of painted small faces (1×1 squares)? Explain why the result equals the surface area of the box.

c) An $n \times n \times n$ cube ($n \geq 3$) is painted and separated in the same way. Find n such that the number of small cubes with exactly 1 painted face equals the number of small cubes with no painted face.

Your work:

Problem 50 ★★

(Extension of Problems 19 and 25.) Ten unit squares (side 1) are joined to form a flat shape such that: adjacent squares share a whole side, and the shape is "in one piece" (from any square one can reach any other square through shared sides). The perimeter of the shape is the total length of the square sides that are not covered by another square.



an example

a) Show that the perimeter is always an even number.

b) What is the greatest possible perimeter?

c) What is the smallest possible perimeter?

d) Which numbers can be the perimeter of such a shape?

Your work:

3 Counting and Probability

The big idea. In this topic every problem comes back to *one core skill*: **counting correctly** – missing nothing and counting nothing twice.

- **Counting** answers the question "*how many?*": how many numbers, how many seating arrangements, how many rectangles in a figure, how many toothpicks...
- **Probability** answers the question "*how likely is it?*". When all outcomes are equally likely, we only need to *count twice*:

$$\text{Probability} = \frac{\text{number of favourable outcomes}}{\text{number of all possible outcomes}}.$$

- Our counting tools: **systematic listing**, **tables**, **tree diagrams**, the **multiplication rule**, **splitting into cases** (the addition rule), **finding a pattern** and **reasoning by elimination**.

Always ask yourself three questions before starting: *(1) How many possible outcomes are there altogether? (2) Are these outcomes "fair" (equally likely)? (3) How many of them satisfy the requirement?*



① Experiments, outcomes, events and probability

Key words.

- A **random experiment** is an action for which we know in advance *all the possible outcomes*, but not *which outcome will happen*. Examples: rolling a die, tossing a coin, spinning a spinner, drawing a marble from a bag without looking.
- The **set of possible outcomes** is the list of *all* outcomes. Rolling a die: $\{1, 2, 3, 4, 5, 6\}$ – 6 outcomes. Tossing a coin: {heads, tails} – 2 outcomes.
- An **event** is something we are interested in, for example "rolling an even number". The outcomes that make the event happen are called **favourable outcomes** for the event. The event "rolling an even number" has 3 favourable outcomes: $2, 4, 6$.
- Outcomes are **equally likely** if none of them is "easier to get" than another (a fair die, a fair coin, equal sectors on a spinner, marbles of identical shape and size).

Probability formula (when the outcomes are equally likely):

$$P(\text{event}) = \frac{\text{number of outcomes favourable to the event}}{\text{number of all possible outcomes}}.$$

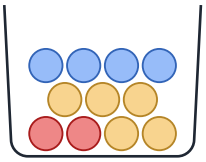
Properties. The number of favourable outcomes is between 0 and the total number of outcomes, so always $0 \le P \le 1$.

- $P = 0$: an *impossible* event (for example rolling a 7 on a die).
- $P = 1$: a *certain* event (for example rolling a number less than 7).

Why is it true? Why a division? Suppose there are n equally likely outcomes. The total "chance" 1 (one of the outcomes is sure to happen) is shared *equally* among n outcomes, so each outcome has probability $\frac{1}{n}$. If the event has k favourable outcomes, we add k equal parts: $\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n} = \frac{k}{n}$. □



Spinner with 6 equal sectors: each sector has probability $\frac{1}{6}$



Bag with 2 red, 5 yellow, 4 blue: $P(\text{red}) = \frac{2}{11}$

Common mistake. Count *objects*, not *kinds*. The bag above has 3 *colours* but 11 *marbles*. The probability of drawing a red marble is $\frac{2}{11}$, **not** $\frac{1}{3}$. Each marble is one equally likely outcome; the three colours are not equally likely (yellow has more marbles, so it is easier to draw).

Common mistake. Use the formula only when the outcomes are equally likely. On a spinner with big and small sectors, the sectors are *not* equally likely. Then the probability of landing in a sector equals *the fraction of the whole circle* that the sector takes up, that is $\frac{\text{central angle of the sector}}{360°}$. For example a sector with angle 90° has probability $\frac{90}{360} = \frac{1}{4}$.

ThinkAlgo Tip. In multiple-choice questions the answer may be written in *unsimplified* form, such as $\frac{2}{6}$ instead of $\frac{1}{3}$. Compare the values of the fractions, not just how they look. On the other hand, when you write your own solution, simplify your final answer.

② Complementary events. Finding a quantity from a probability

Complementary event. The event "A does *not* happen" is called the complement of A. We have

$$P(\text{not } A) = 1 - P(A).$$

Example: rolling a die, $P(\text{a } 6) = \frac{1}{6}$, so $P(\text{not a } 6) = 1 - \frac{1}{6} = \frac{5}{6}$.

Finding a quantity from a probability. From the probability formula we get

$$\text{number of favourable outcomes} = P \times \text{number of all outcomes.}$$

Example: a bag has 20 sweets and the probability of drawing a lemon sweet is $\frac{3}{4}$; then there are $20 \times \frac{3}{4} = 15$ lemon sweets.

When the quantities change (drawn and *not replaced*, eaten, added…): **recount both the numerator and the denominator** before computing the probability for the next draw.

Why is it true? Suppose there are n outcomes, of which k are favourable to A. Each of the remaining $n - k$ outcomes makes A *not* happen. So $P(\text{not } A) = \frac{n-k}{n} = \frac{n}{n} - \frac{k}{n} = 1 - P(A)$. □

Common mistake. A box has 10 red and 10 blue marbles. A student takes out 1 blue marble and does *not put it back*. What is the probability that the next marble is red?

- **Wrong:** "still $\frac{10}{20}$" (forgetting that the box now has one marble fewer).
- **Right:** 19 marbles are left, and all 10 red ones are still there, so the probability is $\frac{10}{19}$.

Note for parents. Do a real experiment with sweets or bottle caps of different colours: put them in a cloth bag, count together "how many are there altogether, how many are red", then take one out and ask the same two questions again. Your child will see for themselves that both the numerator and the denominator change.

③ Systematic listing: lists, tables and tree diagrams

When the number of possibilities is not too large, the safest method is to **list them all**. To list without missing or repeating anything, we need **an order**:

- An **ordered list** (from smallest to largest, alphabetically…). Example: the two-digit numbers with different digits made from 1, 2, 3: 12, 13, 21, 23, 31, 32 – ordered by the tens digit.
- A **two-way table** when the experiment has *two stages* (rolling two dice, choosing a main dish and a dessert…): each row is an outcome of stage 1, each column an outcome of stage 2, each cell a combined outcome.
- A **tree diagram** when there are *two, three, four stages in a row*: each stage branches out; each "path" from the root to the end of a branch is one outcome.

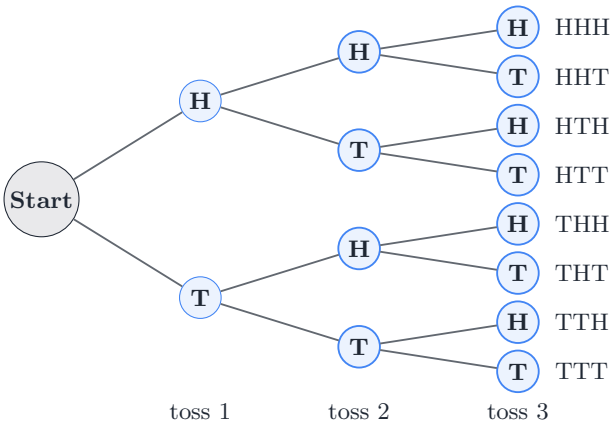
Table for rolling two dice (one red, one blue). Each cell shows the *sum* of the two numbers. There are 6 rows and 6 columns, so $6 \times 6 = 36$ equally likely outcomes.

red ↓ \ blue →	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Orange cells: sum equals 7. There are 6 such cells, so $P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$.

Common mistake. (1, 2) and (2, 1) are two different outcomes! Red 1, blue 2 is different from red 2, blue 1. If you only count "pairs of numbers" without order, the pairs are *no longer* equally likely (the pair $\{1, 2\}$ can happen in 2 ways, the pair $\{1, 1\}$ in only 1 way). That is why the sums $2, 3, \ldots, 12$ are *not* equally likely: a sum of 7 can happen in 6 ways, a sum of 2 in only 1 way.

Tree diagram for tossing a coin three times (H: heads, T: tails). Each toss has 2 branches, so there are $2 \times 2 \times 2 = 8$ paths – that is, 8 equally likely outcomes.



Example: the event "exactly two heads" consists of HHT, HTH, THH – 3 outcomes, so its probability is $\frac{3}{8}$.

ThinkAlgo Tip. A problem like "walk on a grid by tossing a coin" (heads: move up, tails: move right) is really a problem about counting sequences of heads and tails: after 4 tosses there are $2^4 = 16$ equally likely sequences, and the *end point* depends only on the *number of heads*, not on their order.

④ The multiplication rule and the addition rule

Multiplication rule. If a task consists of two successive stages, stage 1 can be done in m ways, and *for each* way of doing stage 1, stage 2 can be done in n ways, then the whole task can be done in $m \times n$ ways. (With three or four stages: keep multiplying.)

Addition rule (splitting into cases). If the ways are split into cases that *do not overlap*: case 1 has a ways, case 2 has b ways, ..., then the total number of ways is $a + b + \ldots$

Common situations with digits.

- Three-digit numbers made from 3 given non-zero digits, *no repetition*: $3 \times 2 \times 1 = 6$ numbers.
- Three-digit numbers made from 3 given non-zero digits, *repetition allowed*: $3 \times 3 \times 3 = 27$ numbers.
- The first (leading) digit **cannot be** 0: 9 choices (1 to 9); the other places have 10 choices (0 to 9).
- A three-digit **palindrome** (reads the same forwards and backwards) has the form $\overline{aba}$: 9 ways to choose a, 10 ways to choose b, and the last digit is "forced" to equal a (only 1 way). There are $9 \times 10 = 90$ such numbers.



Number of three-digit palindromes: $9 \times 10 \times 1 = 90$.

Why is it true? Why is the multiplication rule true? Draw a table with m rows (the ways of doing stage 1) and n columns (the ways of doing stage 2). Each cell of the table is *exactly one* way of doing the whole task, and each way of doing the whole task is in *exactly one* cell. The table has m rows of n cells, so $m \times n$ cells. A tree diagram gives the same result: m big branches, each splitting into n small branches. □

Choosing two objects (order does not matter). Choosing 2 of n students for classroom duty: if we care about order (first person, second person) there are $n \times (n-1)$ ways; but each pair is counted *twice* (Anna–Ben and Ben–Anna), so the number of pairs is

$$\frac{n \times (n-1)}{2}.$$

Example: there are $\dfrac{10 \times 9}{2} = 45$ pairs of different numbers taken from 10 numbers.

Common mistake. Double counting. When the question asks "how many *pairs*", "how many *groups*" (who comes first does not matter) and you use the multiplication rule with order, each group is counted several times – you must divide. On the other hand, when the question is about *numbers* (the order of the digits matters) or *seating in a row*, order matters and you do *not* divide.

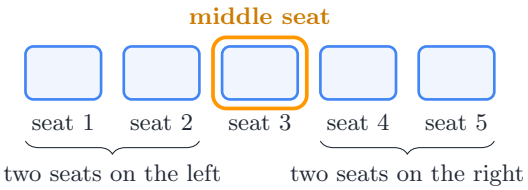
ThinkAlgo Tip. When splitting into cases, ask: "Do the cases *cover* every possibility? Do they *overlap*?" For example, for the sum $m + n$ to be even, m and n must be *both odd* or *both*

even – these two cases are separate and together they cover every possibility.

⑤ Reasoning by elimination: seating and colouring

Some problems do not ask "how many" but "*who can*, *who cannot*", "*what is the least number needed*". Then we use **reasoning**:

- **Draw a diagram** of the positions (seats, cells, faces…) and try filling it in.
- Two people who **must sit together** can be treated as **one block** taking two neighbouring seats.
- To show "*X can* sit in the middle seat": just **give one arrangement** that satisfies all the conditions.
- To show "*X cannot* sit in the middle seat": **suppose** X sits in the middle seat and show that *every* remaining arrangement breaks a condition (leading to a contradiction). This is called **proof by contradiction**.
- A "what is the *least* number" problem has **two parts**: (1) show a way that works with that number; (2) explain why fewer is impossible.



Key observation: once someone sits in the middle seat, the other four seats split into *two pairs of neighbouring seats* (seats 1–2 and seats 4–5). The two seats on either side of the middle seat (seat 2 and seat 4) are *not* next to each other.

Colouring so that neighbouring parts have different colours. Two faces of a cube are "neighbours" if they share an edge. Each face is a neighbour of 4 faces and is *not* a neighbour of exactly 1 face – the **opposite** face. A cube has 3 pairs of opposite faces (top–bottom, front–back, left–right).

- 3 *colours are enough:* paint each pair of opposite faces one colour; two faces of the same colour are opposite, so they are not neighbours.
- *Fewer than* 3 *colours is impossible:* the three faces meeting at a corner are neighbours of each other in pairs, so they need three different colours.

ThinkAlgo Tip. For a multiple-choice question "Who *cannot*…?", you can work backwards: for each option, try to find *one* valid arrangement. Every option for which you find an arrangement is eliminated; the remaining option is the answer – but remember to explain why it really cannot be arranged.

⑥ Counting shapes in a figure

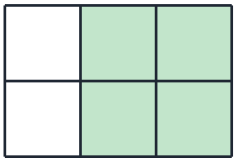
Principle. Counting shapes is a listing problem. To miss nothing and count nothing twice, **split into types** (by size, by position) and then **add up**.

The "choose two lines" trick. A rectangle in a grid is completely determined by *two horizontal lines* (top and bottom sides) and *two vertical lines* (left and right sides). Therefore:

- A strip of 1 row and n squares has $n+1$ vertical lines; the number of rectangles is the number of ways to choose 2 vertical lines: $\frac{(n+1)\times n}{2}$. Example: a strip of 4 squares has $\frac{5\times 4}{2} = 10$ rectangles.
- A grid with a horizontal lines and b vertical lines (all running the full length) has $\frac{a(a-1)}{2} \times \frac{b(b-1)}{2}$ rectangles (multiplication rule!).
- The same counting works for *parallelograms* in a grid whose slanted lines are parallel to each other.



Choose vertical lines 2 and 4
⇒ the orange rectangle

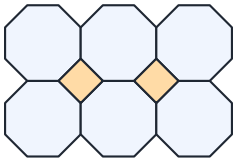


3 horizontal, 4 vertical lines:
$3 \times 6 = 18$ rectangles

Why is it true? Why does a strip of n squares have $\frac{(n+1)\times n}{2}$ rectangles? Choose the left side: $n+1$ ways; choose the right side (different from the left side): n ways – that gives $(n+1)\times n$ *ordered* choices. But choosing "line 2 then line 4" or "line 4 then line 2" gives the same rectangle, so each rectangle is counted 2 times; divide by 2. Counting by size gives the same result: a strip of 4 squares has 4 rectangles of width 1, 3 of width 2, 2 of width 3 and 1 of width 4, in total $4+3+2+1 = 10$. □

Common mistake. When the shapes **partly overlap** (not a grid with full-length lines), the "choose two lines" trick *does not work*, because many segments do not run all the way across. Check each shape: **are all four sides really drawn?** Do not count "imaginary rectangles" that are missing a side.

Counting in a tiling. A tiling made of regular octagons and small squares: if there are m rows with n octagons in each row, there are $m \times n$ octagons. Each small square sits where *four octagons meet*; between n columns there are $n-1$ gaps and between m rows there are $m-1$ gaps, so there are $(m-1)\times(n-1)$ squares.



2 rows, 3 columns: $2 \times 3 = 6$ octagons,
$(2-1)\times(3-1) = 2$ squares.

⑦ Counting with patterns: evenly spaced sequences, constant rates

Counting the whole numbers from a to b (including a and b): there are $b - a + 1$ numbers. For example from 1 to 11 there are 11 numbers, of which 6 are odd $(1, 3, 5, 7, 9, 11)$ and 5 are even.

Evenly spaced sequence (each number is a fixed amount d more than the one before):

$$\text{number of terms} = \frac{\text{last} - \text{first}}{d} + 1, \qquad \text{sum} = \frac{(\text{first} + \text{last}) \times \text{number of terms}}{2}.$$

Quantities that change at a constant rate (moving steadily, flowing steadily, counting steadily…): amount = rate × time. For example, if 3 sheep pass each minute, then in 14 minutes $3 \times 14 = 42$ sheep pass.

Figures made of sticks. Count the *horizontal sticks* and the *vertical sticks* separately and then add – usually easier than counting square by square.

Events that repeat in cycles. If one thing repeats every a minutes and another every b minutes, and they start together, they coincide again every $\text{LCM}(a, b)$ minutes.

Why is it true? Gauss's way of adding. Compute $S = 1 + 2 + 3 + \cdots + 10$. Write the sum twice, once forwards and once backwards:

$$\begin{array}{rcccccc} S & = & 1 & +2 & +3 & +\cdots+ & 10 \\ S & = & 10 & +9 & +8 & +\cdots+ & 1 \\ \hline 2S & = & 11 & +11 & +11 & +\cdots+ & 11 \end{array}$$

There are 10 columns, each with sum 11, so $2S = 10 \times 11 = 110$ and $S = 55$. In general: sum $= \dfrac{(\text{first} + \text{last}) \times \text{number of terms}}{2}$. □

Common mistake. The "off by one" error. From 3 to 10 there are $10 - 3 + 1 = 8$ numbers, *not* 7. A fence 10 m long with a post every 1 m (including both ends) needs 11 posts, not 10. Test a formula on a small example to check it.

Note for parents. For "toothpick" problems, let your child actually build the small figures (1 square, 2 squares, 3 squares…) with toothpicks, write the number of toothpicks in a table, and then ask: "each time we add one square, how many toothpicks do we add?". A child who finds the pattern alone remembers it much longer than a memorised formula.

Summary: read the clues, choose the tool.

Clue in the problem	Tool to use
"choose one at random", "what is the probability"	count all outcomes and favourable outcomes, then divide
probability given, quantity asked	favourable $= P \times$ total
something was taken out/eaten before the next draw	recount numerator and denominator
"at least one", "not"	complement: $1 - P$
two dice, two independent choices	two-way table
tossing a coin several times, several stages in a row	tree diagram
"how many numbers can be made", "how many ways"	multiplication rule (choose restricted positions first)
"how many pairs" (order does not matter)	multiplication rule, then divide by 2
conditions of the form "either... or..."	split into cases, add
"who can / cannot"	try one arrangement / proof by contradiction
"how many shapes..."	split by size, or choose two lines
rows of seats, toothpicks, constant rates	find the pattern, evenly spaced sequences

Example 1

A box contains 3 red marbles, 5 blue marbles and 4 yellow marbles, all of the same size. One marble is drawn at random.

a) Find the probability of drawing a blue marble.

b) Find the probability of drawing a marble that is *not* red.

c) Find the probability of drawing a purple marble.

Guided solution.

Step 1. Count all possible outcomes

Each marble is one outcome; the marbles have the same size, so the outcomes are equally likely. There are $3 + 5 + 4 = 12$ marbles, i.e. 12 possible outcomes.

Step 2. Part a

There are 5 blue marbles, i.e. 5 favourable outcomes. $P(\text{blue}) = \dfrac{5}{12}$.

Step 3. Part b – use the complement

$P(\text{red}) = \dfrac{3}{12} = \dfrac{1}{4}$, so $P(\text{not red}) = 1 - \dfrac{1}{4} = \dfrac{3}{4}$.

Another way (direct counting): 5 blue + 4 yellow = 9 marbles are not red, so $P = \frac{9}{12} = \frac{3}{4}$ – the same result.

Step 4. Part c

There are no purple marbles in the box: 0 favourable outcomes. $P(\text{purple}) = \dfrac{0}{12} = 0$ – an impossible event.

Answer: a) $\frac{5}{12}$; **b)** $\frac{3}{4}$; **c)** 0.

Example 2

A bag contains 30 sweets, some strawberry and some orange. If one sweet is drawn at random, the probability that it is strawberry is $\frac{2}{5}$.

a) How many strawberry sweets and how many orange sweets are in the bag?

b) Anna takes out 2 strawberry sweets and eats them. Then Ben draws one sweet at random from the bag. Find the probability that Ben gets a strawberry sweet.

Guided solution.

Step 1. Find the number of strawberry sweets

> **Recall.** Number of favourable outcomes = probability × number of all outcomes.

Strawberry sweets: $30 \times \frac{2}{5} = \frac{30 \times 2}{5} = \frac{60}{5} = 12$. Orange sweets: $30 - 12 = 18$.

Check: $\frac{12}{30} = \frac{2}{5}$ ✓.

Step 2. Recount after Anna eats 2 sweets

	strawberry	orange	total
At the start	12	18	30
After Anna eats 2 strawberry	$12 - 2 = 10$	18	$30 - 2 = 28$

Step 3. Compute the probability for Ben

$P(\text{strawberry}) = \frac{10}{28} = \frac{5}{14}$.

> **Common mistake.** Do not answer $\frac{10}{30}$: Anna has eaten 2 sweets, so the total number of sweets (the denominator) has gone down too.

Answer: a) 12 strawberry, 18 orange; b) $\frac{5}{14}$.

Example 3

A red die and a blue die (both fair) are rolled together. Find the probability of each event:

a) The sum of the two numbers is 8.

b) Both dice show the same number.

c) The sum of the two numbers is greater than or equal to 10.

Guided solution.

Step 1. Make a table of all outcomes

Rows are the numbers on the red die, columns the numbers on the blue die, and each cell shows the sum. The table has $6 \times 6 = 36$ cells, corresponding to 36 equally likely outcomes.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

sum equals 8 (5 cells)
both numbers equal (6 cells)
sum ≥ 10 (6 cells)

Step 2. **Part a**

The cells with sum 8: $(2,6), (3,5), (4,4), (5,3), (6,2)$ – 5 outcomes (the first number is the red die). $P = \frac{5}{36}$.

Step 3. **Part b**

Equal numbers lie on the diagonal: $(1,1), (2,2), \ldots, (6,6)$ – 6 outcomes. $P = \frac{6}{36} = \frac{1}{6}$.

Step 4. **Part c**

Sum 10: $(4,6), (5,5), (6,4)$; sum 11: $(5,6), (6,5)$; sum 12: $(6,6)$. Altogether $3+2+1 = 6$ outcomes. $P = \frac{6}{36} = \frac{1}{6}$.

ThinkAlgo Tip. The 6×6 table also shows *symmetry*: the number of cells above the diagonal (red smaller than blue) equals the number of cells below the diagonal (red greater than blue).

Answer: a) $\frac{5}{36}$; **b)** $\frac{1}{6}$; **c)** $\frac{1}{6}$.

Example 4

A fair coin is tossed three times in a row. Find the probability of getting: a) exactly two heads; b) at least one head.

Guided solution.

Step 1. Draw a tree diagram and list

Each toss has 2 outcomes (H or T). By the tree diagram in section 3, there are $2 \times 2 \times 2 = 8$ equally likely outcomes:

$$\text{HHH},\ \text{HHT},\ \text{HTH},\ \text{HTT},\ \text{THH},\ \text{THT},\ \text{TTH},\ \text{TTT}.$$

Step 2. Part a

The outcomes with exactly two H's: HHT, HTH, THH – 3 outcomes. $P = \frac{3}{8}$.

Step 3. Part b – complement

The complement of "at least one head" is "no heads at all", which is only TTT (1 outcome). So $P(\text{at least one H}) = 1 - \frac{1}{8} = \frac{7}{8}$.

> **ThinkAlgo Tip.** When you see the words "*at least one*", think immediately of the complement "*none at all*" – it usually has very few outcomes and is easy to count.

Answer: a) $\frac{3}{8}$; b) $\frac{7}{8}$.

Example 5

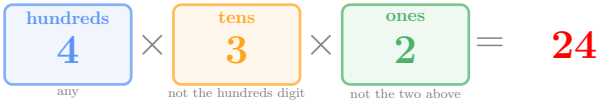
Using the digits $1, 2, 3, 4$, how many three-digit numbers can be made if: a) digits may be repeated; b) the digits must be different; c) the digits are different and the number is even?
d) One of the numbers from part b) is chosen at random. Find the probability that it is even.

Guided solution.

Step 1. Part a – repetition allowed

Hundreds: 4 ways; tens: still 4 ways; ones: 4 ways. By the multiplication rule: $4 \times 4 \times 4 = 64$ numbers.

Step 2. Part b – no repetition



Hundreds: 4 ways. The tens digit must differ from the digit already used: 3 ways. The ones digit must differ from both digits used: 2 ways. There are $4 \times 3 \times 2 = 24$ numbers.

Step 3. Part c – even, no repetition: choose the ones digit FIRST

An even number ends in 2 or 4: 2 ways. Then the hundreds digit is one of the 3 remaining digits: 3 ways; the tens digit: 2 ways. There are $2 \times 3 \times 2 = 12$ numbers.

ThinkAlgo Tip. A position with a *condition* (must be even, must not be 0...) should be **chosen first**. If you choose the hundreds digit first, the number of choices for the ones digit is sometimes 1 and sometimes 2 (depending on whether the hundreds digit used an even or an odd digit) – very easy to get wrong.

Step 4. Part d

$P(\text{even}) = \frac{12}{24} = \frac{1}{2}$.

Check: List the numbers starting with 1: $123, 124, 132, 134, 142, 143$ – exactly $6 = 3 \times 2$ numbers, of which $124, 132, 134, 142$ are even (4 numbers). The four possible first digits behave alike, so there are $4 \times 6 = 24$ numbers, ... matching part b.

Answer: a) 64; b) 24; c) 12; d) $\frac{1}{2}$.

Example 6

Four friends Anna, Ben, Chloe and David sit on four chairs in a row, numbered $1, 2, 3, 4$ from left to right. We know that: David sits on chair 1; Ben and Chloe sit next to each other; Anna does not sit next to Ben; Anna does not sit on chair 4. Who sits on chair 4?

Guided solution.

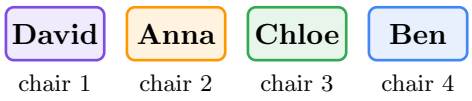
Step 1. Place David

David is on chair 1, leaving chairs $2, 3, 4$ for Anna, Ben and Chloe.

Step 2. Consider Anna's position

Anna does not sit on chair 4, so Anna sits on chair 2 or chair 3.

- *If Anna sits on chair* 3*:* Ben and Chloe take chairs 2 and 4 – these chairs are not next to each other, which breaks the condition "Ben and Chloe sit next to each other". Rejected.
- *If Anna sits on chair* 2*:* Ben and Chloe take chairs 3 and 4 (next to each other, good). Anna must not be next to Ben, and chair 3 is next to chair 2, so Ben is not on chair 3. So Ben is on chair 4 and Chloe on chair 3.



Check: David on chair 1 ✓; Ben (chair 4) next to Chloe (chair 3) ✓; Anna (chair 2) is next to David and Chloe, not next to Ben ✓; Anna is not on chair 4 ✓.

Answer: Ben sits on chair 4.

Example 7

a) The figure on the left is a strip of 4 squares. How many rectangles (including squares) are there altogether?
b) The figure on the right is a 2×2 grid of squares. How many rectangles are there? How many of them are squares?



Guided solution.

Step 1. **Part a – count by width**
Measure the width in squares:

Width	1 square	2 squares	3 squares	4 squares
Number of shapes	4	3	2	1

Total: $4+3+2+1=10$ rectangles. *Check with the "choose two vertical lines" trick:* there are 5 vertical lines, and the number of ways to choose 2 of them is $\dfrac{5\times 4}{2}=10$ ✓.

Step 2. **Part b – choose two horizontal and two vertical lines**
The grid has 3 horizontal and 3 vertical lines. Choose 2 horizontal lines: $\dfrac{3\times 2}{2}=3$ ways; choose 2 vertical lines: 3 ways. By the multiplication rule there are $3\times 3=9$ rectangles.

Step 3. **Count the squares separately**
1×1 squares: 4; 2×2 squares: 1 (the whole grid). There are $4+1=5$ squares.

Check: Counting by size: 1×1: 4; 1×2 (lying flat): 2; 2×1 (standing): 2; 2×2: 1. Total $4+2+2+1=9$ ✓.

Answer: a) 10 rectangles; b) 9 rectangles, of which 5 are squares.

Example 8

Toothpicks are used to make a row of joined squares as shown below.



a) How many toothpicks are needed for a row of 20 squares?
b) Lucy builds, one after another and separately, a row of 1 square, a row of 2 squares, ..., a row of 10 squares. How many toothpicks does Lucy use altogether?

Guided solution.

Step 1. **Find the pattern**

Squares	1	2	3	n
Toothpicks	4	7	10	$3n+1$

Explanation: count each kind of toothpick separately. Horizontal toothpicks: each square has 1 on top and 1 at the bottom, that is $2n$ toothpicks. Vertical toothpicks: n squares have $n+1$ vertical toothpicks (just as n gaps need $n+1$ posts). Total $2n+(n+1)=3n+1$.

Step 2. **Part a**

$n=20$: $3\times 20+1=61$ toothpicks.

Step 3. **Part b – sum of an evenly spaced sequence**

The numbers of toothpicks are $4, 7, 10, \ldots, 31$ (a row of 10 squares needs $3\times 10+1=31$ toothpicks). This is an evenly spaced sequence with step 3 and 10 terms.

> **Recall.** Sum of an evenly spaced sequence $=\dfrac{(\text{first}+\text{last})\times \text{number of terms}}{2}$.

Sum: $\dfrac{(4+31)\times 10}{2}=\dfrac{350}{2}=175$ toothpicks.

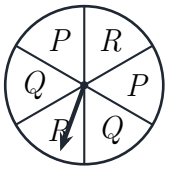
Check: $4+7+10+13+16+19+22+25+28+31$: pair first with last, $4+31=7+28=10+25=13+22=16+19=35$, giving 5 pairs, $5\times 35=175$ ✓.

Answer: a) 61 toothpicks; b) 175 toothpicks.

Core practice

Problem 1

The spinner shown is divided into 6 sectors of equal size, each marked with a letter. The spinner is spun once. What is the probability that the arrow lands on a sector marked with the letter P?



(A) $\frac{3}{6}$ (B) $\frac{4}{6}$ (C) $\frac{5}{6}$ (D) $\frac{2}{6}$ (E) $\frac{1}{6}$

Your work:

Problem 2

Using the three digits 4, 7 and 9, each digit used *at most once*, how many three-digit whole numbers with different digits can be made?

(A) 6 (B) 3 (C) 5 (D) 12 (E) 9

Your work:

Problem 3

An envelope contains 2 sunflower seeds, 3 mung bean seeds and 4 pumpkin seeds. Carrie takes one seed from the envelope at random. What is the probability that Carrie takes a sunflower seed?

(A) $\frac{2}{9}$ (B) $\frac{5}{9}$ (C) $\frac{9}{7}$ (D) $\frac{7}{9}$ (E) $\frac{1}{9}$

Your work:

Problem 4

A cinema has 11 rows of seats, numbered from 1 to 11. Each odd-numbered row has 15 seats and each even-numbered row has 16 seats. How many seats are there in the cinema altogether?

(A) 176 (B) 186 (C) 165 (D) 170 (E) 171

Your work:

Problem 5

At a class party, each student in turn draws a wrapped present at random from a bag. The presents are only books and calculators; there are 27 presents in the bag altogether. Meghan draws first. The probability that Meghan draws a book is $\frac{2}{3}$. How many books are in the bag?

(A) 15 (B) 9 (C) 21 (D) 7 (E) 18

Your work:

Problem 6

A crate of fruit contains 20 apples, 10 oranges and no other fruit. One fruit is taken from the crate at random. What is the probability that it is an orange?

(A) $\frac{1}{10}$ (B) $\frac{1}{20}$ (C) $\frac{1}{30}$ (D) $\frac{1}{3}$ (E) $\frac{2}{3}$

Your work:

Problem 7

A bag contains 2 red balls, 5 yellow balls and 4 blue balls, all the same size. Without looking, one ball is taken from the bag. The probability of taking a yellow ball is

(A) $\frac{2}{11}$ (B) $\frac{5}{11}$ (C) $\frac{4}{11}$ (D) $\frac{6}{11}$ (E) $\frac{7}{11}$

Your work:

Problem 8

You are doing a multiple-choice test. On one question you do not know the answer, so you choose an option at random. If the question has five options (A, B, C, D, E) and only one of them is correct, what is the probability that you guess correctly?

(A) $\frac{1}{5}$ (B) $\frac{5}{5}$ (C) $\frac{4}{5}$ (D) $\frac{2}{5}$ (E) $\frac{3}{5}$

Your work:

Problem 9

A fair six-sided die has the numbers 1 to 6 on its faces. The die is rolled once. What is the probability that it shows a 5?

(A) $\frac{2}{6}$ (B) $\frac{1}{6}$ (C) $\frac{5}{6}$ (D) $\frac{3}{6}$ (E) $\frac{4}{6}$

Your work:

Problem 10

Five friends Andy, Jen, Sally, Mike and Tom sit in a row of five seats. Andy does *not* sit next to Jen, and Sally sits next to Mike. Who *cannot* sit in the middle seat?

(A) Andy (B) Jen (C) Sally (D) Mike (E) Tom

Your work:

Problem 11

A box of jelly beans contains 15 red, 20 blue and 16 green jelly beans. Jack takes a green jelly bean and eats it. Then he takes a blue jelly bean and eats it too. If each jelly bean left in the box is equally likely to be chosen, what is the probability that the next jelly bean Jack takes is red?

(A) $\frac{15}{31}$ (B) $\frac{34}{49}$ (C) $\frac{15}{49}$ (D) $\frac{2}{7}$ (E) $\frac{1}{15}$

Your work:

Problem 12

Nicholas is counting the sheep in a flock as they cross a road. The sheep start crossing at 2:00 p.m., at a steady rate of 3 sheep per minute. After counting 42 sheep, Nicholas falls asleep. He sleeps for exactly an hour and a half and then wakes up, and at that moment the number of sheep that have crossed since 2:00 p.m. is exactly half of the whole flock. How many sheep are in the whole flock?

(A) 630 (B) 621 (C) 582 (D) 624 (E) 618

Your work:

Problem 13

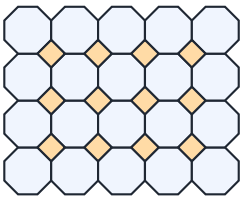
A positive whole number is called a *palindrome* if its digits read the same from left to right as from right to left. For example, 474 and 222 are palindromes. How many palindromes are there between 100 and 1000?

(A) 10 (B) 90 (C) 100 (D) 900 (E) 1000

Your work:

Problem 14

The figure shown is made only of octagons and small squares. The ratio of the number of octagons to the number of squares is



(A) 1 : 1 (B) 2 : 1 (C) 25 : 12

(D) 5 : 0 (E) 5 : 3

Your work:

Problem 15

The figure shown consists of three identical rectangles overlapping each other. How many rectangles of all sizes are there in the figure?



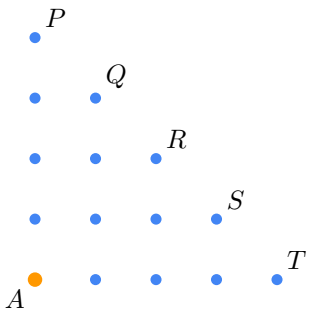
(A) 11 (B) 15 (C) 7

(D) 13 (E) 9

Your work:

Problem 16

On the dot grid shown, Jane stands at dot A. She tosses a fair coin to decide how to move: if the coin shows *heads* she moves *up* one dot; if it shows *tails* she moves *right* one dot. After exactly four tosses, Jane will stop at one of the five dots P, Q, R, S, T. What is the probability that Jane stops at dot R?

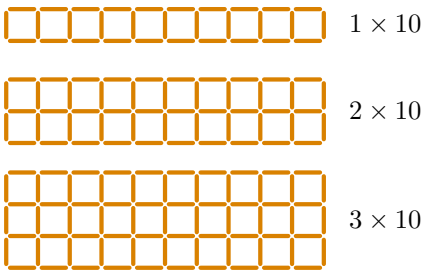


(A) $\frac{1}{2}$ (B) $\frac{3}{8}$ (C) $\frac{9}{16}$ (D) $\frac{7}{16}$ (E) $\frac{5}{16}$

Your work:

Problem 17

Identical toothpicks are used to build rectangular grids of squares, as shown below. For example, the 1×10 grid (1 row of 10 squares) uses 31 toothpicks in total. How many toothpicks are needed for a 43×10 grid (43 rows, each with 10 squares)?



(A) 913 (B) 860 (C) 871 (D) 903 (E) 946

Your work:

Problem 18

In the cafeteria at Braydon's school, the meats are beef and chicken, and the fruits are apple, pear and banana. Braydon is given a random lunch consisting of one meat and one fruit. The probability that Braydon's lunch includes a banana is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{5}$ (E) $\frac{3}{5}$

Your work:

Problem 19

A special six-sided die is made, with one number written on each face (numbers may repeat). When this die is rolled, the probability of getting a multiple of 3 is $\frac{1}{2}$, and the probability of getting an even number is $\frac{1}{3}$. The six numbers on the faces could be

(A) $1, 2, 3, 5, 5, 6$ (B) $1, 2, 3, 3, 5, 6$ (C) $1, 2, 3, 4, 6, 6$

(D) $1, 2, 3, 3, 4, 6$ (E) $2, 3, 3, 3, 5, 6$

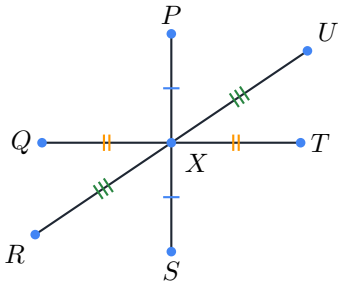
Your work:

Problem 20

Three bus routes run along the roads shown in the figure, all passing through stop X (segments with the same number of small marks are equal: $PX = XS$, $QX = XT$, $RX = XU$).

- Bus A runs the route $P \to X \to S \to X \to P$; each round trip takes 12 minutes.
- Bus B runs the route $Q \to X \to T \to X \to Q$; each round trip takes 20 minutes.
- Bus C runs the route $R \to X \to U \to X \to R$; each round trip takes 28 minutes.

At 1:00 p.m., buses A, B, C leave P, Q, R respectively. Each bus travels at a constant speed, turns around instantly at both ends of its route and runs without stopping until 11:00 p.m. Between 5:00 p.m. and 10:00 p.m., at how many moments are two or more buses at stop X at the same time?



(A) 18 (B) 19 (C) 20 (D) 21 (E) 22

Your work:

Problem 21

Each face of a cube is painted with exactly one colour. What is the smallest number of colours needed to paint the cube so that any two faces sharing an edge have different colours?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Your work:

Problem 22

How many different pairs (m, n) can be formed from the integers $1, 2, 3, \ldots, 20$ such that $m < n$ and $m + n$ is even?

(A) 55 (B) 90 (C) 140 (D) 110 (E) 50

Your work:

Problem 23

Two different two-digit positive integers are called a *reversal pair* if one is obtained from the other by swapping its two digits. For example, 52 and 25 form a reversal pair. The number 2015 has a special property: it is the product of three different prime numbers, two of which form a reversal pair ($2015 = 5 \times 13 \times 31$). Including 2015, how many positive integers less than 10 000 have this same property?

(A) 18 (B) 14 (C) 20 (D) 17 (E) 19

Your work:

Problem 24

The figure is made of 8 identical small parallelograms. Including these 8 small parallelograms, how many parallelograms are there in the figure altogether?



(A) 29 (B) 30 (C) 26 (D) 27 (E) 28

Your work:

Problem 25

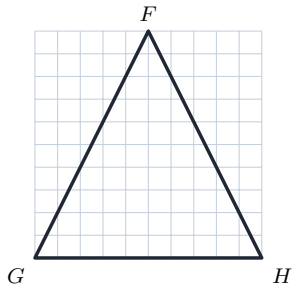
A jar contains 2 red marbles and 2 blue marbles and no other marbles. Angie randomly takes 2 marbles out of the jar at the same time. If the two marbles have the same colour, she throws one away and puts the other back into the jar. If they have different colours, she throws away the red marble and puts the blue marble back into the jar. Angie does this three times in total. The probability that the last marble left in the jar is red is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$ (E) 0

Your work:

Problem 26

On the 10×10 grid of squares shown, triangle FGH is drawn. Point P can be any of the 41 points where grid lines cross that lie *inside* (not on the sides of) triangle FGH. For each position of P, exactly three triangles are formed: FPG, GPH, HPF. Among all $41 \times 3 = 123$ of these triangles, how many have an area equal to exactly half the area of triangle FGH?

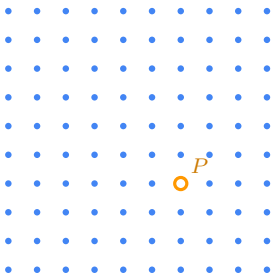


(A) 5 (B) 7 (C) 3 (D) 11 (E) 9

Your work:

Problem 27

A 10×10 square grid is made of 100 points, as shown. Point P is marked. One of the other 99 points is chosen at random to be Q. The probability that segment PQ is horizontal or vertical is



(A) $\frac{2}{11}$ (B) $\frac{1}{5}$ (C) $\frac{1}{10}$ (D) $\frac{4}{25}$ (E) $\frac{5}{33}$

Your work:

Problem 28 *

The sum of all the digits of the integers from 98 to 101 is

$$9+8+9+9+1+0+0+1+0+1=38.$$

What is the sum of all the digits of the integers from 1 to 2008?

(A) 30 054 (B) 27 018 (C) 28 036 (D) 30 036 (E) 28 054

Your work:

Problem 29

Two fair dice are rolled, one red and one blue (each has six faces numbered 1 to 6). The probability that the number on the red die is greater than the number on the blue die is

(A) $\frac{18}{36}$ (B) $\frac{25}{36}$ (C) $\frac{15}{36}$ (D) $\frac{12}{36}$ (E) $\frac{17}{36}$

Your work:

Problem 30

Two fair dice are rolled, one red and one blue. The probability that the *product* of the two numbers shown is even is

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$ (E) $\frac{11}{36}$

Your work:

Extension and challenge

Problem 31

A pencil case holds 6 red pens, 9 blue pens and 5 black pens, identical except for the colour of the ink. One pen is taken at random.

a) Find the probability of taking a blue pen.

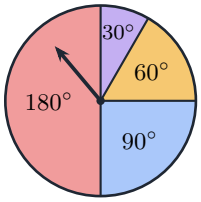
b) Find the probability of taking a pen that is *not* black.

c) How many more black pens must be put into the case so that the probability of taking a black pen is $\frac{1}{2}$?

Your work:

Problem 32

A spinner is divided into four sectors with central angles 180° (red), 90° (blue), 60° (yellow) and 30° (purple), as shown. The spinner is spun once.



a) Find the probability that the arrow lands on blue.

b) Find the probability that the arrow lands on yellow or purple.

c) Leo says: "There are 4 colours, so each colour has probability $\frac{1}{4}$." Is Leo right or wrong? Why?

Your work:

Problem 33

The school canteen has this lunch menu: one main dish out of 3 (chicken rice, spaghetti, pizza); one drink out of 2 (apple juice, orange juice); one dessert out of 2 (ice cream, yoghurt). Each lunch consists of one main dish, one drink and one dessert.

a) How many different lunches are there?

b) Mia is given a lunch at random. Find the probability that it includes spaghetti.

c) Find the probability that Mia's lunch includes both spaghetti and orange juice.

Your work:

Problem 34

A hall has 20 rows of seats. The first row has 12 seats, and each row behind has 2 seats more than the row just in front of it.

a) How many seats are in row 20?

b) Which row has exactly 30 seats?

c) How many seats does the hall have altogether?

Your work:

Problem 35

A box contains only red and blue marbles. If one marble is drawn at random, the probability that it is red is $\frac{3}{5}$. If 5 more blue marbles are put into the box, the probability of drawing a red marble drops to $\frac{1}{2}$. How many red marbles and how many blue marbles were in the box at the start?

Your work:

Problem 36

Recall: a palindrome is a number that reads the same forwards and backwards (for example 3443, 7007).

a) How many four-digit palindromes are there?

b) How many of them are even?

c) A four-digit palindrome is chosen at random. Find the probability that it is even.

d) Show that every four-digit palindrome is divisible by 11.

Your work:

Problem 37

Using the four digits $0, 1, 2, 3$, four-digit numbers with all digits different are made (each digit used exactly once).

a) How many such numbers can be made?

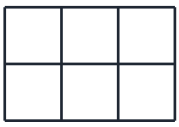
b) How many of them are even?

c) One of the numbers from part a) is chosen at random. Find the probability that it is even.

Your work:

Problem 38

The figure shows a rectangular grid of 2 rows and 3 columns of squares.



a) How many rectangles (including squares) are there in the figure altogether?

b) How many squares are there?

c) One of these rectangles is chosen at random. Find the probability that it is a square.

Your work:

Problem 39 ⋆

Four friends Anna, Ben, Chloe and David are seated at random on four chairs in a row.

a) How many seating arrangements are there?

b) Find the probability that Anna and Ben sit next to each other.

c) Find the probability that Anna and Ben sit at the two ends of the row.

Your work:

Problem 40 ⋆⋆

A box contains 4 red marbles and 6 blue marbles of the same size. Grace draws one marble at random, does *not put it back*, and then draws a second marble at random.

a) Find the probability that both marbles are red.

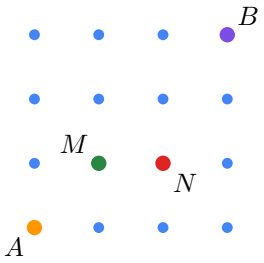
b) Find the probability that the two marbles have the same colour.

c) Find the probability that the two marbles have different colours.

Your work:

Problem 41 *

On a square dot grid, Ruby stands at dot A. She tosses a fair coin 6 times; each time it shows heads she moves *up* one dot, and each time it shows tails she moves *right* one dot (as in problem 16). The dots M, N, B are marked in the figure.



a) Find the probability that after 6 tosses Ruby stops at dot B.

b) Find the probability that Ruby passes through dot M *and* stops at B.

c) Find the probability that Ruby stops at B *without* passing through dot N.

Your work:

Problem 42 **

We build grids of squares from toothpicks as in problem 17. An "$m \times n$" grid has m rows, each with n squares.

a) Find a formula for the number of toothpicks needed for an $m \times n$ grid.

b) A square $k \times k$ grid uses exactly 760 toothpicks. Find k.

c) Sam has exactly 2026 toothpicks and wants to build an $m \times n$ grid (with $1 \le m \le n$) that uses *all* of them. Find all the grids Sam can build.

Your work:

Problem 43 *

Three fair dice coloured red, blue and yellow are rolled together (each is numbered 1 to 6). Find the probability of each event:

a) the three numbers are all different;

b) the number on the red die is less than the number on the blue die, and the number on the blue die is less than the number on the yellow die;

c) at least one die shows a 6;

d) the largest of the three numbers is exactly 4.

Your work:

Problem 44 **

Special six-sided dice are made with one of the numbers $1, 2, 3, 4, 5, 6$ written on each face (repeats allowed), so that, as in problem 19, the probability of rolling a multiple of 3 is $\frac{1}{2}$ and the probability of rolling an even number is $\frac{1}{3}$. Two dice are considered the same if they carry the same six numbers (regardless of the order of the faces).

a) What is the largest possible sum of the six numbers on such a die? What is the smallest?

b) How many different such dice are there?

Your work:

Problem 45 ⋆

Three lights flash: the first flashes every 4 seconds, the second every 6 seconds, the third every 10 seconds. At the start (0 seconds) all three flash together. Consider the next 10 minutes (from second 1 to the end of second 600).

a) How many times do all three lights flash together?

b) At how many moments do *exactly two* lights flash together?

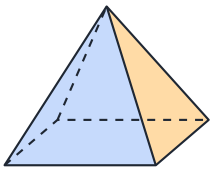
c) At how many moments does at least one light flash?

Your work:

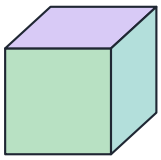
Problem 46 ⋆⋆

a) A square pyramid has 5 faces: a square base and four triangular side faces. What is the smallest number of colours needed to paint the 5 faces (one colour per face) so that any two faces sharing an edge have different colours?

b) There are 6 different colours. We paint the six faces of a cube, one colour per face, using all 6 colours. Two cubes are considered the same if one can be rotated to look exactly like the other. How many different cubes are there?



square pyramid



cube with 6 colours

Your work:

Problem 47 ★★

Consider the integers $1, 2, 3, \ldots, 20$.

a) How many pairs (m, n) with $m < n$ are there such that $m + n$ is divisible by 3?

b) How many pairs (m, n) with $m < n$ are there such that $m + n$ is divisible by 4?

c) How many triples (m, n, p) with $m < n < p$ are there such that $m + n + p$ is divisible by 3?

Your work:

Problem 48 ★

For a two-digit number $n = \overline{ab}$, call $\overline{ba}$ the *reversal* of n (as in problem 23).

a) Find all two-digit numbers n such that n plus its reversal is a perfect square.

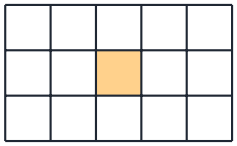
b) Find all two-digit numbers with both digits non-zero such that n minus its reversal is a positive perfect square.

c) For a three-digit number $\overline{abc}$ ($a \neq 0$, $c \neq 0$), prove that the difference $\overline{abc} - \overline{cba}$ is always divisible by 99.

Your work:

Problem 49 ★★

A grid of 3 rows and 5 columns of squares (15 small squares) is shown; the square in the very centre is shaded.



a) How many rectangles (including squares) with sides on the grid lines are there in the figure?

b) How many of them are squares?

c) One of the rectangles in part a is chosen at random. Find the probability that it contains the shaded square.

Your work:

Problem 50 ★★

A box contains 2026 red marbles and 2027 blue marbles. Outside the box there is a very large pile of blue marbles. Felix repeats the following step: take 2 marbles from the box at random.

- If the two marbles have the *same colour* (both red or both blue): throw both away, then take 1 blue marble from the pile outside and put it into the box.
- If the two marbles have *different colours*: throw away the blue one and put the red one back into the box.

Felix stops when only 1 marble is left in the box.

a) How many steps did Felix carry out?

b) Find the probability that the last marble is red.

c) If the box had started with 2025 red marbles and 2027 blue marbles, what would the probability in part b be?

Your work:

4 Number Sense

Big idea. "Number sense" is the ability to *understand* numbers, not just to *compute* with them: knowing which number is larger, whether a result is reasonable, how to round for a quick estimate, and seeing the hidden structure inside a number (its digits, its divisors, its multiples). In this topic you will review and connect:

- **decimals and negative numbers**: comparing and ordering them on the number line;
- the **order of operations** and **estimating** results;
- **fractions, ratios, percentages**: "a fraction of a number", equivalent fractions, discounts;
- **speed** – distance – time and unit conversion (km/h, m/s);
- **divisibility, common divisors, last digits, place value** (solving digit puzzles);
- the **mean, the median** and **number patterns** (Fibonacci, arithmetic sequences).

The general secret: *estimate before you calculate; check after you calculate.* A student with good number sense can often eliminate several unreasonable options in a multiple-choice question without any exact calculation.

① Decimals: reading, comparing, ordering, adding and subtracting

1. Place value. In a decimal, each digit sits in a *place*. After the decimal point come the tenths, the hundredths, the thousandths, ...

Tens	Ones	.	Tenths	Hundredths	Thousandths
	2	.	2	3	
	2	.	3	0	

Example: $2.23 = 2 + \frac{2}{10} + \frac{3}{100}$, while $2.3 = 2 + \frac{3}{10} = 2.30$.

2. Adding or removing zeros at the end of the decimal part does not change the value: $2.3 = 2.30 = 2.300$. Reason: $\frac{3}{10} = \frac{30}{100} = \frac{300}{1000}$ (multiply numerator and denominator by 10).

3. Rule for comparing two positive decimals.

- Compare the *whole-number parts* first: the one with the larger whole part is larger.
- If the whole parts are equal, *add zeros* so that both decimal parts have the same number of digits, then compare place by place from left to right: tenths, then hundredths, ...

Example: compare 0.304 and 0.34. Write $0.34 = 0.340$. Tenths: $3 = 3$; hundredths: $0 < 4$. So $0.304 < 0.340 = 0.34$.

4. Adding and subtracting decimals: write the numbers in a column *with the decimal points lined up* (places lined up), then add/subtract as with whole numbers.

5. Rounding decimals. To round to the nearest hundredth (two digits after the point), look at the digit just to the right of that place (the thousandths digit): if it is less than 5, keep the hundredths digit; if it is 5 or more, add 1 to the hundredths digit; then drop all digits after it.

Example: $4.6875 \approx 4.69$ (the thousandths digit is $7 \geq 5$), while $3.1416 \approx 3.14$ (because $1 < 5$). The sign "$\approx$" reads "is approximately".

Why is it true? Why can't we compare "by length" (the number with more digits is larger)? Write both in thousandths: $0.304 = \frac{304}{1000}$ and $0.34 = \frac{340}{1000}$. The fractions have the same denominator and $304 < 340$, so $0.304 < 0.34$, even though 0.304 "looks longer". Adding zeros is exactly bringing both to the same denominator. □

Common mistake. Many students think $2.23 > 2.3$ because "$23 > 3$". Wrong! Compare place by place: 2.23 has 2 tenths, while 2.3 has 3 tenths, so $2.23 < 2.3$.

ThinkAlgo Tip. To find the smallest (largest) number in a set of decimals, write all of them in *one column* with the decimal points lined up, adding zeros to make them equally long. Then you just compare them like whole numbers.

② Negative numbers and the number line

1. The number line. The integers are placed on a straight line: 0 in the middle, positive numbers to the right, negative numbers to the left.



2. Comparison rule. On a (horizontal) number line, *a number to the left is smaller than a number to the right.* Therefore:

- every negative number is less than 0, and 0 is less than every positive number;
- every negative number is less than every positive number;
- of two negative numbers, the one *farther from* 0 is *smaller*: $-8 < -2$ because $8 > 2$.

3. Everyday pictures. A temperature of $-8°$C is colder than $-2°$C; owing 8 dollars leaves you "poorer" than owing 2 dollars (an amount of -8 compared with -2).

4. The coordinate plane. A point is given by a pair of numbers $(x; y)$: x is the horizontal coordinate (moving right/left), y is the vertical coordinate (moving up/down). A horizontal move changes only x; a vertical move changes only y.

Why is it true? Why is $-8 < -2$? We have $-2 = -8 + 6$: to go from -8 to -2 on the number line we must move 6 units *to the right*. Moving right means increasing, so -2 is larger than -8. □

Common mistake. Do not compare negative numbers "by the feel of positive numbers". -8 is *less* than -2, even though 8 is greater than 2. When in doubt, sketch a quick number line.

ThinkAlgo Tip. **Parity (even – odd)** is a powerful "number sense" tool: even + even = even, odd + odd = even, even + odd = odd; a product of odd numbers is always odd. For instance, the sum of 5 odd numbers is odd, and the sum of 4 odd numbers is even. When a point moves on a

coordinate grid, looking at the parity of its coordinates often tells you at once whether a position can be reached.

③ Order of operations and estimation

1. Order of operations.

1. *Brackets* first: ( ), then [ ], then { }.
2. Next, *powers*.
3. Then *multiplication and division*, from left to right.
4. Finally *addition and subtraction*, from left to right.

Example: $6\times2-3=12-3=9$ (multiply first, then subtract), whereas $6\times(2-3)=6\times(-1)=-6$.
2. Rounding to estimate. To know quickly "roughly how much" a result is, round the factors to the nearest ten or hundred and compute mentally. Example: $99\times9\approx100\times9=900$.
3. A quick-calculation trick: $99\times9=(100-1)\times9=900-9=891$ (multiplication distributes over subtraction).

Why is it true? Why is $(100-1)\times9=100\times9-1\times9$? Imagine 9 rows of 100 sweets each, but 1 sweet is taken from every row. The sweets left are 9 rows × 99 sweets, which also equals (9 × 100 sweets) minus (the 9 sweets taken). This is the *distributive property*: $a\times(b-c)=a\times b-a\times c$. □

Common mistake. Working "from left to right regardless of the operation" is wrong: $2+3\times4\neq5\times4$. The correct value is $2+3\times4=2+12=14$.

ThinkAlgo Tip. In a multiple-choice question asking "which number is closest to…", the options often differ by factors of 10 (10; 100; 1000; …). Then a rough estimate is enough to choose correctly.

④ Fractions: a fraction of a number, equivalent fractions

1. A fraction is a part of a whole. $\frac{a}{b}$ means: split the whole into b equal parts and take a of them.
2. Finding a fraction of a number: to find $\frac{m}{n}$ of a, compute $a\times\frac{m}{n}$ (split a into n parts and take m of them). Example: $\frac{1}{3}$ of 396 is $396\div3=132$.
3. The part that is left. If $\frac{m}{n}$ of a job is done, the part left is $1-\frac{m}{n}=\frac{n-m}{n}$.
4. Equivalent fractions (the basic property). Multiplying (or dividing) the numerator and the denominator by the same non-zero number gives an equal fraction:

$$\frac{3}{7}=\frac{3\times9}{7\times9}=\frac{27}{63},\qquad\frac{12}{36}=\frac{12\div12}{36\div12}=\frac{1}{3}.$$

5. Simplifying a fraction: divide the numerator and the denominator by their greatest common divisor to get a fraction in lowest terms.

6. Two numbers with a known ratio and sum (or difference). If $\frac{x}{y} = \frac{5}{8}$, then there is a number k with $x = 5k$ and $y = 8k$ (because $\frac{5k}{8k} = \frac{5}{8}$). Knowing the sum $x + y$ (or the difference $y - x$), we form an equation in k and solve it. Example: $x + y = 91$ gives $5k + 8k = 91$, i.e. $13k = 91$, $k = 7$; hence $x = 35$, $y = 56$.

7. A fraction of a fraction. "Half of a quarter" is $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$: multiply numerator by numerator, denominator by denominator. This happens often when a shape is halved several times in a row: each halving multiplies the area fraction by $\frac{1}{2}$ again.

8. Comparing fractions. Bring them to a common denominator and compare numerators, or convert them to decimals. Examples: $\frac{2}{3} = \frac{8}{12} > \frac{7}{12}$; $\frac{3}{5} = 0.6 < \frac{3}{4} = 0.75$.

Why is it true? Why does multiplying the numerator and denominator by the same number leave the fraction unchanged? $\frac{3}{7}$ is "cut the cake into 7 parts, take 3". If we cut *each* part into 9 smaller pieces, the cake has $7 \times 9 = 63$ pieces and our share has $3 \times 9 = 27$ pieces. The amount of cake has not changed, so $\frac{3}{7} = \frac{27}{63}$. □

Common mistake. Read the question carefully: does it ask for the part *done* or the part *left*? Many students correctly find $\frac{1}{3}$ of 396 to be 132 and circle it straight away, while the question asks for the pages *left*, $396 - 132 = 264$.

Note for parents. In Grade 7, encourage your child to *use an unknown*: let x stand for the quantity we do not know, write the conditions of the problem as an equation, solve it and check. For example, "read $\frac{1}{3}$ of a 396-page book, x pages left" becomes $132 + x = 396$. This works for every problem, even when the numbers are "awkward" and a picture would not help.

⑤ Percentages, unit price and discounts

1. Percent means "out of one hundred". $80\% = \frac{80}{100} = 0.8$.

2. Finding $p\%$ of a number a: compute $a \times \frac{p}{100}$. Example: 80% of 50 is $50 \times \frac{80}{100} = 40$.

3. Expressing a as a percentage of b: compute $\frac{a}{b} \times 100\%$. Example: 32 correct answers out of 40 questions is $\frac{32}{40} \times 100\% = 80\%$.

4. Unit price. The unit price is the price of *one* unit (one kilogram, one book, ...):

$$\text{Total cost} = \text{unit price} \times \text{quantity}, \qquad \text{unit price} = \text{total cost} \div \text{quantity}.$$

5. A discount of $p\%$. New price = old price $\times \left(1 - \frac{p}{100}\right)$. A 10% discount leaves 90%: multiply by 0.9.

6. Successive discounts. Each time the price is reduced by 10% *of the current price*, we multiply by 0.9 once more. Example: 100 dollars after two reductions: $100 \to 90 \to 81$ (dollars), *not* 80.

Why is it true? Why are two 10% reductions not the same as one 20% reduction? The second reduction is 10% of 90 (that is 9), not 10% of 100 (that is 10). Since the price has already dropped, the next reduction is smaller too: in total the price drops by only $10 + 9 = 19$ dollars, i.e. 19%. □

ThinkAlgo Tip. Finding 10% very fast: just divide by 10 (move the decimal point one place to the left). From 10% you get 5% (halve it), 20% (double it), $80\% = 100\% - 20\%$, ...

⑥ Speed, distance, time and unit conversion

1. Three linked formulas. For motion at a constant speed:

$$s = v \times t, \qquad v = s \div t, \qquad t = s \div v,$$

where s is the distance, v the speed and t the time.

2. Who is faster? The faster person covers a *longer* distance in the *same amount of time*. The safest way: compute each person's speed in the same unit (for example m/s) and compare.

3. Running laps. Running n laps of a track L metres long means a distance of $n \times L$ metres.

4. Converting km/h to m/s. 1 km = 1000 m and 1 hour = 3600 seconds, so

$$1 \text{ km/h} = \frac{1000 \text{ m}}{3600 \text{ s}} = \frac{5}{18} \text{ m/s}, \qquad 18 \text{ km/h} = 18 \times \frac{5}{18} = 5 \text{ m/s}, \qquad 1 \text{ m/s} = 3.6 \text{ km/h}.$$

Why is it true? Why $v = s \div t$ and $t = s \div v$? It is just finding an unknown factor in the equation $s = v \times t$: dividing both sides by t (non-zero) gives $v = \frac{s}{t}$; dividing both sides by v (non-zero) gives $t = \frac{s}{v}$. □

Common mistake. The quantities must use *matching units*: if the distance is in metres and the time in seconds, the speed comes out in m/s; do not divide metres by hours. Before comparing two speeds, convert them to the same unit.

⑦ Divisibility, common divisors, last digits and place value

1. Multiples and divisors. a is a *multiple* of b (or b is a *divisor* of a) if $a = b \times k$ for some whole number k. Example: $45 = 9 \times 5$, so 45 is a multiple of 9.

2. Divisibility rules.

- by 2: the last digit is $0, 2, 4, 6, 8$; by 5: the last digit is 0 or 5; by 10: the last digit is 0;
- by 3: the digit sum is divisible by 3; by 9: *the digit sum is divisible by* 9.

3. Common divisors. A number that divides both a and b is a common divisor of a and b. Every common divisor is a divisor of the *greatest common divisor* $\gcd(a, b)$. Example: $\gcd(40, 72) = 8$, so the common divisors of 40 and 72 are $1, 2, 4, 8$.

4. The last digit of a product depends only on the last digits of the factors: ... $3 \times$ ... 7 ends in the same digit as $3 \times 7 = 21$, namely 1.

5. Place value. A number written with digits can be expanded by place:

$$\overline{abc} = 100a + 10b + c, \qquad \overline{PQQ} = 100P + 10Q + Q = 100P + 11Q.$$

(The bar on top shows that this is the number with digits a, b, c, not the product $a \times b \times c$.)

Why is it true? Why does the divisibility rule for 9 work? Take the number $\overline{abc} = 100a+10b+c$ and split it:

$$100a + 10b + c = (99a + 9b) + (a + b + c) = 9 \times (11a + b) + (a + b + c).$$

The part $9 \times (11a + b)$ is always divisible by 9. So $\overline{abc}$ is divisible by 9 exactly when the remaining part $a + b + c$ (the digit sum) is divisible by 9. For numbers with more digits the argument is the same, because $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$, ... □

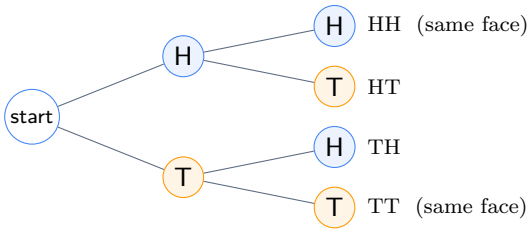
ThinkAlgo Tip. For a "digit puzzle" (each letter stands for a digit) such as $\overline{PQQ} + \overline{PPQ} + \overline{QQQ} = 876$, work *from the ones column to the left*, watching out for carries. The ones column usually has the fewest possibilities, so the first digit is found very quickly. You can also use place value to turn the puzzle into one equation. When you have to count or find all possibilities (sets of digits, outcomes of tossing coins, ...), *list them systematically* (in increasing order, or with a table/tree diagram) so that nothing is missed or repeated.

Systematic listing and chance.

- To count the three-digit numbers whose digits have a given *product*, find the sets of three digits (written in increasing order so as not to repeat), then count the arrangements of each set: three different digits can be arranged in $3 \times 2 \times 1 = 6$ ways; two equal digits and one other in 3 ways. Example: product 12 with the set $\{1; 3; 4\}$ gives 6 numbers $134, 143, 314, 341, 413, 431$; the set $\{2; 2; 3\}$ gives 3 numbers $223, 232, 322$.
- When all outcomes are *equally likely*, the probability of an event is

$$\frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

Example: tossing 2 coins has 4 equally likely outcomes: HH, HT, TH, TT (H: heads, T: tails). The probability that both coins show the same face is $\frac{2}{4} = \frac{1}{2}$. Use a tree diagram to list them:



Common mistake. "The digit sum is divisible by 9" is the rule for 9; it does *not* work for 4, 6 or 8. And a number divisible by 3 is not necessarily divisible by 9: 12 is divisible by 3 but not by 9.

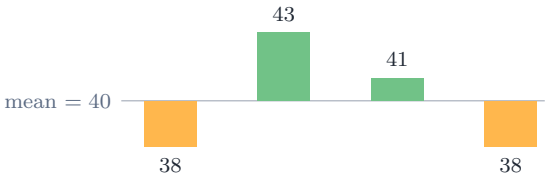
⑧ Number patterns, the mean and the median

1. The Fibonacci sequence. $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$ Each number (from the third on) is *the sum of the two numbers just before it*: $2 = 1 + 1$, $3 = 1 + 2$, $5 = 2 + 3$, ...
2. Arithmetic sequences (equally spaced). Each number (from the second on) equals the previous number plus a fixed number d (the common difference). Example: $1, 4, 7, 10$ (with $d = 3$). The n-th number is $a_1 + (n - 1) \times d$.
3. The mean (average). The mean of n numbers = (their sum) $\div n$. Hence

$$\text{Sum} = \text{mean} \times \text{how many numbers.}$$

4. The "deviation" method. The deviations from the mean always add up to 0: the "excess" of the numbers above the mean exactly equals the "shortfall" of the numbers below the mean.
5. The median. Arrange the numbers in increasing order; the number in the middle is the median (when there is an odd number of numbers). Example: the median of $42, 43, 46, 49, 50$ is 46. With an even number of numbers, the median is the mean of the two middle numbers: the median of $6, 7, 9, 10$ is $(7 + 9) \div 2 = 8$. Note: when a number is added or changed, the "middle" position can move, so re-order the list.
6. Consecutive integers. For an odd number of consecutive integers, the mean is the middle number: the mean of $19, 20, 21, 22, 23$ is 21.

Why is it true? Why do the deviations add up to 0? Suppose 4 numbers have mean 80, so their sum is $4 \times 80 = 320$. Write each number as 80+ (its deviation). Adding up: 320+ (sum of deviations) $= 320$, so the sum of the deviations must be 0. □



The four numbers $38, 43, 41, 38$ have mean 40. Deviations from 40: -2, $+3$, $+1$, -2; sum $= 0$.

Note for parents. You can illustrate the mean with stacks of coins: "level out" the tall stacks into the short ones until all stacks are the same height. That common height is the mean. The coins moved out of the tall stacks exactly equal the coins added to the short ones – this is the idea "the deviations add up to 0".

Worked examples

Example 1

Consider the set $\{0.5;\ 0.45;\ 0.405;\ 0.054;\ 0.54\}$.

a) Arrange the numbers in increasing order.

b) Find the sum of the largest and the smallest number.

Guided solution.

Step 1. Write the numbers in a column, adding zeros to get three digits after the point

Adding zeros at the end of the decimal part does not change the value, so:

$$0.5 = 0.500; \quad 0.45 = 0.450; \quad 0.405; \quad 0.054; \quad 0.54 = 0.540.$$

Step 2. Compare them like whole numbers

Dropping "0.", we compare $500;\ 450;\ 405;\ 54;\ 540$:

$$54 < 405 < 450 < 500 < 540.$$

So the increasing order is: $0.054 < 0.405 < 0.45 < 0.5 < 0.54$.

Step 3. Add the largest and the smallest (decimal points lined up)

$$\begin{array}{r} 0.540 \\ +\ 0.054 \\ \hline 0.594 \end{array}$$

Check: $0.594 - 0.54 = 0.054$. Correct.

Answer: a) $0.054 < 0.405 < 0.45 < 0.5 < 0.54$; **b)** 0.594

Example 2

Emma cycles along a 45 km route. In the morning she covers $\frac{2}{5}$ of the route. How many kilometres does Emma still have to ride in the afternoon, and what fraction of the whole route is that?

Guided solution.

Step 1. Distance covered in the morning

Recall. To find $\frac{m}{n}$ of a number a, compute $a \times \frac{m}{n}$.

Morning: $45 \times \frac{2}{5} = \frac{45 \times 2}{5} = \frac{90}{5} = 18$ (km).

Step 2. Introduce an unknown and form an equation

Let x (km) be the distance still to ride in the afternoon ($x > 0$). The morning distance plus the afternoon distance is the whole route:

$$18 + x = 45.$$

Step 3. Solve the equation
$x = 45 - 18 = 27$ (km).

Step 4. The fraction left
Method 1: $1 - \frac{2}{5} = \frac{5}{5} - \frac{2}{5} = \frac{3}{5}$. Method 2: $\frac{27}{45} = \frac{27 \div 9}{45 \div 9} = \frac{3}{5}$. Both methods give the same result.

Check: $18 + 27 = 45$ km and $45 \times \frac{3}{5} = 27$ km. Correct.

Answer: 27 km, which is $\frac{3}{5}$ of the route

Example 3

a) Evaluate $20 - 4 \times 3 + 8 \div 2$ and $(20 - 4) \times (3 + 8) \div 2$.
b) Without calculating exactly, decide which number 101×49 is closest to: 50; 500; 5000; 50 000? Then compute it exactly with a quick method.

Guided solution.

Step 1. No brackets: multiply and divide first; add and subtract after

$$20 - \underbrace{4 \times 3}_{12} + \underbrace{8 \div 2}_{4} = 20 - 12 + 4 = 8 + 4 = 12.$$

(Subtraction and addition are done from left to right: $20 - 12 = 8$, then $8 + 4 = 12$.)

Step 2. With brackets: brackets first

$$(20 - 4) \times (3 + 8) \div 2 = 16 \times 11 \div 2 = 176 \div 2 = 88.$$

Step 3. Estimate
$101 \approx 100$ and $49 \approx 50$, so $101 \times 49 \approx 100 \times 50 = 5000$. Choose 5000.

Step 4. Exact value using the distributive property
$101 \times 49 = (100 + 1) \times 49 = 4900 + 49 = 4949$, indeed very close to 5000.

Answer: a) 12 and 88; b) closest to 5000; $101 \times 49 = 4949$

Example 4

a) Apples cost \$1.80 per kilogram. Grace buys 15 kg and gets a 10% discount on the whole bill. How much does Grace pay?
b) A shirt costs \$40. After every week the shop reduces the *current price* by 10%. After at least how many weeks is the price of the shirt below \$30?

Guided solution.

Step 1. Total before the discount
Total cost = unit price × quantity $= 1.80 \times 15 = 27$ (dollars).

Step 2. Subtract 10%
10% of 27 is $27 \div 10 = 2.70$ (dollars). Grace pays $27 - 2.70 = 24.30$ (dollars). (Or: 90% remains, i.e. $27 \times 0.9 = 24.3$.)

Step 3. Successive discounts: multiply by 0.9 every week
Let g_n be the price of the shirt (in dollars) after n weeks. Then $g_0 = 40$ and $g_n = g_{n-1} \times 0.9$.

Time	Start	After 1 week	After 2 weeks	After 3 weeks
Price (dollars)	40	36	32.40	29.16
Reduction (dollars)	–	4	3.60	3.24

After 2 weeks the price is \$32.40 > \$30; after 3 weeks it is \$29.16 < \$30.

Common mistake. Do not think "the price drops by \$4 every week" (because 10% of 40 is 4). Each week the reduction is 10% of *the price at that time*, so the reductions get smaller and smaller: 4, then 3.60, then 3.24, ...

Answer: a) \$24.30; b) 3 weeks

Example 5

a) Leo cycles 3 laps around a stadium with a perimeter of 400 m in 4 minutes. Mia cycles at 15 km/h. Who is faster?
b) A car travels at 72 km/h. How many metres does it travel in 3 seconds?

Guided solution.

Step 1. Leo's speed (m/s)
Distance: $3 \times 400 = 1200$ (m). Time: 4 minutes $= 4 \times 60 = 240$ (seconds).

$$v_{\text{Leo}} = 1200 \div 240 = 5 \text{ (m/s)}.$$

Step 2. Convert Mia's speed to m/s

Recall. $1 \text{ km/h} = \dfrac{1000 \text{ m}}{3600 \text{ s}} = \dfrac{5}{18} \text{ m/s}.$

$v_{\text{Mia}} = 15 \times \dfrac{5}{18} = \dfrac{75}{18} \approx 4.17$ (m/s). Since $5 > 4.17$, **Leo is faster**. (Another way: 5 m/s $= 5 \times 3.6 = 18$ km/h > 15 km/h.)

Step 3. The car
72 km/h $= 72 \times \dfrac{5}{18} = 20$ m/s. In 3 seconds it travels $20 \times 3 = 60$ (m).

Answer: a) Leo is faster (5 m/s against about 4.17 m/s); b) 60 m

Example 6

a) Which of the numbers 72; 84; 117; 135 are divisible by 9?
b) Find the digit x such that the number $\overline{3x5}$ is divisible by 9.
c) How many whole numbers are divisors of both 24 and 36?

Guided solution.

Step 1. Use the rule for 9: compute the digit sum
72: $7 + 2 = 9$ (divisible by 9). 84: $8 + 4 = 12$ (not divisible by 9).
117: $1 + 1 + 7 = 9$ (divisible). 135: $1 + 3 + 5 = 9$ (divisible).
So 72, 117, 135 are divisible by 9; 84 is not ($84 = 9 \times 9 + 3$).

Step 2. Find the digit x
The digit sum of $\overline{3x5}$ is $3+x+5=8+x$. Since x is a digit, $8 \le 8+x \le 17$. In this range only 9 is divisible by 9, so $8+x=9$, i.e. $x=1$. The number is $315=9\times 35$.

Step 3. Common divisors of 24 and 36
Divisors of 24: $1, 2, 3, 4, 6, 8, 12, 24$. Divisors of 36: $1, 2, 3, 4, 6, 9, 12, 18, 36$.
The numbers in both lists: $1, 2, 3, 4, 6, 12$ – these are exactly the divisors of $\gcd(24, 36) = 12$.

Answer: a) 72; 117; 135; **b)** $x=1$; **c) 6 numbers**

Example 7

In the calculation $\overline{AB}+\overline{AB}+\overline{AB}=\overline{1AB}$, each letter stands for a digit. Find A and B.

Guided solution.

Method 1 – Place value. We have $\overline{1AB}=100+\overline{AB}$. So

$$3\times\overline{AB}=100+\overline{AB} \;\Rightarrow\; 2\times\overline{AB}=100 \;\Rightarrow\; \overline{AB}=50.$$

So $A=5$, $B=0$.

Method 2 – Column by column. Ones column: $B+B+B=3B$ ends in the digit B, so $3B-B=2B$ ends in 0, hence $B=0$ or $B=5$.

- If $B=5$: $3\times 5=15$, write 5 carry 1. Tens column: $3A+1=\overline{1A}=10+A$, so $2A=9$ – no digit works. Rejected.
- If $B=0$: no carry. Tens column: $3A=10+A$, so $2A=10$, $A=5$.

Check: $50+50+50=150=\overline{1AB}$ with $A=5$, $B=0$. Correct.

Answer: $A=5$, $B=0$

Example 8

a) Noah's four test scores are 6; 9; 10 and x. The mean of the four scores is 8. Find x.
b) The mean of five consecutive whole numbers is 30. Find the smallest of them.

Guided solution.

Step 1. Form an equation from the definition of the mean

Recall. Mean = sum ÷ how many numbers, so sum = mean × how many numbers.

$$\frac{6+9+10+x}{4}=8 \;\Rightarrow\; 25+x=32 \;\Rightarrow\; x=32-25=7.$$

Step 2. Using deviations (faster)
Deviations from 8: -2, $+1$, $+2$, total $+1$. All deviations must add up to 0, so $x=8-1=7$.

Step 3. Five consecutive numbers
Let the smallest number be n; the five numbers are $n, n+1, n+2, n+3, n+4$, with sum $5n+10$. So $5n+10=5\times 30=150$, hence $5n=140$, $n=28$. The five numbers are $28, 29, 30, 31, 32$; the smallest is 28.

Check: $28+29+30+31+32=150=5\times 30$. Correct.

Answer: a) $x=7$; **b)** 28

Core practice

Problem 1

The smallest number in the set $\{3.2;\ 2.3;\ 3;\ 2.23;\ 3.22\}$ is

(A) 3.2 (B) 2.3 (C) 3 (D) 2.23 (E) 3.22

Your work:

Problem 2

Keegan goes on a 36 km kayak trip. Before lunch he paddles the first 12 km. After lunch, what fraction of the whole trip does Keegan still have to paddle?

(A) $\frac{1}{2}$ (B) $\frac{5}{6}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$ (E) $\frac{3}{5}$

Your work:

Problem 3

Which of the following statements is **true**?

(A) 0 is less than -5 (B) 7 is less than -1 (C) 10 is less than $\frac{1}{4}$

(D) -1 is less than -3 (E) -8 is less than -2

Your work:

Problem 4

The value of $6 \times 2 - 3$ is

(A) 9 (B) -6 (C) 12 (D) 15 (E) 10

Your work:

Problem 5

A test is marked out of 50. Mukesh scores 80%. How many marks does Mukesh get?

(A) 40 (B) 62.5 (C) 10 (D) 45 (E) 35

Your work:

Problem 6

Ground beef sells for $5.00 per kilogram. How much do 12 kg of ground beef cost?

(A) $5.00 (B) $12.00 (C) $60.00 (D) $17.00 (E) $2.40

Your work:

Problem 7

The value of $(8 \times 4) + 3$ is

(A) 96 (B) 15 (C) 56 (D) 35 (E) 28

Your work:

Problem 8

Which of the following integers is closest to 99×9?

(A) 10 000 (B) 100 (C) 100 000 (D) 1000 (E) 10

Your work:

Problem 9

Joshua is reading a 396-page book. He has read only the first $\frac{1}{3}$ of the book. How many pages does he still have to read to finish the book?

(A) 264 (B) 124 (C) 250 (D) 199 (E) 244

Your work:

Problem 10

Which of the following numbers is a multiple of 9?

(A) 50 (B) 40 (C) 35 (D) 45 (E) 55

Your work:

Problem 11

Consider the set $\{0.34;\ 0.304;\ 0.034;\ 0.43\}$. The sum of the smallest and the largest numbers in the set is

(A) 0.77 (B) 0.734 (C) 0.077 (D) 0.464 (E) 0.338

Your work:

Problem 12

Keiko and Leah run on a circular track 150 m long. Keiko runs 3 laps in 120 seconds, and Leah runs 5 laps in 160 seconds. Who is the faster runner, and approximately what is her speed?

(A) Keiko, 3.75 m/s (B) Keiko, 2.4 m/s (C) Leah, 3.3 m/s

(D) Leah, 4.69 m/s (E) Leah, 3.75 m/s

Your work:

Problem 13

In the addition shown, P and Q each represent a digit. The value of $P + Q$ is

$$\begin{array}{r} P\ Q\ Q \\ P\ P\ Q \\ +\ Q\ Q\ Q \\ \hline 8\ 7\ 6 \end{array}$$

(A) 3 (B) 5 (C) 7 (D) 6 (E) 4

Your work:

Problem 14

The mean (average) of the four integers 78; 83; 82 and x is 80. Which of the following statements is true?

(A) x is 2 greater than the mean

(B) x is 1 less than the mean

(C) x is 2 less than the mean

(D) x is 3 less than the mean

(E) x is equal to the mean

Your work:

Problem 15

In the Fibonacci sequence $1, 1, 2, 3, 5, \ldots$, each number from the 2 onwards is the sum of the two numbers just before it. For example, the next number in the sequence is $3 + 5 = 8$. Which of the following numbers is in the sequence?

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Your work:

Problem 16

The mean (average) of five consecutive integers is 21. The smallest of the five integers is

(A) 17 (B) 21 (C) 1 (D) 18 (E) 19

Your work:

Problem 17

How many positive whole numbers, including 1, divide exactly into both 40 and 72?

(A) 9 (B) 12 (C) 4 (D) 2 (E) 5

Your work:

Problem 18

The number that goes into the box to make $\frac{3}{7} = \frac{\square}{63}$ true is

(A) 27 (B) 9 (C) 59 (D) 63 (E) 3

Your work:

Problem 19

A fraction is equal to $\frac{5}{8}$. Its numerator and denominator add up to 91. What is the difference between the denominator and the numerator of this fraction?

(A) 21 (B) 3 (C) 33 (D) 13 (E) 19

Your work:

Problem 20

Sally is driving at 80 km/h along a road with equally spaced power poles. Her car passes a pole every 4 seconds. Which of the following is closest to the distance between two neighbouring poles?

(A) 50 m (B) 60 m (C) 70 m (D) 80 m (E) 90 m

Your work:

Problem 21

Each of the integers 334 and 419 has digits whose product is 36 (since $3 \cdot 3 \cdot 4 = 36$ and $4 \cdot 1 \cdot 9 = 36$). How many 3-digit positive integers have digits whose product is 36?

(A) 21 (B) 15 (C) 18 (D) 24 (E) 12

Your work:

Problem 22

An *arithmetic sequence* is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, $2, 4, 6, 8$ and $1, 4, 7, 10$ are arithmetic sequences. In the grid shown, the numbers in each row (read from left to right) must form an arithmetic sequence, and the numbers in each column (read from top to bottom) must form an arithmetic sequence. The value of x is

1			
4			25
7			x
10		36	

(A) 37 (B) 28 (C) 36 (D) 43.75 (E) 46

Your work:

Problem 23

In the six-digit number $\overline{1ABCDE}$, each letter represents a digit. Given that $\overline{1ABCDE} \times 3 = \overline{ABCDE1}$, the value of $A + B + C + D + E$ is

(A) 29 (B) 26 (C) 22 (D) 30 (E) 28

Your work:

Problem 24

On a coordinate grid, a dot starts at the point $(20, 19)$. On its first move it goes 1 unit horizontally or vertically, that is, to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$ or $(20, 18)$. On its second move it goes 2 units in a direction perpendicular to the first move (either way). Each move after that is 1 unit longer than the previous one (3 units, 4 units, 5 units,...) and must be perpendicular to the direction of the move just before it. The dot stops after 10 moves. Which of the following final positions is **not** possible?

(A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Your work:

Problem 25

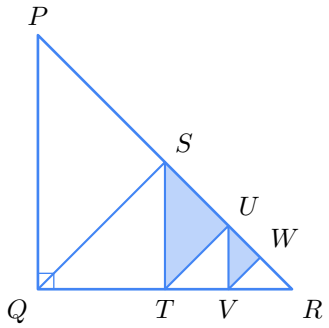
In the "Coin Game", you toss three coins at the same time. You win only if the 3 coins all show heads or all show tails. If you play the game once only, what is the probability of winning?

(A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{2}{27}$ (D) $\frac{2}{3}$ (E) $\frac{1}{3}$

Your work:

Problem 26

In the right-angled triangle PQR, the right angle is at Q and $PQ = QR$. The segments QS, TU and VW are perpendicular to PR, and the segments ST and UV are perpendicular to QR, as shown. What fraction of the area of triangle PQR is shaded?



(A) $\frac{3}{16}$ (B) $\frac{3}{8}$ (C) $\frac{5}{16}$ (D) $\frac{5}{32}$ (E) $\frac{7}{32}$

Your work:

Problem 27

Emily went to a second-hand market where, every 15 minutes, the price of every item was reduced by 10% of its *current* price. At 9:00 a.m., the price of a rug was \$10.00. At 9:15 a.m., the price was reduced to \$9.00. As soon as the price of the rug fell *below* \$8.00, Emily bought it. At what time did Emily buy the rug?

(A) 9:45 a.m. (B) 9:15 a.m. (C) 9:30 a.m.

(D) 10:15 a.m. (E) 10:00 a.m.

Your work:

Problem 28

The number 2018 is used to create six-digit positive integers: each of them must contain the digits $2, 0, 1, 8$ *together and in this order*. For example, 720 186 is allowed, but 209 318 and 210 893 are not. How many of these six-digit integers are divisible by 9?

(A) 28 (B) 27 (C) 31 (D) 34 (E) 22

Your work:

Problem 29

The number N is the product of all positive odd integers from 1 to 99 that do *not* end in the digit 5. That is,

$$N = 1 \times 3 \times 7 \times 9 \times 11 \times 13 \times 17 \times 19 \times \cdots \times 91 \times 93 \times 97 \times 99.$$

The units digit of N is

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Your work:

Problem 30

Five students wrote a quiz with a maximum score of 50. The scores of four of the students were 42, 43, 46 and 49. The score of the fifth student was N. The mean of the five students' scores was the same as the median of the five students' scores. The number of possible values of N is

(A) 3 (B) 4 (C) 1 (D) 0 (E) 2

Your work:

Extension and challenge

Problem 31

Consider the numbers: -1.5; 0.25; -0.75; 1.2; -2.

a) Arrange the numbers in increasing order.

b) Which numbers lie between -1 and 1 on the number line?

c) Find the sum of the largest and the smallest number.

Your work:

Problem 32

Consider the expression $2 + 3 \times 4 - 1$.

a) Evaluate the expression.

b) Insert *one* pair of brackets ( ) into the expression (without changing the order of the numbers or the signs) to get the largest possible value. What is that largest value?

c) Same question as b), but for the smallest value.

Your work:

Problem 33

Anna's storybook has 360 pages. On the first day Anna reads $\frac{1}{4}$ of the book. On the second day Anna reads $\frac{1}{3}$ *of the remaining pages*. After two days, how many pages has Anna not read yet, and what fraction of the book is that?

Your work:

Problem 34

The first test has 40 questions, and Ben answers 32 of them correctly.

a) What percentage of the questions in the first test did Ben answer correctly?

b) The second test has 60 questions. At least how many questions must Ben answer correctly in the second test so that, over both tests together, his correct answers make up at least 85% of all the questions?

Your work:

Problem 35

A shop sells cashew nuts in packs: a 3 kg pack costs \$45, a 5 kg pack costs \$70 (no loose sales).

a) Which pack is cheaper per kilogram?

b) Grace needs to buy *exactly* 15 kg. How should she buy them to spend as little as possible? How much does it cost?

c) How much does it cost to buy *exactly* 13 kg?

Your work:

Problem 36

The five members of a group weigh (in kg) 38; 41; 40; 37 and x. Their mean weight is 40 kg.

a) Find x. Is the fifth member heavier or lighter than the mean, and by how many kilograms?

b) Find the median weight of the group.

c) A sixth member joins the group, and the mean weight of all six becomes 41 kg. How much does the sixth member weigh?

Your work:

Problem 37

A running track around the schoolyard is 200 m long. Mia runs 6 laps in 4 minutes. Tom runs at a constant 18 km/h.

a) Who is faster?

b) Keeping the same speeds, how many minutes does each of them need to run 3 km?

c) They start at the same time from the same point of the track, but run in *opposite directions*. After how many seconds do they meet for the first time?

d) One day Tom increases his speed to 21.6 km/h while Mia runs as before. They start at the same time from the same point, running in the *same direction*. How long is it until Tom is exactly one lap ahead of Mia?

Your work:

Problem 38

a) Find the digits a, b such that the number $\overline{4a7b}$ is divisible by 2, 5 and 9.

b) How many four-digit numbers of the form $\overline{5a3b}$ are divisible by 9? List them.

Your work:

Problem 39 ⋆

In the addition below, M and N are two different non-zero digits. Find all pairs $(M; N)$ that work.

$$\begin{array}{cccc} & & M & N \\ & & N & M \\ + & & N & N \\ \hline & 1 & 8 & 7 \end{array}$$

Your work:

Problem 40 ⋆⋆

A sequence is built by the Fibonacci rule: from the third number on, each number is the sum of the two numbers just before it.

a) The first number is 2 and the fifth number is 19. Find the second and the eighth numbers of the sequence.

b) Find the mean of the first five numbers of the sequence in part a).

c) In the familiar Fibonacci sequence $1, 1, 2, 3, 5, 8, \ldots$, is the 2026th number even or odd? Explain.

Your work:

Problem 41

Sums of consecutive numbers (extends Problem 16).

a) Find *all* the ways of writing 100 as a sum of at least two consecutive positive integers.

b) Prove that 64 cannot be written as a sum of at least two consecutive positive integers.

Your work:

Problem 42

Common divisors and counting divisors (extends Problem 17).

a) How many positive integers divide both 720 and 1080?

b) Find the smallest positive integer with exactly 12 divisors.

Your work:

Problem 43

How does a fraction change? (extends Problems 18 and 19).

a) When the same whole number n is added to both the numerator and the denominator of $\frac{5}{8}$, the result is a fraction equal to $\frac{4}{5}$. Find n.

b) Let $\frac{a}{b}$ be a fraction with $0 < a < b$. Prove that for every positive integer n, $\frac{a+n}{b+n} > \frac{a}{b}$.

c) How many fractions equal to $\frac{3}{7}$ have a two-digit denominator?

Your work:

Problem 44

Poles, a bicycle and a train (extends Problem 20).

a) Leo cycles steadily at 18 km/h along a road whose power poles are equally spaced 50 m apart. How many minutes pass from the moment Leo passes the first pole to the moment he passes the 25th pole?

b) A train travelling at constant speed takes 9 seconds to pass a power pole and 30 seconds to cross a bridge 350 m long (from the moment the front of the train reaches the bridge until the end of the train has just left it). Find the length of the train and its speed in km/h.

Your work:

Problem 45 ★★

Products of digits (extends Problem 21).

a) How many four-digit numbers have digits whose product is 36?

b) Find the smallest positive integer whose digits have product 360.

c) Find the largest positive integer with *all different* digits whose product is 360.

Your work:

Problem 46 ★

Arithmetic grids (extends Problem 22). In a square grid, the numbers in every row and in every column form arithmetic sequences.

a) A 4×4 grid has these four corner numbers: top left 1, top right 10, bottom left 19, bottom right 37. Fill in the whole grid.

b) Find the sum of all 16 numbers in the grid of part a). Compare it with the sum of the four corners: what do you notice? Explain.

c) Prove that in a 3×3 grid of this kind, the number in the centre always equals the mean of the four corner numbers.

1			10
19			37

Your work:

Problem 47 **

"Rotating" numbers (extends Problem 23). In each part, every letter represents a digit, and $A \neq 0$.

a) Find the six-digit number $\overline{2ABCDE}$ such that $\overline{2ABCDE} \times 3 = \overline{ABCDE2}$.

b) Find all digits d (from 1 to 9) for which there is a six-digit number $\overline{dABCDE}$ with $\overline{dABCDE} \times 3 = \overline{ABCDEd}$.

c) Prove that there is no six-digit number $\overline{1ABCDE}$ with $\overline{1ABCDE} \times 2 = \overline{ABCDE1}$.

Your work:

Problem 48 **

A frog on the number line (extends Problem 24). A frog sits at the point 0 on the number line. On its k-th jump ($k = 1, 2, 3, \dots$), the frog jumps exactly k units, to the left or to the right as it likes.

a) After exactly 10 jumps, can the frog be back at 0? Why?

b) After exactly 10 jumps, which points can the frog be at? How many such points are there?

c) What is the smallest number of jumps the frog needs to reach the point 20?

Your work:

Problem 49 *

Numbers containing the block 2025 (extends Problem 28). Consider the six-digit numbers that contain the four digits $2, 0, 2, 5$ *together and in this order* (for example: 720 251, 202 513, 132 025). How many of these numbers are divisible by 45?

Your work:

Problem 50 **

Mean and median (extends Problem 30). A list of five positive integers (repeats allowed) has mean 12 and median 10.

a) What is the largest possible value of the largest number in the list?

b) What is the smallest possible value of the largest number in the list?

c) Answer both questions again if the five numbers must all be different.

Your work:

5 Others: Logic and Puzzles

Big idea. Logical thinking is *the art of drawing certain conclusions from what we already know.* A logic puzzle usually needs no hard calculation: the difficulty lies in *reading carefully*, *recording information systematically* and *not missing any case.* In this chapter you will get to know the following "tools":

- **truth-tellers and liars**, "half-true" statements, and how to negate a statement;
- **case analysis** and **proof by contradiction** (assume, then look for a contradiction);
- **ordering** with arrow diagrams and number lines; telling "*must be true*" apart from "*could be true*";
- **shift ciphers** (the Caesar cipher) and **remainders** on division by 26 (or by 7 for the days of the week);
- **magic squares**, **balance scales** (which are really equations), **sums on figures** and **dice**;
- **Latin squares** and 4×4 **sudoku**.

Four golden habits: *(1) write it down – use symbols; (2) draw it – diagrams, tables; (3) consider every case; (4) check the answer against* ***all*** *the conditions of the problem.*

① True and false statements: truth-tellers and liars

1. Statements. A sentence that is either *true* or *false* (it cannot be both) is called a statement. Examples: "7 is odd" (true), "Paris is the capital of Italy" (false).

2. Truth-tellers and liars. In puzzles, everything a *truth-teller* says is true, and everything a *liar* says is false. Therefore:

- if we know who the speaker is, we know whether the statement is true or false;
- conversely, if we know whether the statement is true or false, we know who the speaker is.

3. Negating a statement (saying "the opposite").

Statement	Negation (true exactly when the statement is false)
Anna *and* Ben are both liars	*At least one* of the two tells the truth
Every apple is red	*There is at least one* apple that is not red
The flower is a red rose	The flower is not red, *or* it is not a rose (or both)

4. "Half-true" statements. If a statement has two parts and we know that *exactly one* part is true, then: first part true ⇒ second part false; first part false ⇒ second part true. Decide true/false for *one* part, and all the other parts follow like a row of dominoes.

5. Lying on certain days. Someone who "lies on Tuesdays" says only false things on Tuesday. To find on which days they *could* make a given statement, make a table: for each day, decide whether the statement is true or false, and compare with whether the person tells the truth or lies on that day.

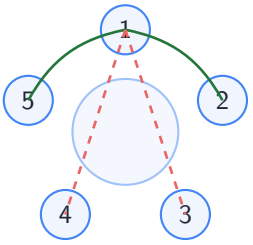
Why is it true? Why can nobody say "I am a liar"? If the speaker were a truth-teller, the statement would be true, so they would be a liar – impossible. If the speaker were a liar, the statement would be false, so they would *not* be a liar – impossible too. Both cases lead to a contradiction, so the sentence can never be spoken (on an "island" with only truth-tellers and liars). □

Common mistake. The negation of "Anna and Ben are both liars" is *not* "Anna and Ben both tell the truth". As soon as *one* of them tells the truth, the original statement is already false.

ThinkAlgo Tip. For a truth-and-lies puzzle, draw a small table: each row is one possibility (who tells the truth, who lies), each column is one statement. Fill each cell with T/F; the row with no contradiction is the answer.

② Case analysis and proof by contradiction

1. Case analysis. When something is unknown, we split into *all* the possibilities (missing none) and examine them one by one. A possibility that leads to something impossible (a contradiction with the problem) is ruled out. The possibility that remains is the answer.
2. Proof by contradiction. To show that a statement P is true, we *assume* that P is false and keep reasoning. If we arrive at something impossible, the assumption was wrong, so P is true.
3. Two contradictory statements. If two statements cannot both be true and cannot both be false (for example Leo says "Helen is lying" about Helen's statement), then *exactly one* of them is true. This is the "key" to very many puzzles: we immediately know that one statement of the pair is true and the other false.
4. Seating around a round table. Rotating the whole table does not change who sits next to whom. So we may *fix* the seat of one person and only consider the seats of the others. Around a round table with 5 seats, each person has exactly 2 neighbours, and exactly 2 people who are *not* next to them.



solid: seat 1 is next to seats 2 and 5

dashed: seat 1 is not next to seats 3, 4

Why is it true? Why is proof by contradiction valid reasoning? A statement can only be true or false. If "P is false" leads to something impossible, then the possibility "P is false" cannot happen. Only the possibility "P is true" remains. This is just case analysis with two cases: P true or P false. □

ThinkAlgo Tip. Start with the "tightest" piece of information – the one that allows the fewest possibilities. For instance, "Jack and Kate do *not* sit next to each other" at a 5-seat table means there is *exactly one* person between them on one side. Fewer possibilities means fewer cases to check!

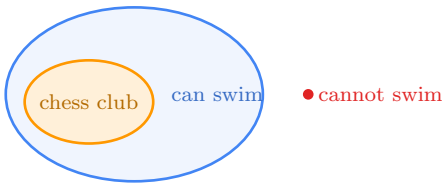
③ Ordering; "must be true" and "could be true"

1. Use symbols. Translate each sentence into an inequality: "Ryan runs faster than Henry" becomes $R > H$; "Kayla ate less than Max" becomes $K < M$ (or $M > K$).
2. Transitivity. If $A > B$ and $B > C$, then $A > C$. This lets us chain inequalities together.
3. Diagrams. Draw each person as a point on a *vertical axis*: whoever is "more" goes higher; or draw an arrow from the "lesser" person to the "greater" one. Two people who are not joined by a chain of arrows in the same direction *cannot be compared* yet.
4. Position in a line. If someone is k-th from the front and m-th from the back of a line, the line has $k + m - 1$ people.
5. "Must be true" and "could be true".

- A statement that **must be true** (is always true): true in *every* situation that satisfies the problem.
- A statement that **could be true**: true in *at least one* situation that satisfies the problem.
- To refute a "must be true" claim, *one counterexample* is enough: a situation that satisfies the problem but in which the claim is false.

6. "Every A is a B". From this we may conclude: "whoever is *not* a B is *not* an A". But we may *not* conclude "whoever is a B is an A".





Why is it true? Why does being k-th from the front and m-th from the back give $k + m - 1$ people? Counting from the front up to that person gives k people, counting from that person to the back gives m people. Adding gives $k + m$, but that person has been counted *twice*, so we subtract 1. Why does "every A is a B" give "not a B means not an A"? Look at the right-hand picture: the "chess club" oval lies *entirely* inside the "can swim" oval. A point outside the big oval is certainly outside the small one. A point inside the big oval, however, may be inside or outside the small one. □

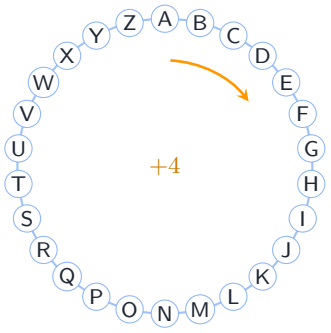
Common mistake. When you see "$A > B$" and "$A > C$", do not jump to conclusions about B and C: they *cannot be compared* yet. A claim like "$B > C$" then only *could* be true; it is not *certainly* true.

④ Shift ciphers and remainders

1. The Caesar cipher. Arrange the 26 letters A, B, C, …, Z of the English alphabet in a circle (after Z comes A again). To encrypt with *shift* k: replace each letter by the letter k places after it, clockwise. To decrypt: shift *back* k places.
2. Numbering the letters. Number them A $= 0$, B $= 1$, …, Z $= 25$. To encrypt letter number x with shift k: compute $x + k$; if the result is ≥ 26, subtract 26 (that is, take the *remainder* of $x + k$ on division by 26).

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

For example, with shift 4: Y $= 24$, $24 + 4 = 28 \geq 26$, $28 - 26 = 2 =$ C. So Y $\to$ C.
3. Shifting several times. Shifting by a and then by b is the same as shifting once by $a + b$. Shifting by 26 brings every letter back to itself, so only the *remainder* of the number of steps on division by 26 matters.
4. Days of the week "go round" too. After Sunday comes Monday again: the days repeat in a cycle of 7. Which day it will be "in n days" depends only on the remainder of n on division by 7.



Shift 4:
A $\to$ E, B $\to$ F, …
W $\to$ A, X $\to$ B,
Y $\to$ C, Z $\to$ D.

Why is it true? Why does a shift of 26 bring us back to the start? The circle has exactly 26 letters, so 26 steps is one full turn. Likewise, a shift of $30 = 26 + 4$ is one full turn followed by 4 more steps: exactly the same as a shift of 4. And 4 is precisely the remainder when 30 is divided by 26. □

Common mistake. The most common mistake is forgetting to "wrap around" at the end of the alphabet: W shifted by 4 does *not* give some strange symbol after Z, but A (W $\to$ X $\to$ Y $\to$ Z $\to$ A). The second mistake: decrypting by shifting *forwards* instead of *backwards*.

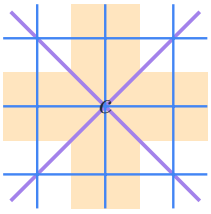
⑤ Magic squares

1. Definition. A 3×3 magic square is a grid of 9 cells filled with 9 different numbers so that the sums of the numbers in every row, every column and both diagonals are equal. This common sum is called the **magic sum**, written S.
2. The magic sum. The three rows cover the whole grid, each cell exactly once, so

$$3S = \text{sum of all 9 numbers}, \qquad S = \frac{\text{sum of all 9 numbers}}{3}.$$

Example: for the numbers $1, 2, \ldots, 9$ the total is 45, so $S = 15$.
3. The centre number. In a 3×3 magic square: **centre number** $= S \div 3$ (the average of the 9 numbers). For the numbers $1, \ldots, 9$ the centre is 5.
4. Quick filling trick. Once you know S and a row (column, diagonal) already has two numbers, the missing number is $S-$ (sum of the two known numbers).

2	7	6
9	5	1
4	3	8



4 lines through the centre

Why is it true? Why is the centre number equal to $S \div 3$? Exactly 4 lines pass through the centre: the middle row, the middle column and the two diagonals (right-hand picture). Adding the sums of these 4 lines gives $4S$. In that sum the centre c is counted 4 times, while every other cell is counted exactly once (check on the picture: each outer cell lies on exactly one of the four lines). Hence

$$4S = (\text{sum of the 9 numbers}) + 3c = 3S + 3c.$$

Subtract $3S$ from both sides: $S = 3c$, that is, $c = S \div 3$. □

ThinkAlgo Tip. Magic squares may use negative numbers too. With the numbers $-6, -5, \ldots, 2$: the total is -18, so $S = -6$ and the centre is -2. The formula works just the same!

⑥ A balance scale is an equation

1. Balanced scale = equality. If a scale balances, the total masses on the two pans are equal. Giving each kind of object a letter for its mass, we can write an equality (an equation).
2. Removing identical objects. Removing identical objects from *both* pans keeps the scale balanced (subtract the same number from both sides).
3. Substitution. If a group of objects A balances a group B, then on any other scale we may replace A by B and the scale stays balanced.
4. Reduce to one unknown. Let the mass of *one* kind of object be x, then use the equalities to express the mass of every other object in terms of x. Comparing two groups of objects then

becomes comparing two expressions in x.



$4O = 3T$. Let $O = x$: $3T = 4x$, $T = \dfrac{4x}{3}$.

Why is it true? Why may we do this? A balanced scale means that the two sides of the equation $4O = 3T$ are equal. Dividing both sides by the same non-zero number still gives a true equation: $T = 4O \div 3$. So however many grams a circle really weighs, a triangle always weighs 4/3 of a circle. For instance, if $x = 30$ g then $T = 40$ g; check: $4 \cdot 30 = 120 = 3 \cdot 40$. □

Common mistake. Never remove an object from *one* pan without removing an identical object from the other. And only remove objects that are *identical* (have the same mass).

⑦ Clever counting: sums on figures, dice, fair games

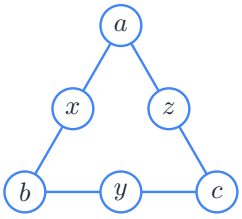
1. How many times is each number counted? When numbers are written in circles along the sides of a figure and every side has the same sum S, add up the sums of *all the sides*. A number at a *corner* belongs to two sides, so it is counted twice; a number in the *middle of a side* is counted once. For a triangle:

$$3S = (\text{sum of all the numbers}) + (\text{sum of the three corner numbers}).$$

2. Lower and upper bounds. The sum of the three corners cannot be smaller than the sum of the three smallest numbers, nor larger than the sum of the three largest. This pins down S; then check each possible value.

3. Dice. On a standard die, *opposite faces add up to* 7: 1 – 6, 2 – 5, 3 – 4. The six faces add up to $1+2+3+4+5+6 = 21$. Therefore: sum of the *hidden* faces $= 21 -$ sum of the visible faces.

4. Fair games. Rolling a die has 6 equally likely outcomes. A game "you win if ..." is *fair* when the chance of winning equals the chance of losing, that is, there are exactly 3 winning and 3 losing outcomes.



left side: $a + x + b$
base: $b + y + c$
right side: $c + z + a$
a, b, c appear twice.



visible $3 + 6 + 2 = 11$
hidden $21 - 11 = 10$

Why is it true? Why is $3S = (\text{sum of all the numbers}) + (a + b + c)$? Add the three sides: $(a + x + b) + (b + y + c) + (c + z + a) = (a + b + c + x + y + z) + (a + b + c)$. Each middle number x, y, z appears once; each corner number a, b, c appears twice, so $a + b + c$ must be added once more. □

Common mistake. For a die standing on a table, the *bottom* face is hidden as well. Do not forget it! The safest method is to use "sum of the 6 faces = 21" and subtract the visible faces.

⑧ Latin squares and sudoku

1. 4×4 Latin square. Write the digits $1, 2, 3, 4$ in a 4×4 grid so that every row and every column contains all four digits (no repeats). A 4×4 **sudoku** adds one more condition: each 2×2 box (thick border) also contains all four digits.

2. Three "regions". Each cell belongs to exactly three regions: one row, one column, one 2×2 box. The digit in the cell must be *different* from every digit already present in all three regions.

3. Two ways of reasoning.

- *Elimination in one cell:* cross out every digit already in the cell's row, column and box; if exactly one digit is left, write it in.
- *Finding a place for a digit:* if, in some region, digit d can go in only one cell, then d must go there.

4. Write "candidates". If a cell still has two possibilities, write both in small (e.g. $\{1, 4\}$) and move on; information elsewhere will settle it.

1		3	
	2		
			4

Cell in row 1, column 2: row 1 already has $1, 3$;
the 2×2 box already has $1, 2$.
The only digit left is 4.

Common mistake. Do not "guess" and write the digit straight in. Every digit you write must have a reason (all other possibilities are ruled out). If you really must try something, write "suppose" clearly and be ready to go back when you meet a contradiction.

Note for parents. Logic puzzles are great to do together with your child: read the problem aloud, let your child retell it in their own words, then draw a diagram or a table together. When your child gives an answer, do not say at once whether it is right or wrong, but ask: "Have you checked it against *each* condition of the problem?". The habit of checking is worth more than any trick.

Example 1 (Truth-tellers and liars)

On an island, every person either always tells the truth or always lies. You meet two people, Peter and Quinn. Peter says: "Quinn is a liar." Quinn says: "Peter and I are the same type of person." Who tells the truth and who lies?

Guided solution.

Step 1. Case: Peter tells the truth

Then Peter's statement is true: Quinn is a liar. So Quinn's statement must be false. Quinn says "we are the same type"; in fact Peter is truthful and Quinn lies – *different* types, so Quinn's

statement is indeed false. **No contradiction.**

Step 2. Case: Peter lies

Then Peter's statement is false: Quinn is *not* a liar, so Quinn tells the truth. Then Quinn's statement must be true: "Peter and Quinn are the same type". But Peter lies and Quinn tells the truth – different types. So Quinn's statement is false after all. **Contradiction!** This case is ruled out.

Step 3. Summary in a table

Peter	Quinn	Peter's statement	Quinn's statement	Conclusion
truthful	liar	true (fine)	false (fine)	**OK**
liar	truthful	false (fine)	false (impossible!)	ruled out

(The cases "both truthful" and "both liars" are ruled out straight away by Peter's statement: if Peter is truthful then Quinn lies; if Peter lies then Quinn is truthful.)

Answer: Peter tells the truth, Quinn lies.

Example 2 (Each person is half right)

Three teams, Red, Blue and Yellow, play a tug-of-war; no two teams share a place. Before the match, three fans predict:

- Grace: "Red first, Blue second." Jack: "Blue first, Yellow second." Lucy: "Blue third, Yellow first."

In the end, each person got exactly one part right. Find the places of the three teams.

Guided solution.

Step 1. Pick one part to split into cases

Each prediction has exactly one true part. Look at Grace's: either "Red first" is true or "Blue second" is true.

Step 2. Case 1: "Blue second" is true (and "Red first" is false)

If Blue is second, Blue is not first, so Jack's part "Blue first" is false; hence his other part "Yellow second" must be true. But Blue is already second, so Yellow cannot be second as well. **Contradiction**, ruled out.

Step 3. Case 2: "Red first" is true (and "Blue second" is false)

Blue is not first (Red is) and not second, so **Blue is third**. That leaves Yellow second.

Step 4. Check all three people

Person	First part	Second part
Grace	Red first: true	Blue second: false
Jack	Blue first: false	Yellow second: true
Lucy	Blue third: true	Yellow first: false

Each person has exactly one part right.

Answer: Red first, Yellow second, Blue third.

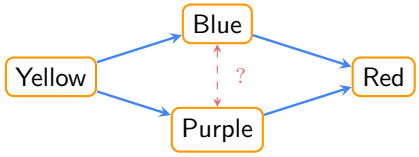
Example 3 (Ordering – must or could?)

There are four coloured pencils, red, blue, yellow and purple, all of different lengths. We know: the red pencil is longer than the blue one; the yellow pencil is shorter than the blue one; the purple pencil is longer than the yellow one but shorter than the red one.
a) Which pencil is certainly the longest? Which is certainly the shortest?
b) Is the statement "The purple pencil is the second longest" certainly true, possibly true, or certainly false?

Guided solution.

Step 1. Use symbols
Call the lengths R, B, Y, P. We are told: $R > B$; $B > Y$; $P > Y$; $R > P$.

Step 2. Draw an arrow diagram (arrows point from the shorter pencil to the longer one)



Step 3. Answer part a
From $R > B > Y$ and $R > P > Y$: the red pencil is longer than all three others and the yellow pencil is shorter than all three others. So **red is certainly the longest** and **yellow is certainly the shortest**.

Step 4. Answer part b: find two situations
No condition compares Blue with Purple. Try two sets of lengths (cm), both satisfying every condition:

- Red 15, Purple 13, Blue 12, Yellow 8: purple is second longest – the statement is *true*;
- Red 15, Blue 12, Purple 10, Yellow 8: purple is third longest – the statement is *false*.

So "The purple pencil is the second longest" is only **possibly true**, not certainly true.

Answer: a) Red is the longest, yellow the shortest (certainly). b) Possibly true (not certain).

Example 4 (Shift cipher – finding the shift)

Tom encrypts a word with a Caesar cipher and gets **JVKL**. He tells us that with this encryption the letter A becomes H.
a) Find the original word. b) After this shift, what is the smallest number of further steps (in the same direction) that brings every letter back to its original letter?

Guided solution.

Step 1. Find the shift
A = 0, H = 7. A becomes H, so the shift is $k = 7$.

Step 2. Decrypt: shift back 7 (subtract 7; if the result is negative, add 26)

Cipher letter	J	V	K	L
Number	9	21	10	11
Subtract 7	2	14	3	4
Original letter	**C**	**O**	**D**	**E**

The original word is **CODE**.

Common mistake. Subtracting 7 does not always give a negative number; we add 26 (wrap around past A) only when the number of the cipher letter is less than 7. Here all four letters have numbers ≥ 7, so no wrapping is needed.

Step 3. Part b

A total shift of 26 steps goes once round the circle. We have shifted 7 steps, so we need $26 - 7 = 19$ more steps.

Check: Encrypt CODE again with shift 7: C(2) $\to 9 =$ J, O(14) $\to 21 =$ V, D(3) $\to 10 =$ K, E(4) $\to 11 =$ L. We get JVKL.

Answer: a) CODE. b) 19 steps.

Example 5 (Completing a magic square)

Place the even numbers $2, 4, 6, \ldots, 18$ (each once) in the grid so that every row, column and diagonal has the same sum.

4		12
	6	

Guided solution.

Step 1. Find the magic sum S

Sum of the nine numbers: $2+4+\cdots+18 = 2\times(1+2+\cdots+9) = 2\times 45 = 90$. So $S = 90 \div 3 = 30$.

Step 2. The centre number

Recall. In a 3×3 magic square the centre number equals $S \div 3$.

Centre $= 30 \div 3 = 10$.

Step 3. Fill in the lines that already have two numbers

- Top row: middle cell $= 30 - 4 - 12 = 14$.
- Diagonal 4 – 10 – ?: bottom-right corner $= 30 - 4 - 10 = 16$.
- Diagonal 12 – 10 – ?: bottom-left corner $= 30 - 12 - 10 = 8$.
- Left column: $30 - 4 - 8 = 18$; right column: $30 - 12 - 16 = 2$.

4	14	12
18	10	2
8	6	16

Check: Bottom row: $8 + 6 + 16 = 30$; middle row: $18 + 10 + 2 = 30$; middle column: $14 + 10 + 6 = 30$. Every even number from 2 to 18 is used exactly once.

Answer: Top row $4, 14, 12$; middle row $18, 10, 2$; bottom row $8, 6, 16$.

Example 6 (A balance-scale problem)

Both scales below are balanced; identical shapes have equal masses. How many circles does one square weigh?



Guided solution.

Step 1. Write equalities

Let the masses of a square, a triangle and a circle be V, T, O. Scale 1: $2V = 3T$. Scale 2: $T = 2O$.

Step 2. Substitute

On scale 1, replace each triangle by two circles: $2V = 3 \times 2O = 6O$.

Step 3. Halve both pans

$2V = 6O$, so $V = 3O$.

Check: Take $O = 1$, $T = 2$, $V = 3$: scale 1: $2 \times 3 = 6 = 3 \times 2$; scale 2: $2 = 2 \times 1$. Correct.

Answer: One square weighs as much as 3 circles.

Example 7 (Sums along the sides of a triangle)

Place the numbers $1, 2, 3, 4, 5, 6$ in the six circles (three at the corners, three in the middles of the sides) so that the three numbers on each side add up to the same number S. Find all possible values of S.

Guided solution.

Step 1. Count how many times each number is counted

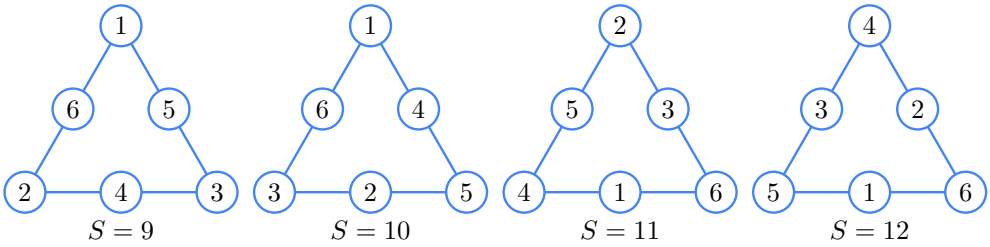
> **Recall.** When adding the three side sums, corner numbers are counted twice and middle numbers once.

$3S = (1 + 2 + \cdots + 6) + (\text{sum of the three corners}) = 21 + (\text{sum of the three corners})$.

Step 2. Pin it down

The smallest possible corner sum is $1 + 2 + 3 = 6$, the largest is $4 + 5 + 6 = 15$. So $27 \le 3S \le 36$, that is, $9 \le S \le 12$.

Step 3. Check each value with an actual arrangement



For instance, with $S = 10$: $1 + 6 + 3 = 10$, $3 + 2 + 5 = 10$, $5 + 4 + 1 = 10$.

Answer: S **can be** 9, 10, 11 **or** 12.

Example 8 (4×4 sudoku)

Complete the sudoku: every row, every column and every thick-bordered 2×2 box must contain the digits $1, 2, 3, 4$.

2	1		
		1	
	4		
			3

Guided solution.

Step 1. **Top-left box**

It already has $2, 1$ in row 1, so row 2, columns 1 and 2, are $\{3, 4\}$. Column 2 already has a 4 (in row 3), so row 2 column 2 is 3 and row 2 column 1 is 4.

Step 2. **Row 2 and the top-right box**

Row 2: $4, 3, 1, _ \Rightarrow$ the last cell is 2. Row 1, columns 3 and 4, are $\{3, 4\}$; column 4 already has a 3 (row 4), so row 1 column 4 is 4 and row 1 column 3 is 3.

Step 3. **Column 2 and the bottom-left box**

Column 2: $1, 3, 4, _ \Rightarrow$ row 4 column 2 is 2. Column 1 is still missing $\{1, 3\}$; row 4 already has a 3, so row 4 column 1 is 1 and row 3 column 1 is 3.

Step 4. **The last two rows**

Row 3: $3, 4, _, _$ needs $\{1, 2\}$; column 3 already has a 1 (row 2), so row 3 column 3 is 2 and column 4 is 1. Row 4: $1, 2, _, 3 \Rightarrow$ the empty cell is 4.

2	1	3	4
4	3	1	2
3	4	2	1
1	2	4	3

Check: Every row, column and 2×2 box contains all of $1, 2, 3, 4$.

Answer: **Row** 1: 2 1 3 4; **row** 2: 4 3 1 2; **row** 3: 3 4 2 1; **row** 4: 1 2 4 3.

Core practice

Problem 1

Mum has just bought a new pot of flowers. Three children guess what flower it is. Anna guesses: "a red rose". Ben guesses: "a purple daisy". Chloe guesses: "a red dahlia". Mum says that each child got *exactly one* of the two things right – either the colour or the kind of flower. The flower Mum bought is a

(A) purple dahlia (B) purple rose (C) red dahlia

(D) yellow rose (E) red daisy

Your work:

Problem 2

The 26 letters of the English alphabet are written in order A, B, C, ..., Z clockwise around a circle. To *encrypt* a message, each letter is replaced by the letter 4 places after it, clockwise (this is called a *Caesar cipher*). For example, the word ZAP is encrypted as DET. The word WIN is encrypted as

(A) ALN (B) ZLN (C) AMR (D) AMQ (E) ZMQ

Your work:

Problem 3

Five friends, Ryan, Henry, Felix, Tom and Oscar, run a race; no two finish at the same time. Ryan runs faster than Henry and Felix. Henry runs slower than Felix. Tom runs faster than Ryan but slower than Oscar. Who finishes fourth?

(A) Felix (B) Henry (C) Oscar (D) Ryan (E) Tom

Your work:

Problem 4

In a magic square, the numbers in every row, every column and both diagonals have the same sum. The magic square shown uses each of the integers from -6 to 2 exactly once. The value of Y is

1		Y
-4		
-3		-5

(A) -1 (B) 0 (C) -6

(D) 2 (E) -2

Your work:

Problem 5

A game is called *fair* if your chance of winning equals your chance of losing. You roll a fair six-sided die. How many of the following four games are fair?

- You win if you roll a 2.
- You win if you roll an even number.
- You win if you roll a number less than 4.
- You win if you roll a number divisible by 3.

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Your work:

Problem 6

Five friends, Jack, Kate, Leo, Mia and Noah, sit on the 5 chairs around a round table. Leo and Mia sit next to each other. Jack and Kate do *not* sit next to each other. The two people sitting on either side of Noah are

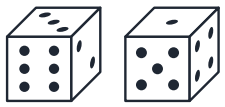
(A) Jack and Leo (B) Jack and Kate (C) Kate and Mia

(D) Leo and Mia (E) Mia and Jack

Your work:

Problem 7

On each die, the numbers of pips on opposite faces always add up to 7. Two dice stand on a table as shown. The total number of pips on all the faces of the two dice that *cannot be seen* is



(A) 14 (B) 20 (C) 21

(D) 24 (E) 30

Your work:

Problem 8

Chris lies on Fridays, Saturdays and Sundays and tells the truth on the other days. Mark lies on Tuesdays, Wednesdays and Thursdays and tells the truth on the other days. On which day of the week could both of them say: "Tomorrow I will lie"?

(A) Monday (B) Thursday (C) Friday (D) Sunday (E) Tuesday

Your work:

Problem 9

Both scales below are balanced; identical shapes have equal masses.



Which of the following groups has the same mass as the group ●▲▲▲ (one circle and three triangles)?

(A) ■■■ (B) ▲▲■■ (C) ●●●●

(D) ●●●■ (E) ●●●

Your work:

Problem 10

Five friends, Chris, Max, Brandon, Kayla and Tanya, go strawberry picking; no two of them pick the same number of strawberries. Chris picks more than Max. Brandon picks fewer than Kayla. Kayla picks fewer than Max but more than Tanya. Who picks the second most?

(A) Brandon (B) Chris (C) Kayla (D) Max (E) Tanya

Your work:

Problem 11

Three friends, Amy, Helen and Tom, count the comics and the science books they read over the summer (no two of them read the same number of comics, and no two read the same number of science books). Amy read more comics than Helen but fewer science books than Helen. Tom read more comics than Helen but fewer science books than Amy. Which of the following statements **must** be true?

(A) Tom read the most comics.

(B) Amy read the most comics.

(C) Helen read the fewest comics.

(D) Amy read the fewest science books.

(E) Tom read the most science books.

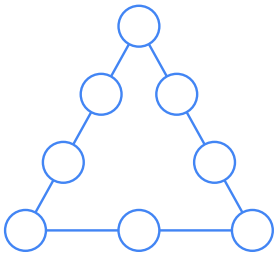
Your work:

Problem 12

Each of the numbers $1, 2, 3, 4, 5, 6, 7, 8$ is placed in a different circle of the triangle shown, so that the sums of the numbers on the three sides of the triangle (the two slanted sides have 4 circles each, the base has 3 circles) are all equal to the same number S. The sum of all the different possible values of S is

(A) 85 (B) 99 (C) 66

(D) 81 (E) 67



Your work:

Problem 13

The 4×4 grid shown is to be filled so that each row and each column contains the digits $1, 2, 3, 4$. The grid is divided into four 2×2 squares (thick borders), and each of these squares must also contain the digits $1, 2, 3, 4$. Which digit goes in cell P?

1		3	
	2		
	P		
			4

(A) 1

(B) 2

(C) 3

(D) 4

(E) The digit cannot be determined

Your work:

Problem 14

On an island, every person either always tells the truth or always lies. You meet two people, Alice and Ben. Alice says: "We are both liars." Which of the following is correct?

(A) Both tell the truth.

(B) Both are liars.

(C) Alice tells the truth, Ben lies.

(D) Alice lies, Ben tells the truth.

(E) It cannot be determined.

Your work:

Problem 15

A vase has been broken. Exactly one of the four friends David, Helen, Leo and Mia broke it. When asked, they answer:

- David: "Helen broke it." Helen: "Leo broke it."
- Leo: "Helen is lying." Mia: "I did not break it."

We know that **exactly one** of them is telling the truth. Who broke the vase?

(A) David (B) Helen (C) Leo (D) Mia (E) Cannot tell

Your work:

Problem 16

Kate encrypts a short wish with a Caesar cipher with shift 5 (each letter is replaced by the letter 5 places after it on the circle of letters A–Z, where A follows Z). She gets: **MFAJ KZS**. The original wish is

(A) HAVE FUN (B) RKFO PEX (C) IBWF GVO

(D) HAVE FUM (E) GZUD ETM

Your work:

Problem 17

Nora encrypts the letter M with a Caesar cipher three times in a row: first with shift 9, then with shift 11 (applied to the result of the first step), and finally with shift 9 again. After the three steps, which letter has M become?

(A) V (B) G (C) J (D) P (E) O

Your work:

Problem 18

Five friends, Nina, Olivia, Peter, Quinn and Sam, all have different heights. Nina is taller than Olivia. Peter is shorter than Quinn but taller than Nina. Sam is shorter than Olivia. When the five friends line up by height, who stands *exactly in the middle*?

(A) Nina (B) Olivia (C) Peter (D) Quinn (E) Sam

Your work:

Problem 19

In a queue for tickets, Henry is 7th from the front and 12th from the back. How many people are in the queue?

(A) 17 (B) 18 (C) 19 (D) 20 (E) 5

Your work:

Problem 20

Place the numbers $1, 2, 3, \ldots, 9$ (each once) in the grid shown to make a magic square (the numbers in every row, every column and both diagonals have the same sum). Three numbers are already filled in. Complete the grid. What is the number in the bottom-left corner?

2	7	
		1

Your work:

Problem 21

The numbers $3, 4, 5, \ldots, 11$ (nine consecutive whole numbers) are placed in the grid shown, each once, to make a magic square. The number x in the bottom-right corner is

(A) 6 (B) 8 (C) 10

(D) 3 (E) 5

4	9	
11		
		x

Your work:

Problem 22

Both scales below are balanced. The circles all have the same mass.



How many grams does the square ▢ weigh?

Your work:

Problem 23

There are four kinds of boxes: red, yellow, blue, purple; boxes of the same colour have equal masses. The following three scales are all balanced:



How many yellow boxes does one purple box weigh?

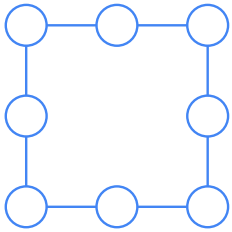
(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Your work:

Problem 24

The numbers $1, 2, 3, 4, 5, 6, 7, 8$ are placed in the eight circles of the square shown (four at the corners, four in the middles of the sides), each once, so that the three numbers on each side add up to 13. The sum of the four corner numbers is

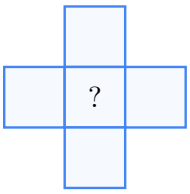
(A) 12 (B) 13 (C) 16 (D) 20 (E) 36



Your work:

Problem 25

Place the numbers $1, 2, 3, 4, 5$ in the five cells of the plus-shaped figure shown, each once, so that the sum of the three numbers in the horizontal row equals the sum of the three numbers in the vertical column. How many different values can the centre cell take?

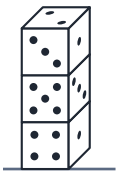


(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Your work:

Problem 26

Three standard dice (opposite faces add up to 7) are stacked on a table as shown. The top face shows 2 pips. The total number of pips on the five hidden *horizontal* faces (the faces pressed together between two dice and the bottom face resting on the table) is



(A) 14 (B) 17 (C) 19 (D) 21 (E) 23

Your work:

Problem 27

The 4×4 grid shown is to be filled so that every row, every column and every thick-bordered 2×2 box contains the digits $1, 2, 3, 4$. The digit in cell Q is

	3		2
			1
4			
		2	Q

(A) 1

(B) 2

(C) 3

(D) 4

(E) It cannot be determined

Your work:

Problem 28

In class 7A, *all* members of the chess club can swim. Leo cannot swim. Nora can swim. Which of the following statements **must** be true?

(A) Leo is in the chess club.

(B) Leo is not in the chess club.

(C) Nora is in the chess club.

(D) Nora is not in the chess club.

(E) Everyone in the class who can swim is in the chess club.

Your work:

Problem 29

Mr Fox lies on Mondays, Tuesdays and Wednesdays and tells the truth on the other days of the week. One day, Fox says: "Yesterday I lied." Which day could it be?

(A) Only Monday.

(B) Only Thursday.

(C) Monday or Thursday.

(D) Wednesday or Thursday.

(E) Sunday or Monday.

Your work:

Problem 30

Four friends, Amy, Ben, Chloe and David, run a race; no two finish at the same time. Three spectators make predictions:

- Henry: "Amy comes first, Ben comes second."
- Kate: "Chloe comes first, David comes fourth."
- Lucy: "Ben comes third, David comes second."

In the end, each spectator got *exactly one* part right. Find the finishing order of the four friends.

Your work:

Extension and challenge

Problem 31

On an island, every inhabitant is either a *truth-teller* (always tells the truth) or a *liar* (always lies).

a) You meet two inhabitants, Kate and Leo. Kate says: "Both of us are liars." What type is each of them?

b) You meet three inhabitants, Anna, Ben and Chloe.
Anna says: "Ben is a liar."
Ben says: "Chloe is a liar."
Chloe says: "Anna and Ben are both liars."
What type is each of them?

Your work:

Problem 32

Mia lies on Mondays, Tuesdays and Wednesdays and tells the truth on the other days of the week. Henry lies on Thursdays, Fridays and Saturdays and tells the truth on the other days. One day, both of them say: "Yesterday was a day on which I lie." Which day of the week is it?

Your work:

Problem 33

Six friends Lucy, Max, Noah, Olivia, Peter and Quinn run a race. No two of them finish at the same time. We know that:

- Max finishes after Lucy but before Noah;
- Olivia finishes after Noah;
- Peter finishes *immediately after* Lucy;
- Quinn is neither first nor last, and Quinn finishes before Max.

Find the finishing order of the six friends. Who finishes third?

Your work:

Problem 34

Five friends Hannah, Kevin, Lily, Maya and Tom sit in a row of 5 chairs numbered $1, 2, 3, 4, 5$ from left to right. We know that: Hannah and Kevin sit next to each other; Lily and Maya do *not* sit next to each other; Tom sits at one end of the row (chair 1 or chair 5).

a) How many seating arrangements satisfy these conditions?

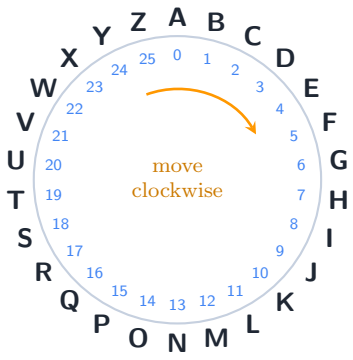
b) Which of the following statements is *certainly true*?
(1) Lily sits at one end of the row.
(2) Hannah does not sit at an end of the row.
(3) Kevin sits in chair 3.

c) If we also know that Lily sits in chair 1 and Hannah sits to the left of Kevin, find each friend's seat.

Your work:

Problem 35

Number the letters $A, B, C, \ldots, Z$ (the 26-letter English alphabet) as $0, 1, 2, \ldots, 25$. The *Caesar cipher with key k* replaces each letter with number x by the letter whose number is the *remainder* of $x + k$ when divided by 26 (spaces are kept).



a) Encrypt the text MATH CLUB with key $k = 5$.

b) Decrypt the text FVB DPU, knowing that the key is $k = 7$.

c) In a message, the word CAT was encrypted as RPI. Find the key k (with $0 \leq k \leq 25$), then decrypt the word SDV from the same message.

d) A message is encrypted with key 10, and the result is encrypted once more with key 20. These two encryptions together are the same as *one* encryption with which key?

e) Sam encrypts a word with key 4, then encrypts the result with key 4 again, and so on. After how many encryptions, at least, does Sam get back exactly the original word?

Your work:

Problem 36

Again number $A = 0, B = 1, \ldots, Z = 25$. A *multiplicative cipher* replaces the letter with number x by the letter whose number is the remainder of $3x + 2$ on division by 26.

a) Encrypt the word MAP.

b) Decrypt the word EGW.

c) Nathan suggests using $2x + 2$ instead of $3x + 2$. Find two different letters with the same cipher letter. Why does that make decryption impossible?

d) Explain why, with $3x + 2$, two different letters always get two different cipher letters.

Your work:

Problem 37

In a 3×3 *magic square*, the numbers in every row, every column and both diagonals have the same sum (called the *magic sum* M).

a) Let c be the number in the centre cell. Prove that $M = 3c$.

b) Fill in the empty cells of the magic square below.

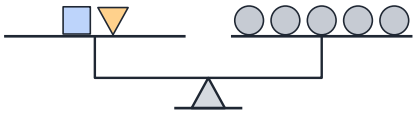
9	19	
		13

Your work:

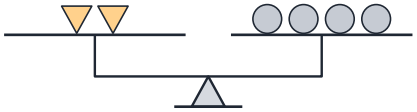
Problem 38

The three equal-arm balances below are all balanced. Objects of the same shape have the same mass.

(1)



(2)



(3)



a) How many ◯ does each of ▽, □, ☆ weigh?

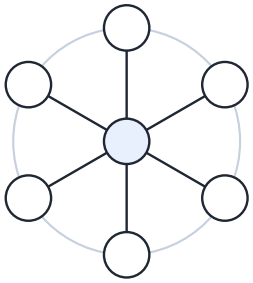
b) How many ▽ are needed to balance ☆☆□□?

c) Each ◯ weighs 12 g. How many grams do the three objects ☆▽□ weigh altogether?

Your work:

Problem 39

The "wheel" below has 7 circles: one at the hub and 6 on the rim, forming 3 straight "spokes" of 3 circles each (two opposite circles on the rim and the hub). Place the numbers $1, 2, 3, 4, 5, 6, 7$, one in each circle, so that the three numbers on each spoke have the same sum S.



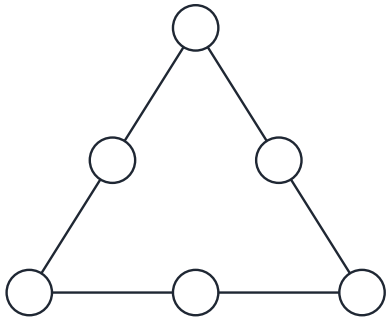
a) Let c be the number at the hub. Prove that $3S = 28 + 2c$.

b) Deduce that the hub can only be 1, 4 or 7. In each case find S and give one arrangement.

Your work:

Problem 40

Place the numbers $1, 2, 3, 4, 5, 6$, one in each circle on the sides of the triangle (three circles at the vertices, three at the midpoints of the sides), so that the three numbers on each side have the same sum S.



a) Let V be the sum of the three numbers at the vertices. Prove that $3S = 21 + V$. Hence find all possible values of S.

b) For each value of S, give one arrangement.

c) When $S = 10$, which three numbers are at the vertices? Explain why there is no other possibility.

Your work:

Problem 41 ★★

Magic squares with the numbers 1 to 9. Place the numbers $1, 2, \ldots, 9$ (each exactly once) in a 3×3 grid so that the three numbers in every row, every column and both diagonals have the same sum.

a) Find the magic sum M and prove that the centre cell must contain 5.

b) Prove that the number 9 cannot be in a corner of the grid.

c) How many such magic squares are there altogether? Explain.

Your work:

Problem 42 ★

4×4 **Sudoku.** Each row, each column and each 2×2 square (thick border) must contain all the digits $1, 2, 3, 4$.

a) Complete grid (I) below (only 4 given digits!). Explain each step.

b) Grid (II) has first row $1, 2, 3, 4$ and first column $1, 3, 2, 4$. In how many ways can grid (II) be completed?

	2		
		1	
			3
4			

(I)

1	2	3	4
3			
2			
4			

(II)

Your work:

Problem 43

6×6 **Sudoku.** Fill in the digits 1 to 6 so that every row, every column and every 2×3 block (thick border) contains all six digits $1, 2, 3, 4, 5, 6$. The grid has only 9 given digits.

a) Fill in the first row and explain why each cell can be filled in only one way.

b) Complete the whole grid.

3				2	6
				6	
		4	1		
		1			
6				3	

Your work:

Problem 44

Finding the fake (lighter) coin. There are 9 identical-looking coins; exactly one of them is fake and *lighter* than the others (the genuine coins all have the same mass). We only have an equal-arm balance (no weights); each weighing only tells us which side is heavier or that both sides are equal.

a) Show how to find the fake coin with only 2 weighings.

b) Explain why 1 weighing is not enough.

c) With 3 weighings, among at most how many coins can we find the (lighter) fake coin? Describe the weighings.

Your work:

Problem 45

The famous 12-coin problem. There are 12 identical-looking coins, exactly one of which is fake. The fake coin has a different mass from the genuine ones, but we do *not* know whether it is heavier or lighter. We may only use an equal-arm balance (no weights).

a) How many "possibilities" must be distinguished? Why are 2 weighings certainly not enough?

b) Show how to use exactly 3 weighings to find the fake coin *and* tell whether it is heavier or lighter.

Your work:

Problem 46

A mini football tournament. Four teams A, B, C, D play a single round robin (each team plays every other team exactly once). A win gives 3 points, a draw 1 point, a loss 0 points. At the end, team A has 7 points, B has 5, C has 3 and D has 1.

a) How many matches were drawn?

b) Determine the result (win/draw/loss) of every match.

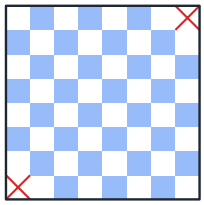
Your work:

Problem 47 *

Invariants.

a) The numbers $1, 2, 3, \ldots, 10$ are written on the board. In each turn, Oscar erases any two numbers a and b and writes the number $|a - b|$ (the larger minus the smaller). After 9 turns, only one number is left on the board. Can it be 0? Can it be 1?

b) Two opposite corner squares are cut off an 8×8 chessboard (see the figure). Can the remaining 62 squares be covered exactly by 31 dominoes, each covering two adjacent squares?





one domino

Your work:

Problem 48 ⋆

An Einstein-style puzzle. Four houses are numbered $1, 2, 3, 4$ from left to right. Each house is painted a different colour (red, yellow, green, white), has one inhabitant (Anna, Ben, Chloe, David), and each inhabitant keeps a different animal (dog, cat, fish, rabbit). We know that:

(1) Anna lives in the red house.

(2) The green house is *immediately to the right* of the white house.

(3) Ben keeps the dog.

(4) The fish owner lives in house 1.

(5) David lives right next to the cat owner.

(6) House 2 is white.

(7) Anna keeps the rabbit.

Who keeps the cat? Whose house is yellow?

Your work:

Problem 49 ⋆

Clocks and calendars.

a) 1 January 2026 is a Thursday. On which day of the week is 2 September 2026? On which day is 1 January 2027?

b) At 3:40, what angle do the hour hand and the minute hand make (the angle not exceeding 180°)?

c) After exactly 3 o'clock, how many minutes later does the minute hand first catch up with the hour hand?

d) ⋆ Prove that every year has at least one Friday the 13th.

Your work:

Problem 50 ★★

The pigeonhole principle.

a) Class 7A has 40 students. Prove that at least 4 of them were born in the same month.

b) Prove that among any 5 integers there are always two whose difference is divisible by 4.

c) Choose any 6 different numbers from $1, 2, \ldots, 10$. Prove that two of the chosen numbers always add up to 11. If only 5 numbers are chosen, is this still certain?

d) At a meeting there are 10 friends; some pairs of friends shake hands (each pair at most once). Prove that two friends have shaken the same number of hands.

Your work: